TCP

transmission control protocol (TCP)

implements reliable streams of bytes

similar mechanism to what we've described

TCP extras/differences

bidirectional —

- separate sequence numbers in each direction
- can combine data (from A to B) with acknowledgment (from B to A)

sequence numbers are byte numbers —

- can retransmit data in different sized packets
- sequence numbers = index of *first byte* sent
- acknowledgment numbers = 1 + index of *last byte* acknowledged

dynamic/variable window sizes

- we'll discuss strategies later

offical name for packets = *segments*

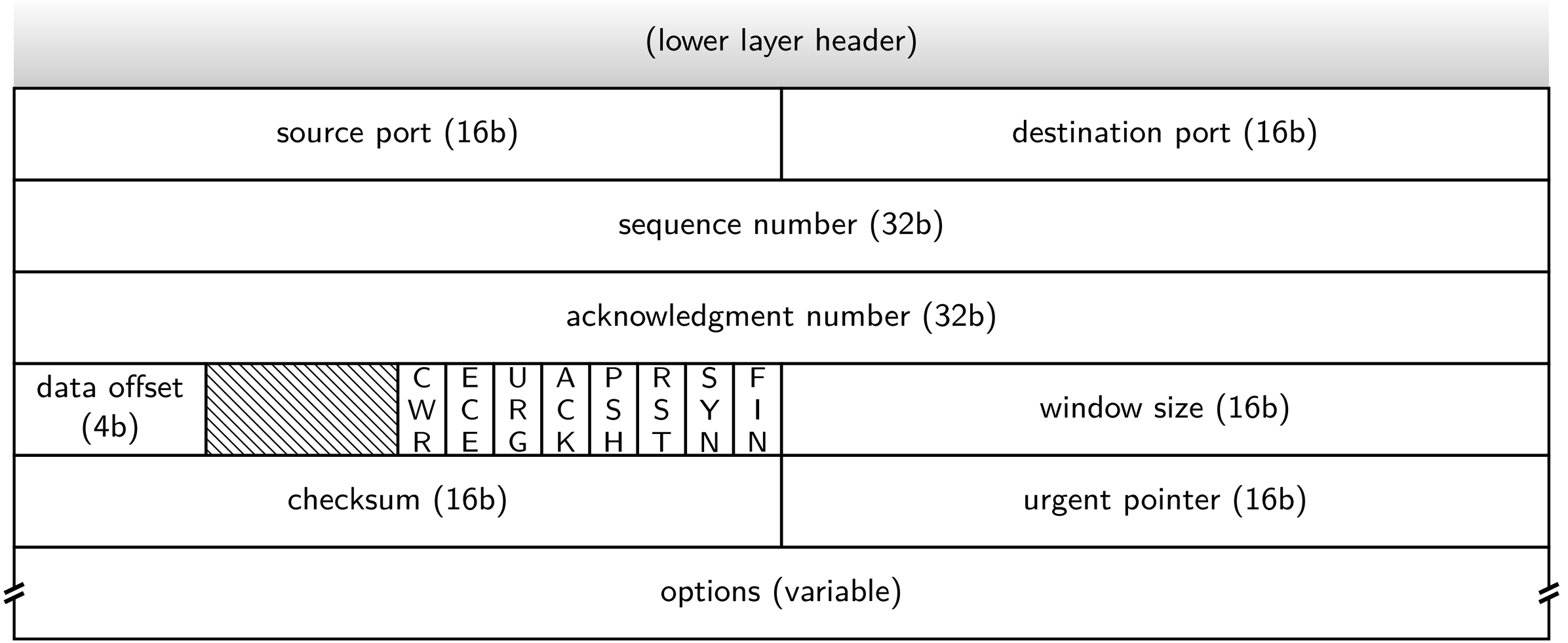
exercise (1)

direction	seq	ack	data len
→	100	200	200
→	300	200	300
←	300	600	0
←	300	600	100
→	?	?	0

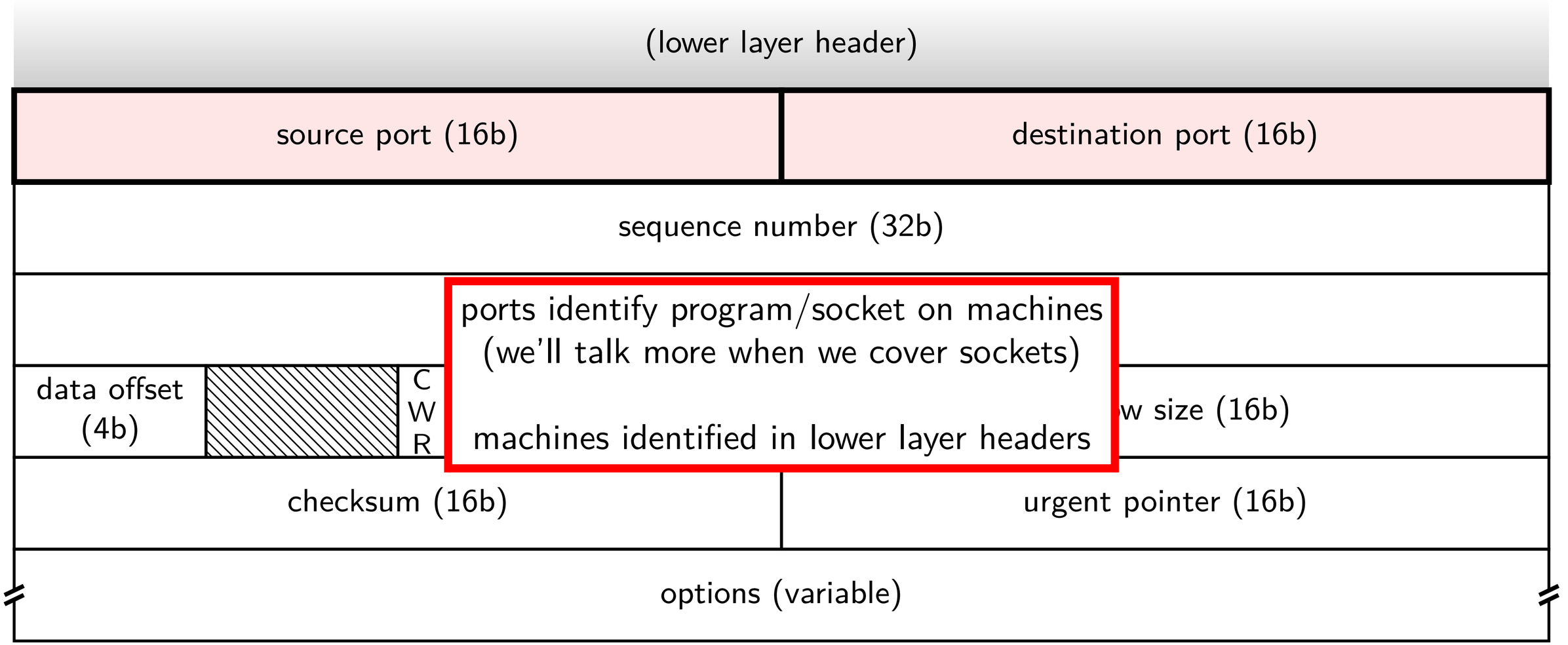
exercise (2)

direction	seq	ack	data len
→	100	200	200
→	300	200	300
←	?	?	100
→	600	300	0

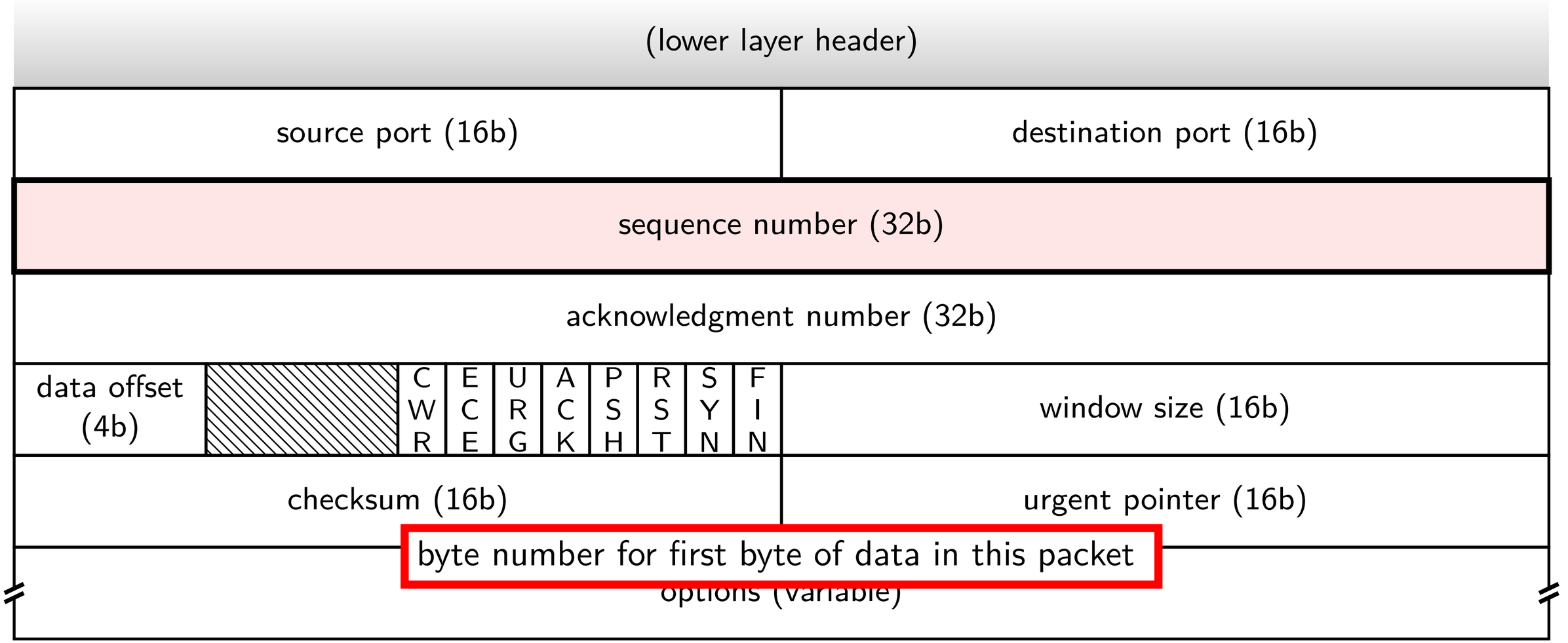
TCP segment format



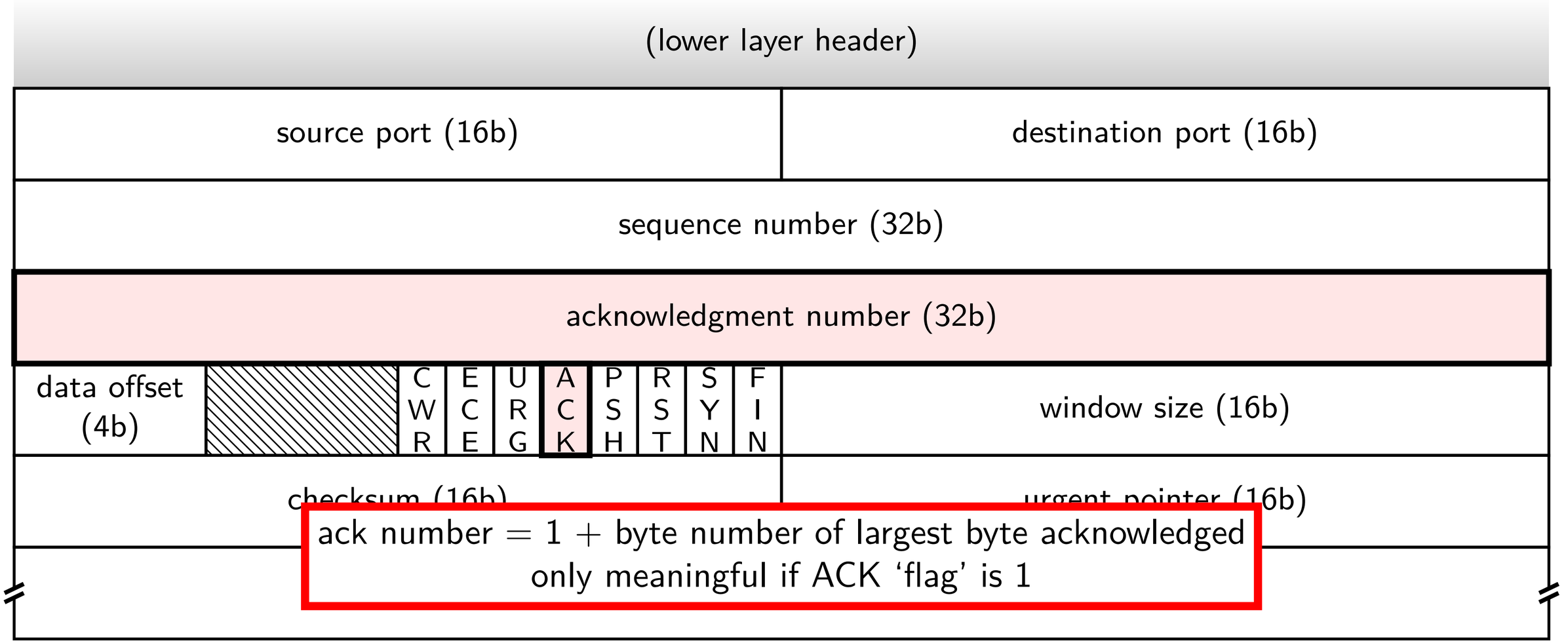
TCP segment format



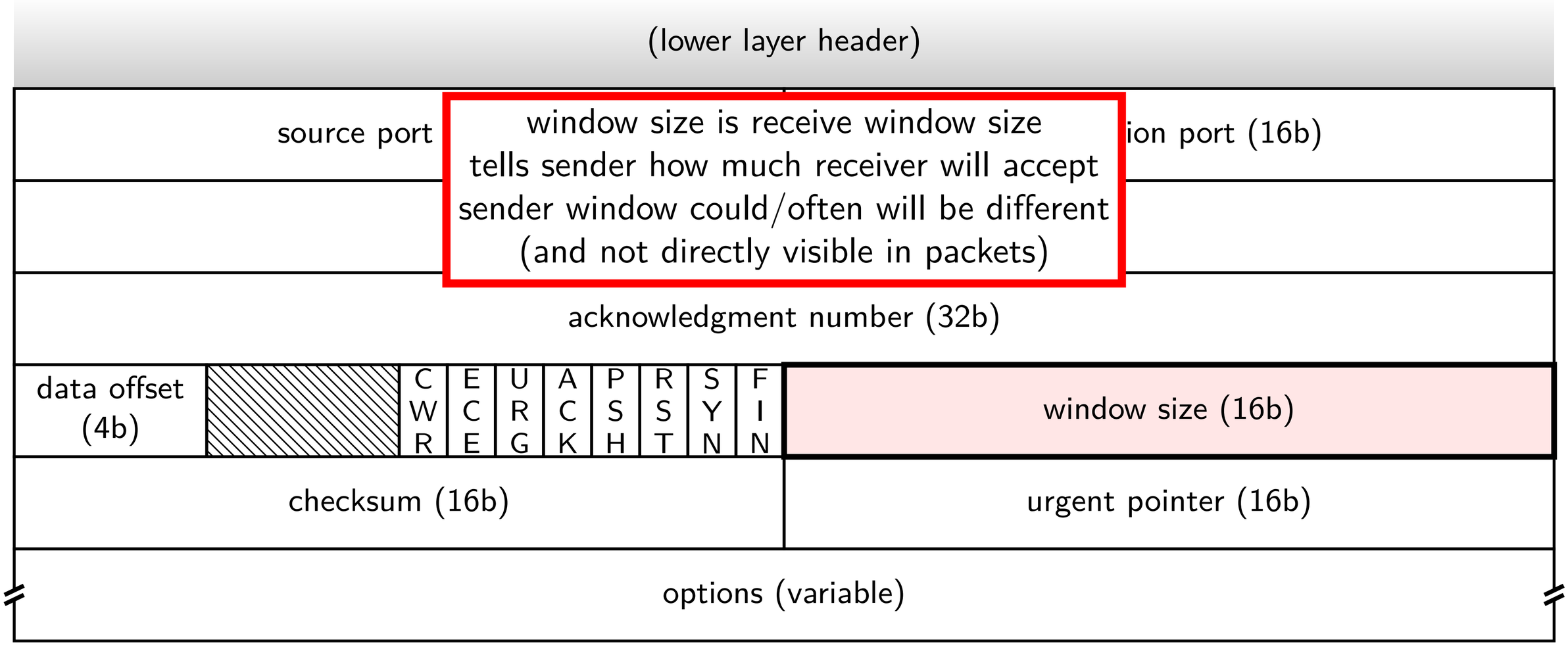
TCP segment format



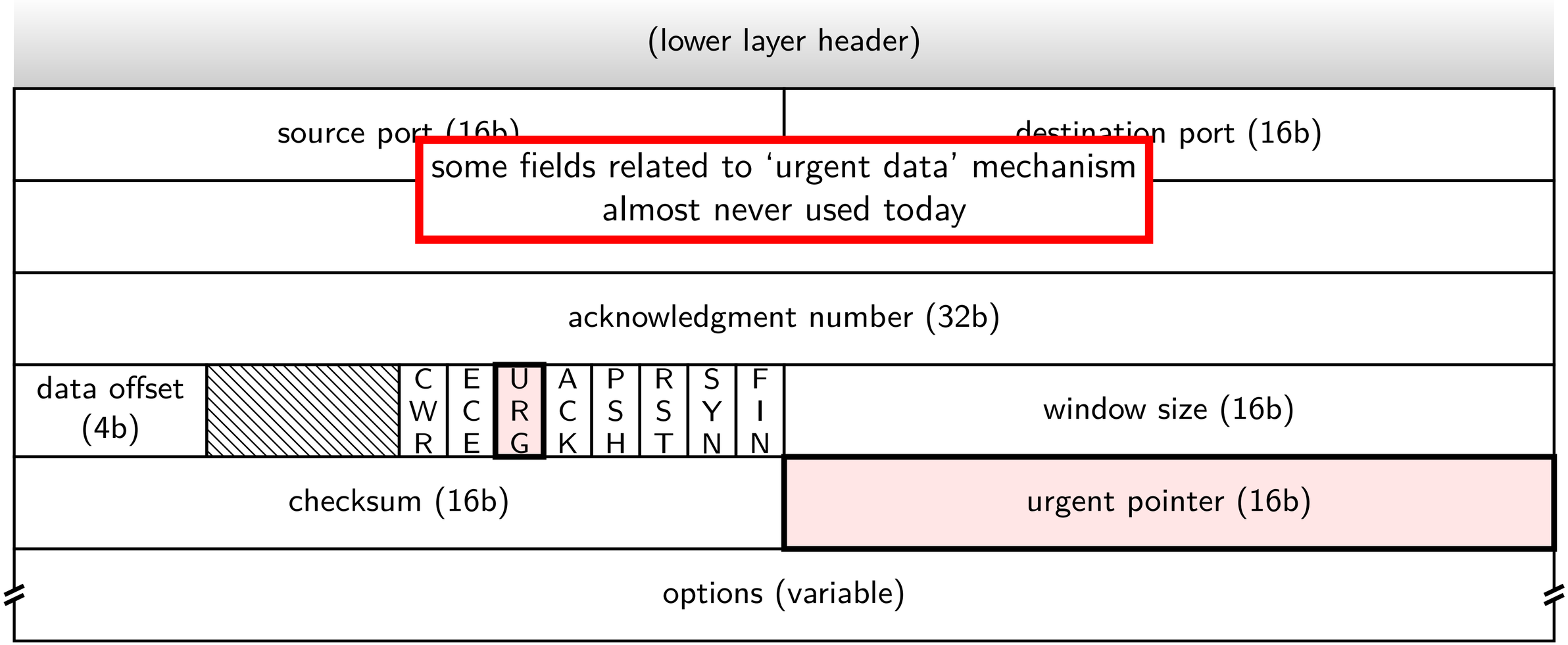
TCP segment format



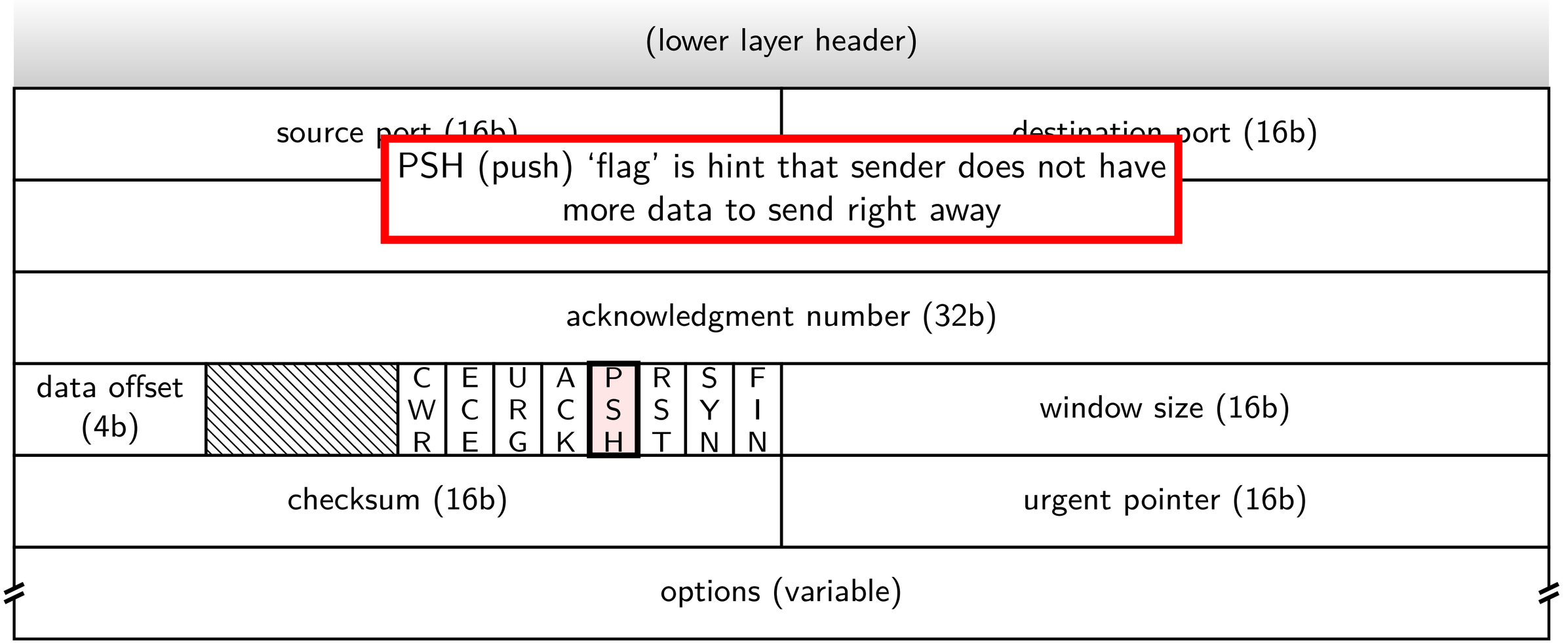
TCP segment format



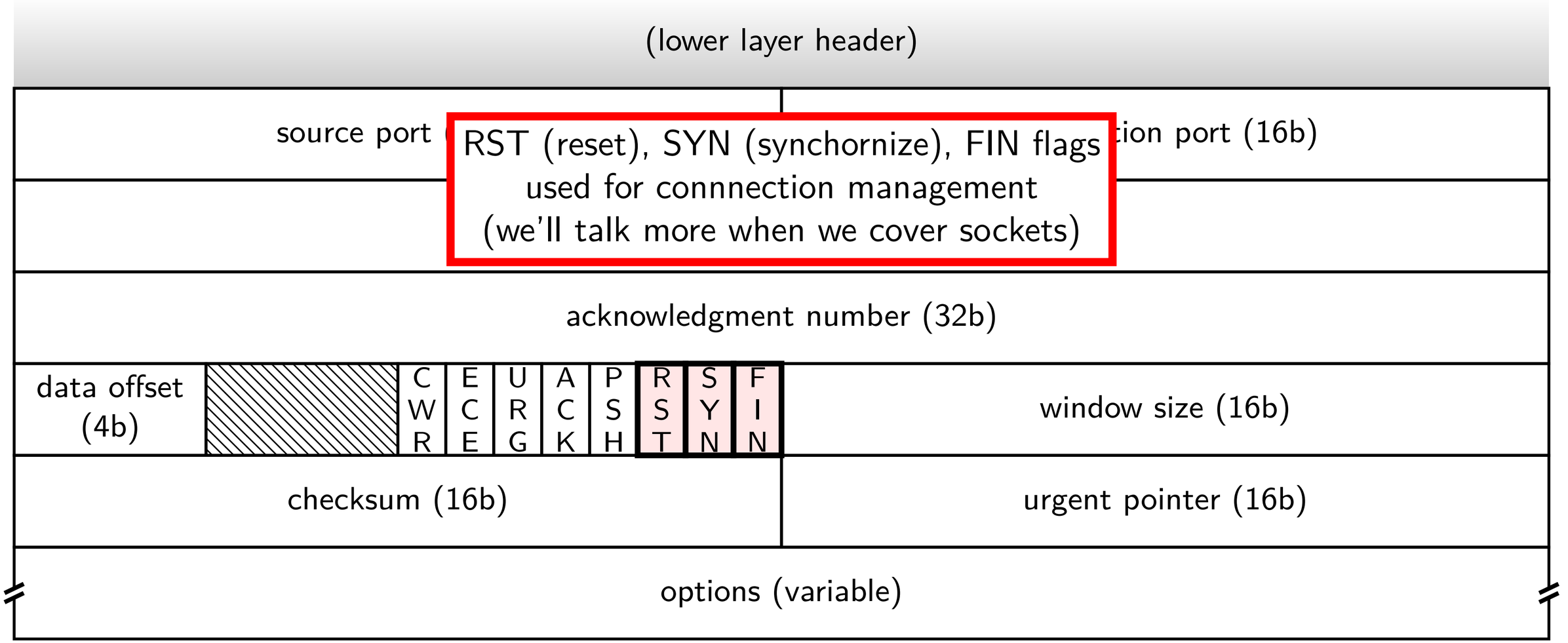
TCP segment format



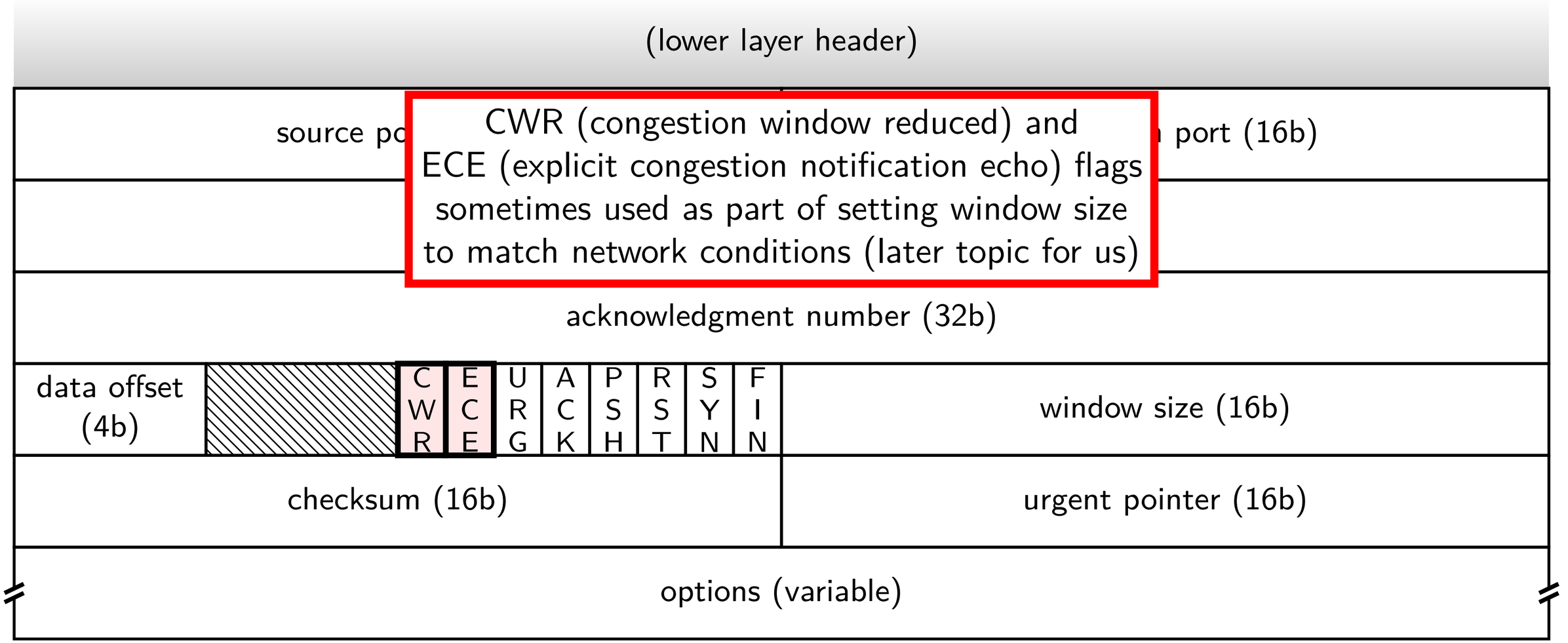
TCP segment format



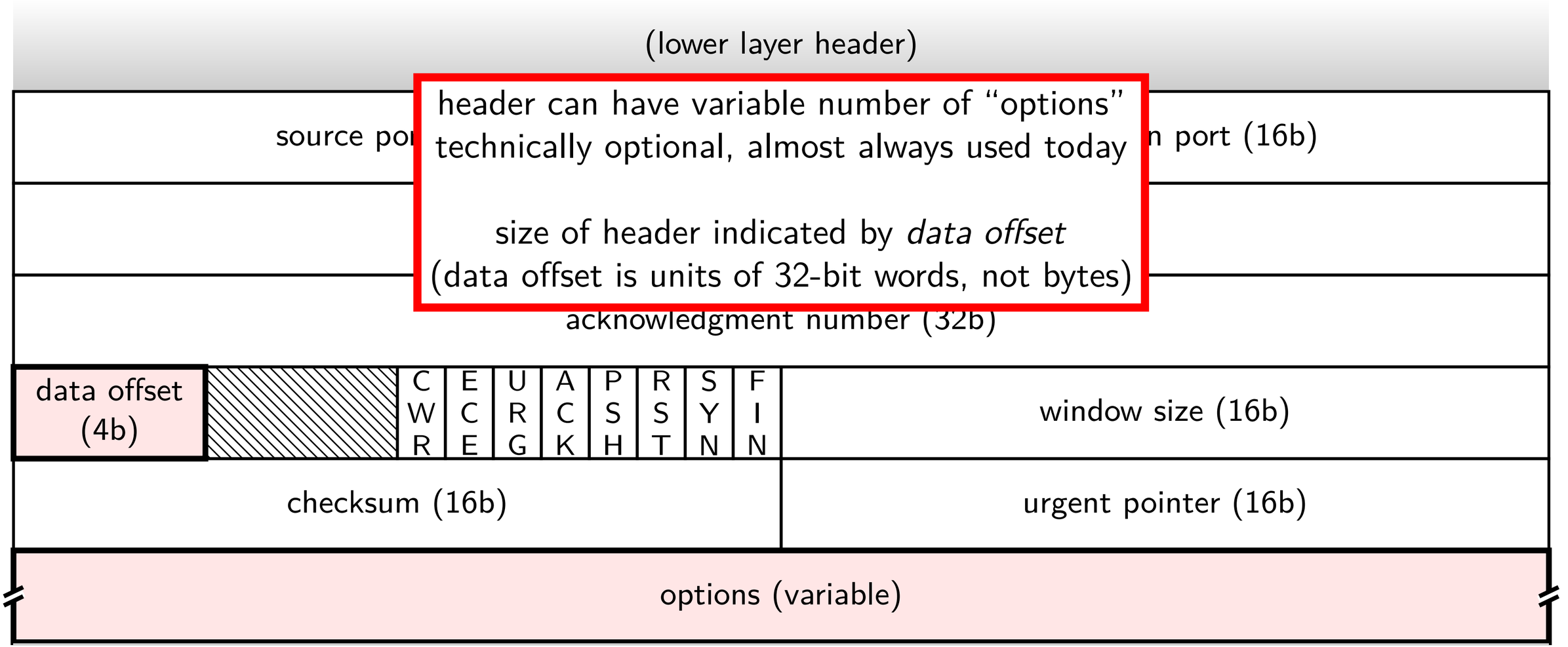
TCP segment format



TCP segment format



TCP segment format



exercise: maximum throughput

let's say we have a receiver window size of 65535 bytes

and a round-trip time of 100 ms

if we want to avoid sending data the receiver will reject as outside its window, maximum throughput?

A. around 32kbyte/sec
B. around 64kbyte/sec
C. around 128kbyte/sec
D. around 320kbyte/sec
E. around 640kbyte/sec
F. around 1280kbyte/sec
G. something else

selected TCP options

window size scale factor

- allow receiver window sizes greater than 64k
- needed to get reasonable bandwidth on modern networks

timestamps

- allow figuring out round trip time to estimate timeout
- extend 32-bit sequence number, which is too small for multi-gigabit networks

selective acknowledgements

- allow providing information about ‘holes’ in received data
- example: I got bytes 1–5000, 6000–7000, 8000–9000
- without it would only say 5000

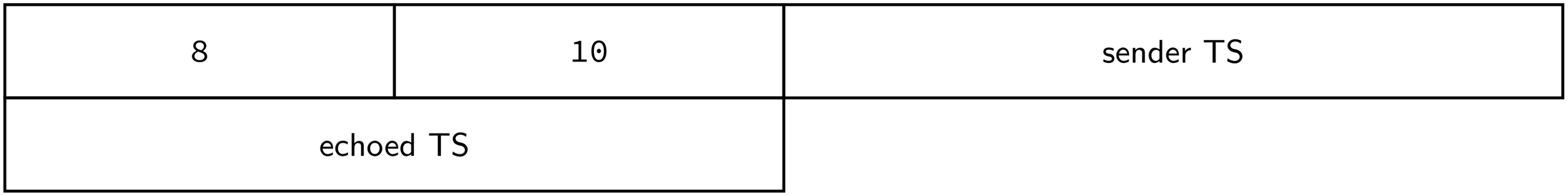
selected TCP option formats



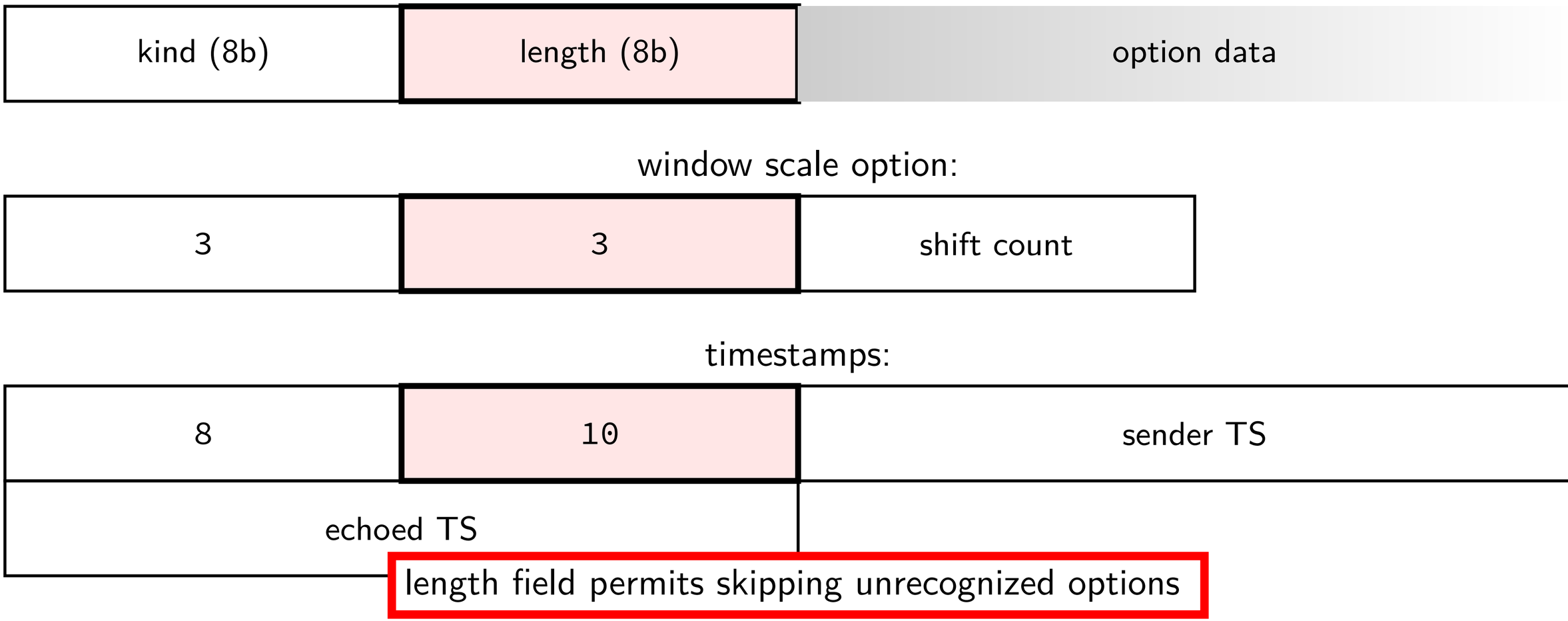
window scale option:



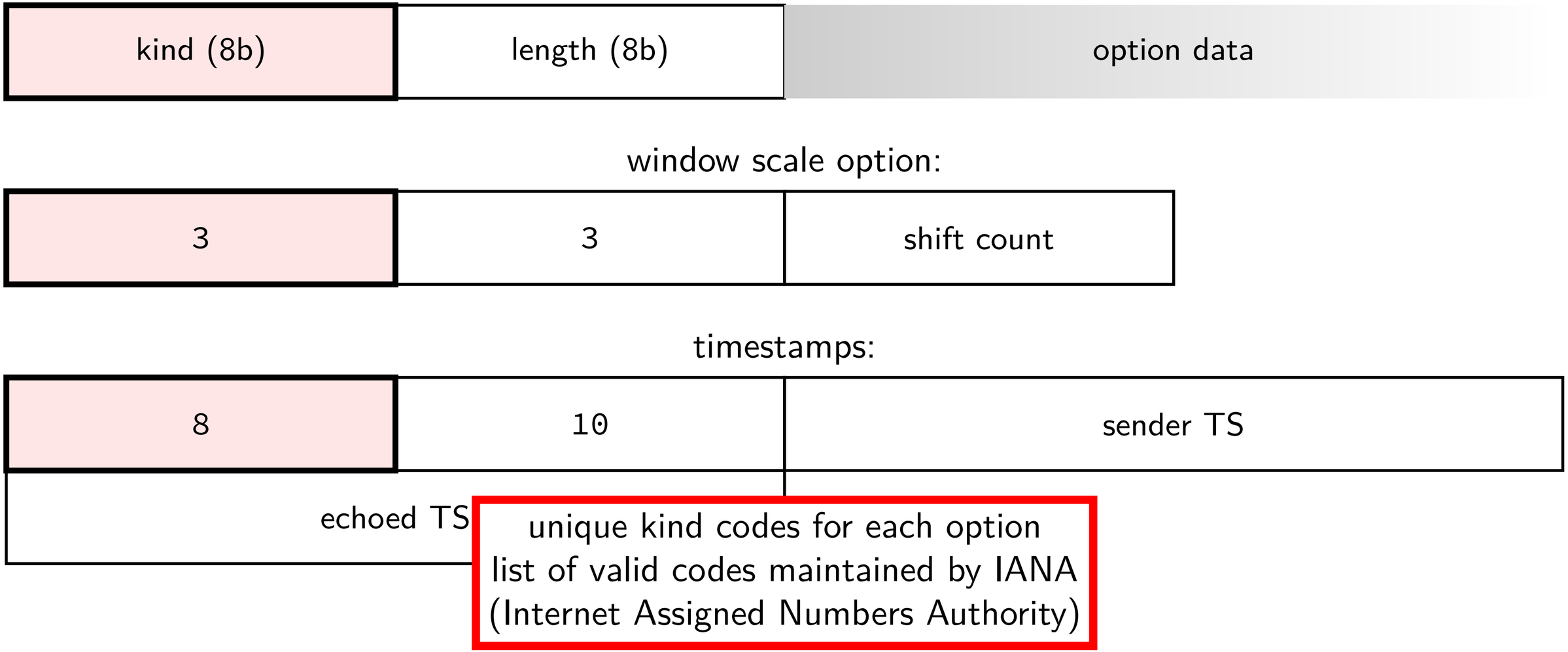
timestamps:



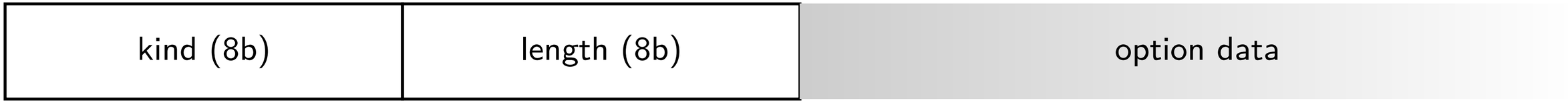
selected TCP option formats



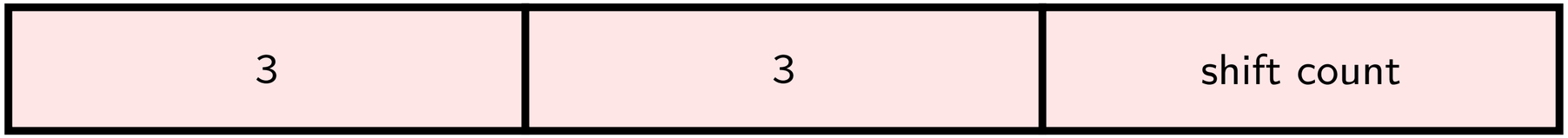
selected TCP option formats



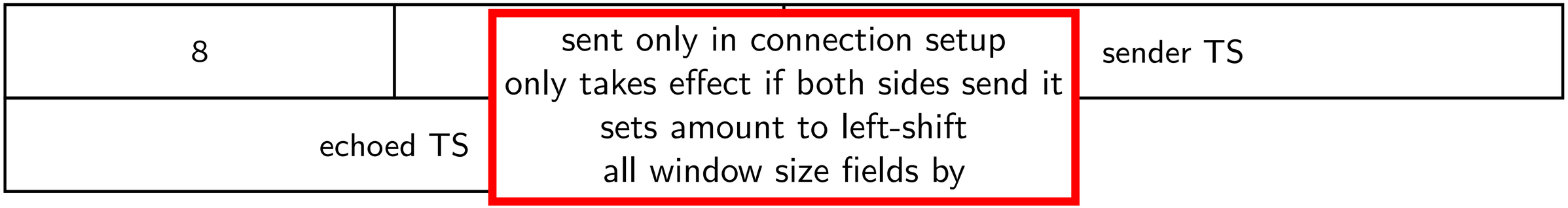
selected TCP option formats



window scale option:



timestamps:

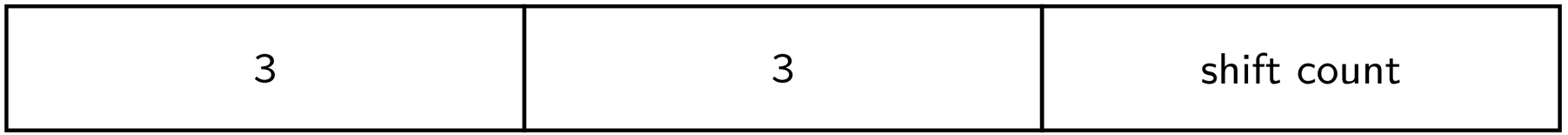


sent only in connection setup
only takes effect if both sides send it
sets amount to left-shift
all window size fields by

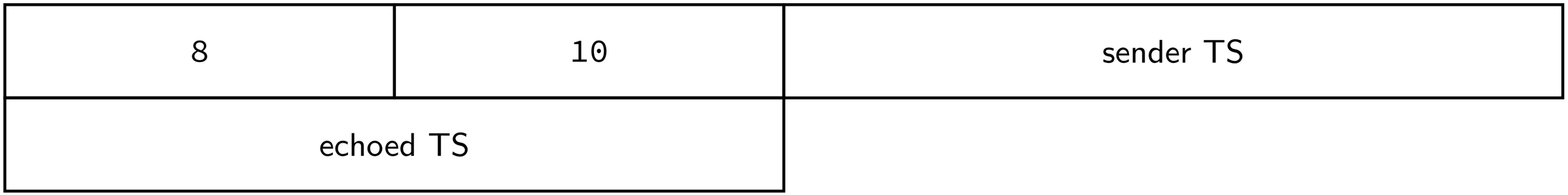
selected TCP option formats



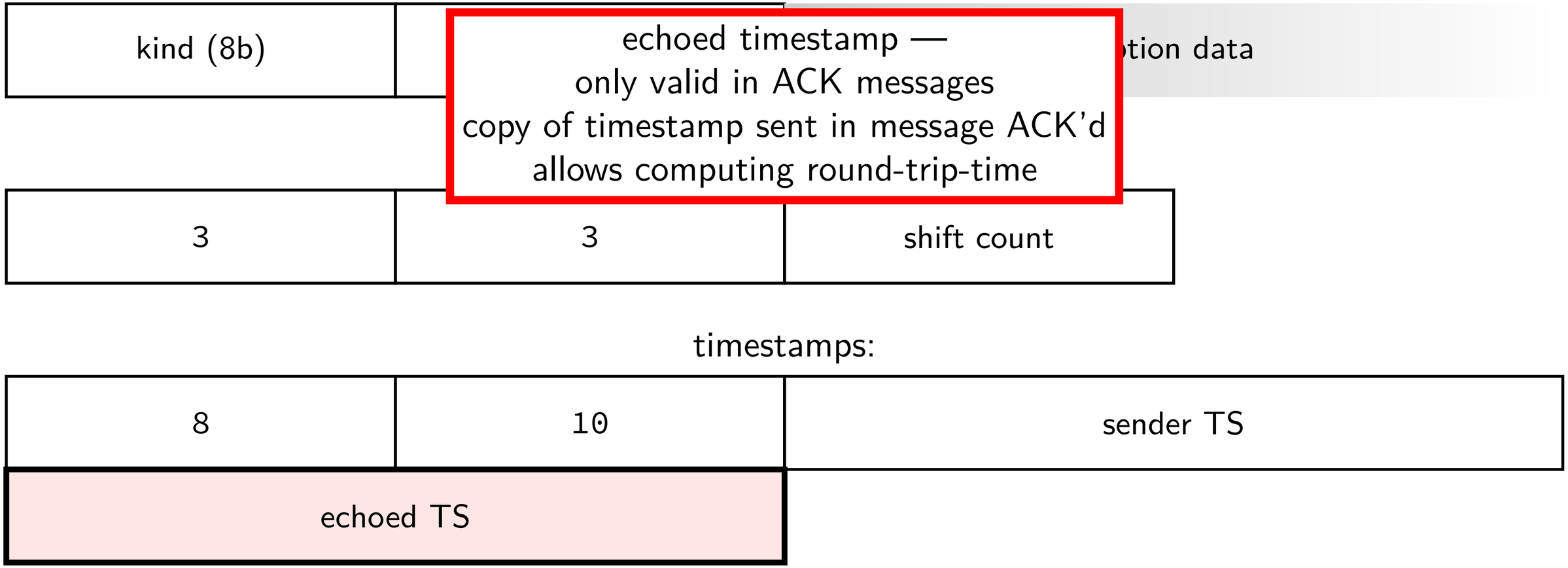
window scale option:



timestamps:



selected TCP option formats



TCP connection setup preview

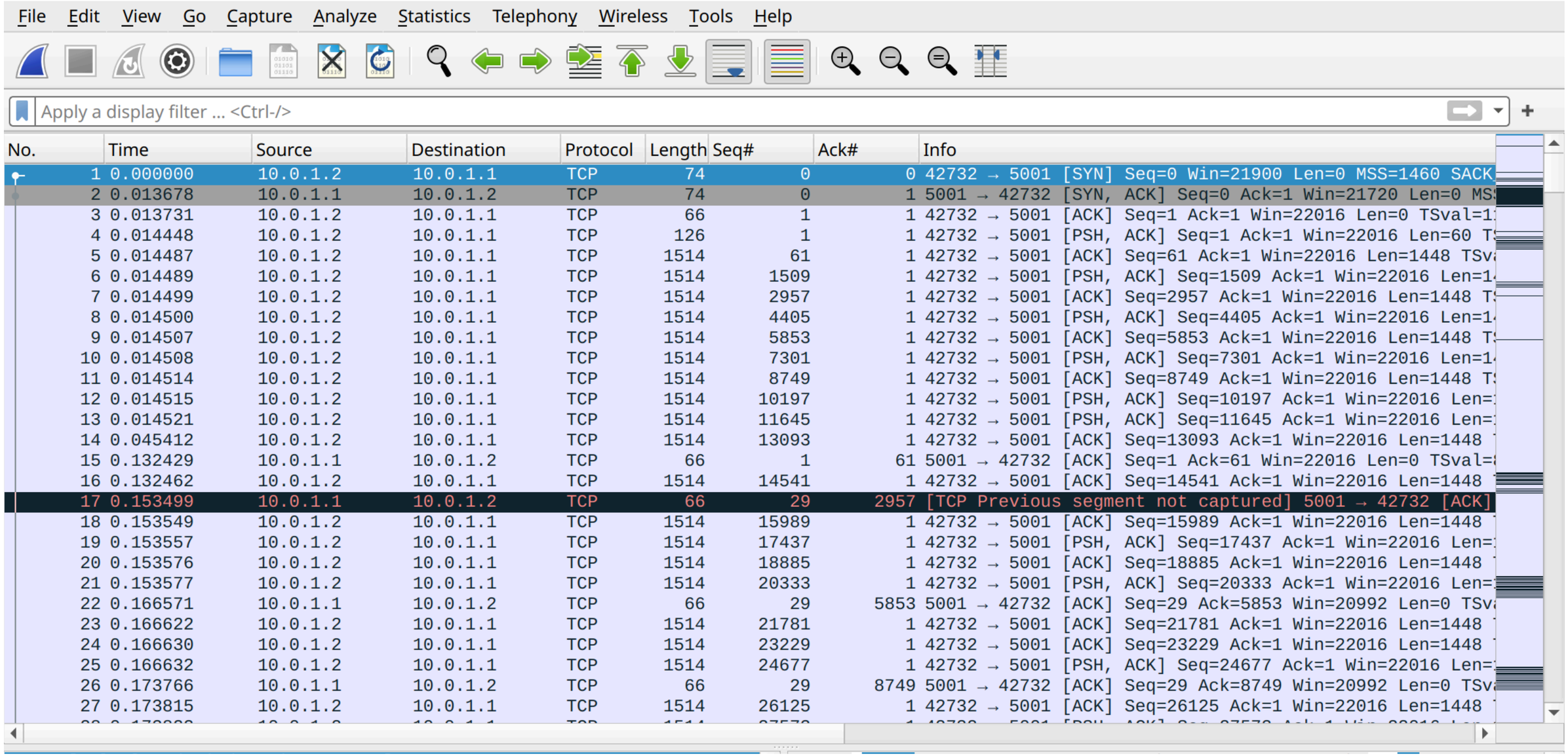
client→server: SYN flag + seq=randomA [no data]

server→client: SYN+ACK flags + seq=randomB ack=randomA+1 [no data]

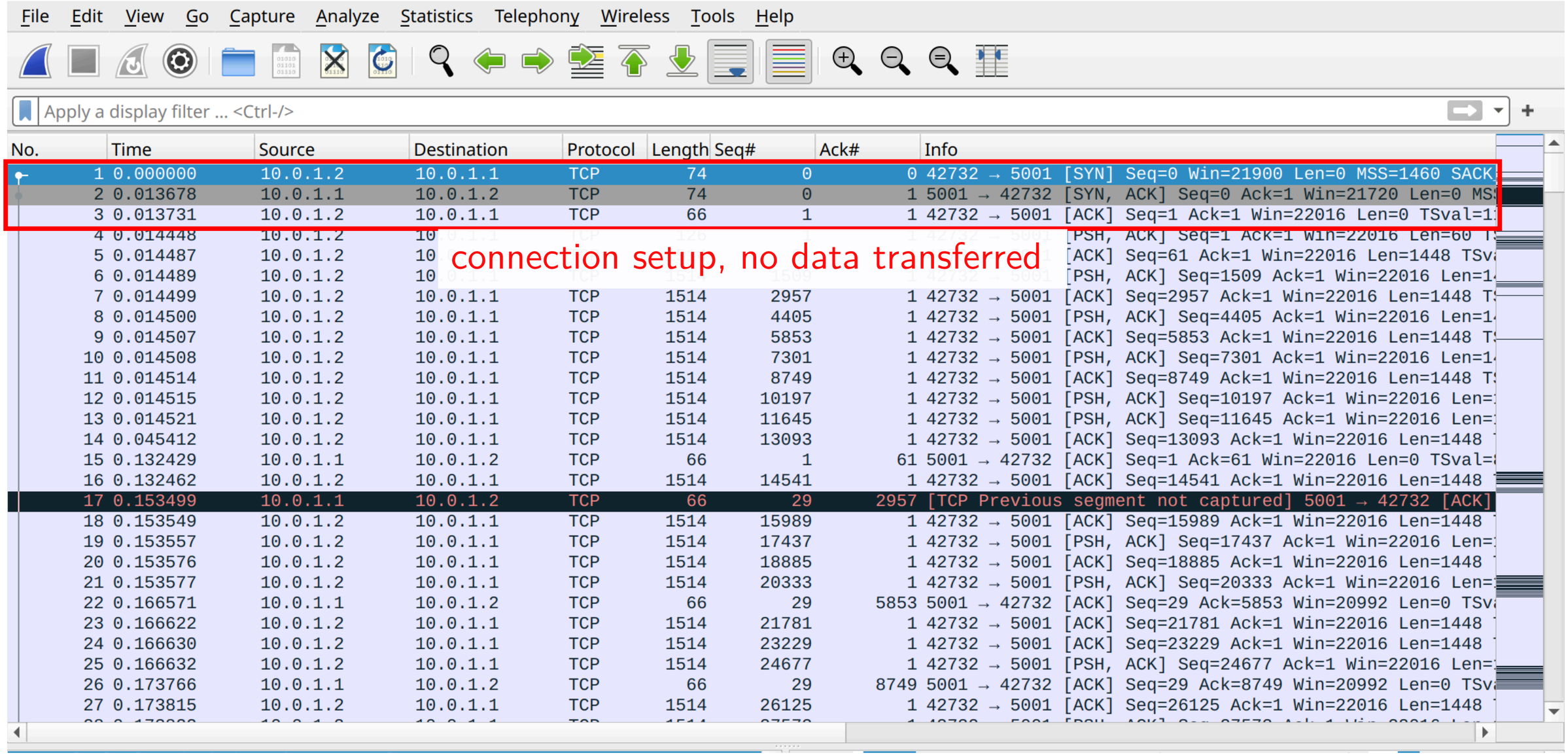
client→server: ACK flags + seq=randomA+1 ack=randomB+1 [no data]

a TCP connection

a TCP connection

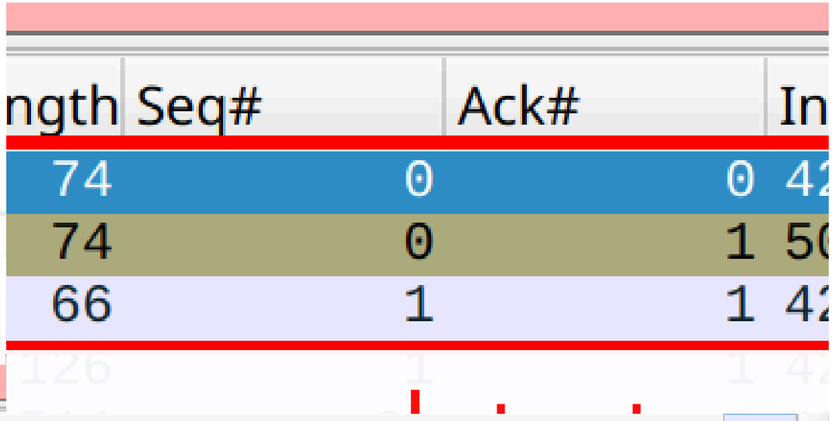


a TCP connection



a TCP connection

server+client sequence numbers advance by 1 to indicate where in setup



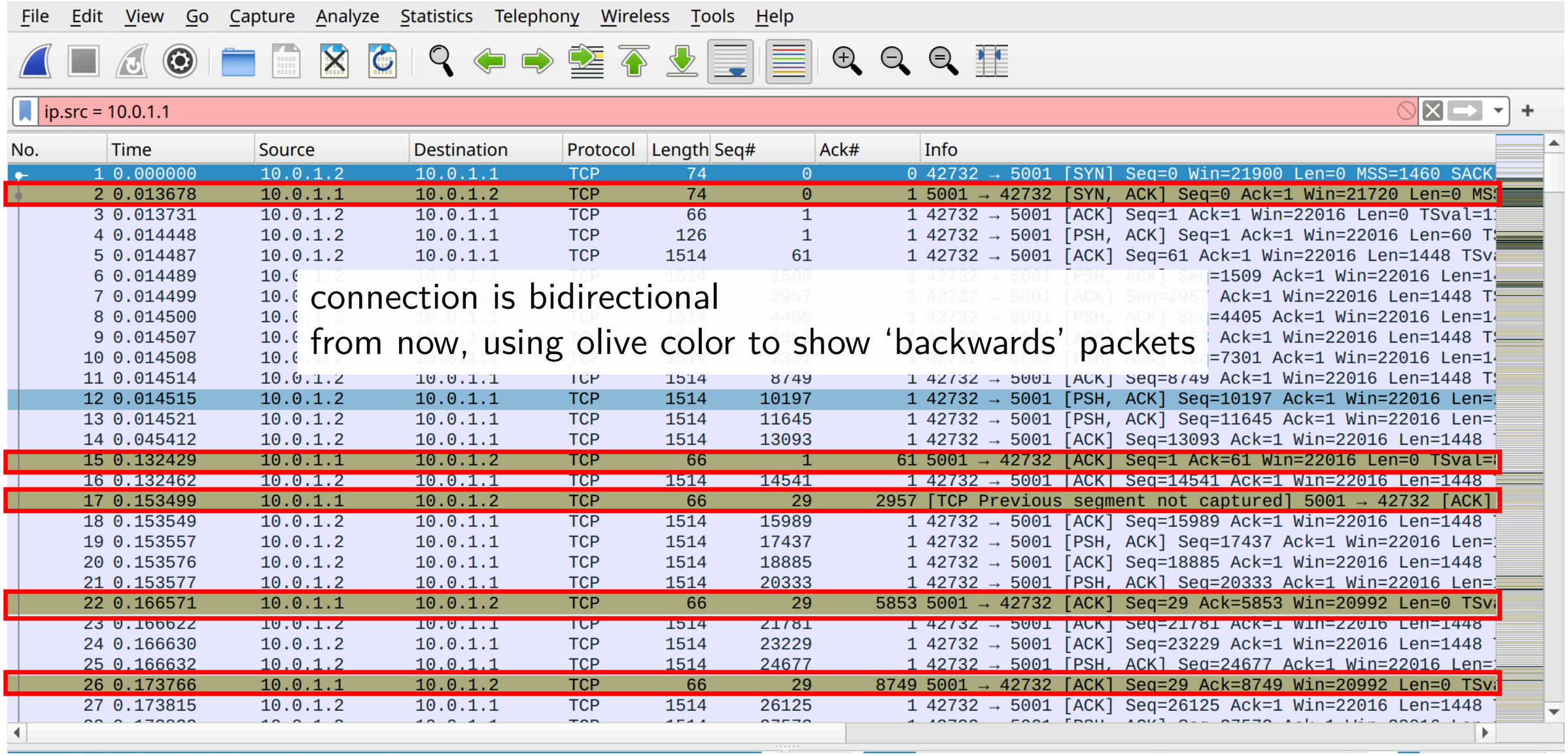
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

ip.src = 10.0.1.1

No.	Time	Source	Destination	Protocol	Length	Seq#	Ack#	Info
1	0.000000	10.0.1.2	10.0.1.1	TCP	74	0	0	42732 → 5001 [SYN] Seq=0 Win=21900 Len=0 MSS=1460 SACK
2	0.013678	10.0.1.1	10.0.1.2	TCP	74	0	1	5001 → 42732 [SYN, ACK] Seq=0 Ack=1 Win=21720 Len=0 MS
3	0.013731	10.0.1.2	10.0.1.1	TCP	66	1	1	42732 → 5001 [ACK] Seq=1 Ack=1 Win=22016 Len=0 TSval=1
4	0.014448	10.0.1.2	10					[PSH, ACK] Seq=1 Ack=1 Win=22016 Len=60 T
5	0.014487	10.0.1.2	10					[ACK] Seq=61 Ack=1 Win=22016 Len=1448 TSv
6	0.014489	10.0.1.2	10					[PSH, ACK] Seq=1509 Ack=1 Win=22016 Len=1
7	0.014499	10.0.1.2	10.0.1.1	TCP	1514	2957	1	42732 → 5001 [ACK] Seq=2957 Ack=1 Win=22016 Len=1448 T
8	0.014500	10.0.1.2	10.0.1.1	TCP	1514	4405	1	42732 → 5001 [PSH, ACK] Seq=4405 Ack=1 Win=22016 Len=1
9	0.014507	10.0.1.2	10.0.1.1	TCP	1514	5853	1	42732 → 5001 [ACK] Seq=5853 Ack=1 Win=22016 Len=1448 T
10	0.014508	10.0.1.2	10.0.1.1	TCP	1514	7301	1	42732 → 5001 [PSH, ACK] Seq=7301 Ack=1 Win=22016 Len=1
11	0.014514	10.0.1.2	10.0.1.1	TCP	1514	8749	1	42732 → 5001 [ACK] Seq=8749 Ack=1 Win=22016 Len=1448 T
12	0.014515	10.0.1.2	10.0.1.1	TCP	1514	10197	1	42732 → 5001 [PSH, ACK] Seq=10197 Ack=1 Win=22016 Len=
13	0.014521	10.0.1.2	10.0.1.1	TCP	1514	11645	1	42732 → 5001 [PSH, ACK] Seq=11645 Ack=1 Win=22016 Len=
14	0.045412	10.0.1.2	10.0.1.1	TCP	1514	13093	1	42732 → 5001 [ACK] Seq=13093 Ack=1 Win=22016 Len=1448
15	0.132429	10.0.1.1	10.0.1.2	TCP	66	1	61	5001 → 42732 [ACK] Seq=1 Ack=61 Win=22016 Len=0 TSval=
16	0.132462	10.0.1.2	10.0.1.1	TCP	1514	14541	1	42732 → 5001 [ACK] Seq=14541 Ack=1 Win=22016 Len=1448
17	0.153499	10.0.1.1	10.0.1.2	TCP	66	29	2957	[TCP Previous segment not captured] 5001 → 42732 [ACK]
18	0.153549	10.0.1.2	10.0.1.1	TCP	1514	15989	1	42732 → 5001 [ACK] Seq=15989 Ack=1 Win=22016 Len=1448
19	0.153557	10.0.1.2	10.0.1.1	TCP	1514	17437	1	42732 → 5001 [PSH, ACK] Seq=17437 Ack=1 Win=22016 Len=
20	0.153576	10.0.1.2	10.0.1.1	TCP	1514	18885	1	42732 → 5001 [ACK] Seq=18885 Ack=1 Win=22016 Len=1448
21	0.153577	10.0.1.2	10.0.1.1	TCP	1514	20333	1	42732 → 5001 [PSH, ACK] Seq=20333 Ack=1 Win=22016 Len=
22	0.166571	10.0.1.1	10.0.1.2	TCP	66	29	5853	5001 → 42732 [ACK] Seq=29 Ack=5853 Win=20992 Len=0 TSv
23	0.166622	10.0.1.2	10.0.1.1	TCP	1514	21781	1	42732 → 5001 [ACK] Seq=21781 Ack=1 Win=22016 Len=1448
24	0.166630	10.0.1.2	10.0.1.1	TCP	1514	23229	1	42732 → 5001 [ACK] Seq=23229 Ack=1 Win=22016 Len=1448
25	0.166632	10.0.1.2	10.0.1.1	TCP	1514	24677	1	42732 → 5001 [PSH, ACK] Seq=24677 Ack=1 Win=22016 Len=
26	0.173766	10.0.1.1	10.0.1.2	TCP	66	29	8749	5001 → 42732 [ACK] Seq=29 Ack=8749 Win=20992 Len=0 TSv
27	0.173815	10.0.1.2	10.0.1.1	TCP	1514	26125	1	42732 → 5001 [ACK] Seq=26125 Ack=1 Win=22016 Len=1448

connection setup, no data transferred

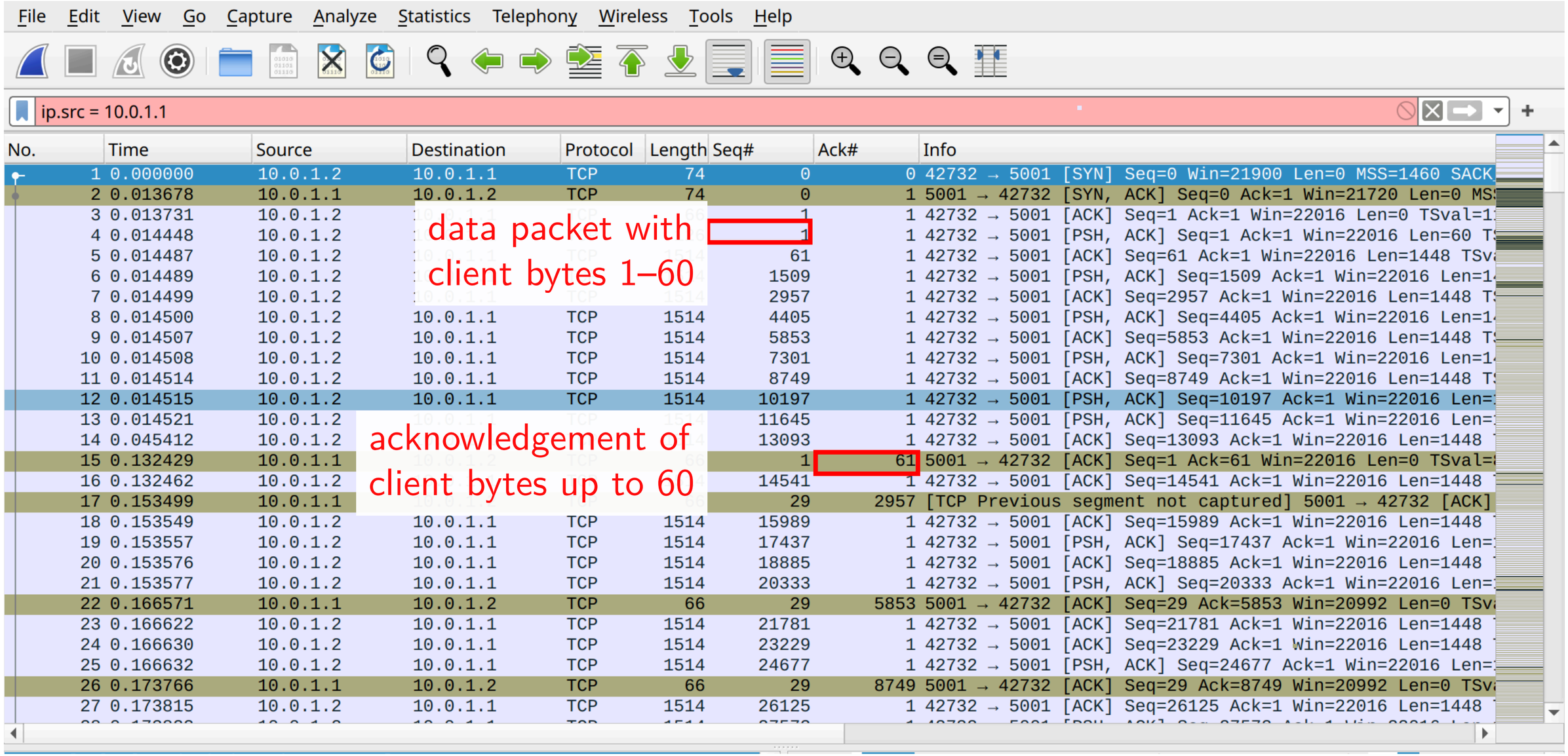
a TCP connection



connection is bidirectional

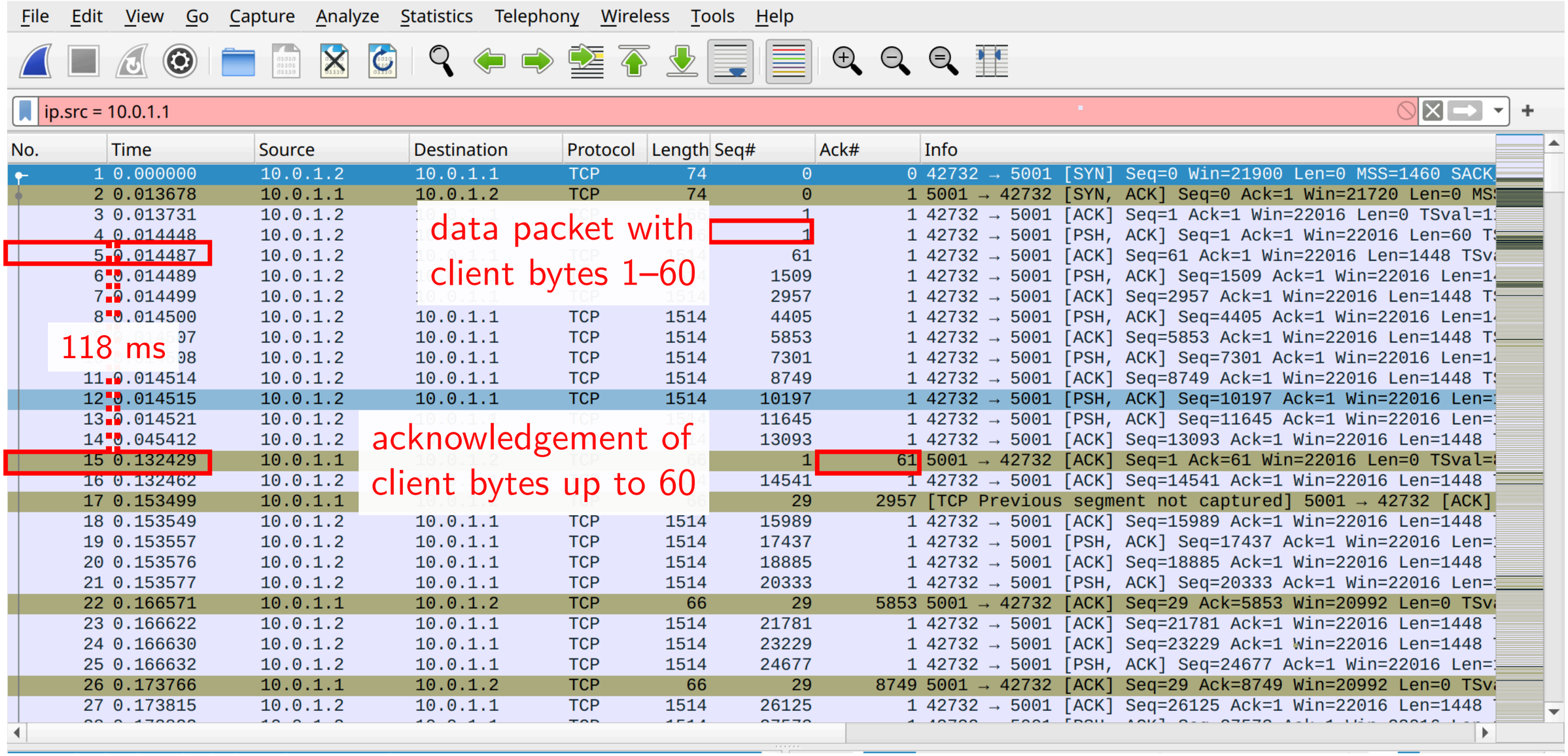
from now, using olive color to show ‘backwards’ packets

a TCP connection

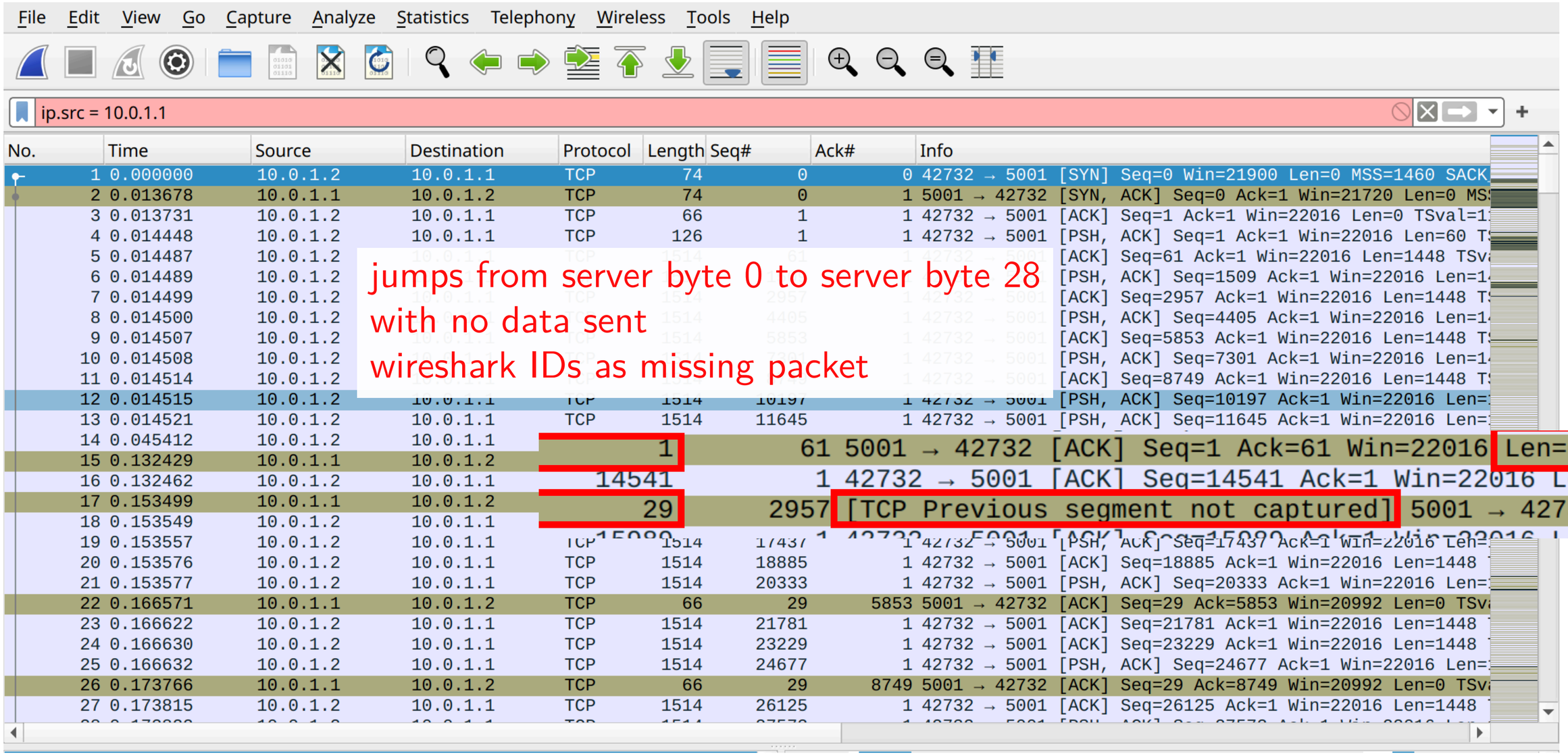


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3	0.013731	10.0.1.2				1	1	42732 → 5001 [ACK] Seq=1 Ack=1 Win=22016 Len=0 TSval=1
4	0.014448	10.0.1.2				1	1	42732 → 5001 [PSH, ACK] Seq=1 Ack=1 Win=22016 Len=60 T
5	0.014487	10.0.1.2				61	1	42732 → 5001 [ACK] Seq=61 Ack=1 Win=22016 Len=1448 TSv
6	0.014489	10.0.1.2				1509	1	42732 → 5001 [PSH, ACK] Seq=1509 Ack=1 Win=22016 Len=1
7	0.014499	10.0.1.2				2957	1	42732 → 5001 [ACK] Seq=2957 Ack=1 Win=22016 Len=1448 T
8	0.014500	10.0.1.2	10.0.1.1	TCP	1514	4405	1	42732 → 5001 [PSH, ACK] Seq=4405 Ack=1 Win=22016 Len=1
9	0.014507	10.0.1.2	10.0.1.1	TCP	1514	5853	1	42732 → 5001 [ACK] Seq=5853 Ack=1 Win=22016 Len=1448 T
10	0.014508	10.0.1.2	10.0.1.1	TCP	1514	7301	1	42732 → 5001 [PSH, ACK] Seq=7301 Ack=1 Win=22016 Len=1
11	0.014514	10.0.1.2	10.0.1.1	TCP	1514	8749	1	42732 → 5001 [ACK] Seq=8749 Ack=1 Win=22016 Len=1448 T
12	0.014515	10.0.1.2	10.0.1.1	TCP	1514	10197	1	42732 → 5001 [PSH, ACK] Seq=10197 Ack=1 Win=22016 Len=
13	0.014521	10.0.1.2				11645	1	42732 → 5001 [PSH, ACK] Seq=11645 Ack=1 Win=22016 Len=
14	0.045412	10.0.1.2				13093	1	42732 → 5001 [ACK] Seq=13093 Ack=1 Win=22016 Len=1448
15	0.132429	10.0.1.1				1	61	5001 → 42732 [ACK] Seq=1 Ack=61 Win=22016 Len=0 TSval=
16	0.132462	10.0.1.2				14541	1	42732 → 5001 [ACK] Seq=14541 Ack=1 Win=22016 Len=1448
17	0.153499	10.0.1.1				29	2957	[TCP Previous segment not captured] 5001 → 42732 [ACK]
18	0.153549	10.0.1.2	10.0.1.1	TCP	1514	15989	1	42732 → 5001 [ACK] Seq=15989 Ack=1 Win=22016 Len=1448
19	0.153557	10.0.1.2	10.0.1.1	TCP	1514	17437	1	42732 → 5001 [PSH, ACK] Seq=17437 Ack=1 Win=22016 Len=
20	0.153576	10.0.1.2	10.0.1.1	TCP	1514	18885	1	42732 → 5001 [ACK] Seq=18885 Ack=1 Win=22016 Len=1448
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23	0.166622	10.0.1.2	10.0.1.1	TCP	1514	21781	1	42732 → 5001 [ACK] Seq=21781 Ack=1 Win=22016 Len=1448
24	0.166630	10.0.1.2	10.0.1.1	TCP	1514	23229	1	42732 → 5001 [ACK] Seq=23229 Ack=1 Win=22016 Len=1448
25	0.166632	10.0.1.2	10.0.1.1	TCP	1514	24677	1	42732 → 5001 [PSH, ACK] Seq=24677 Ack=1 Win=22016 Len=
26	0.173766	10.0.1.1	10.0.1.2	TCP	66	29	8749	5001 → 42732 [ACK] Seq=29 Ack=8749 Win=20992 Len=0 TSv
27	0.173815	10.0.1.2	10.0.1.1	TCP	1514	26125	1	42732 → 5001 [ACK] Seq=26125 Ack=1 Win=22016 Len=1448

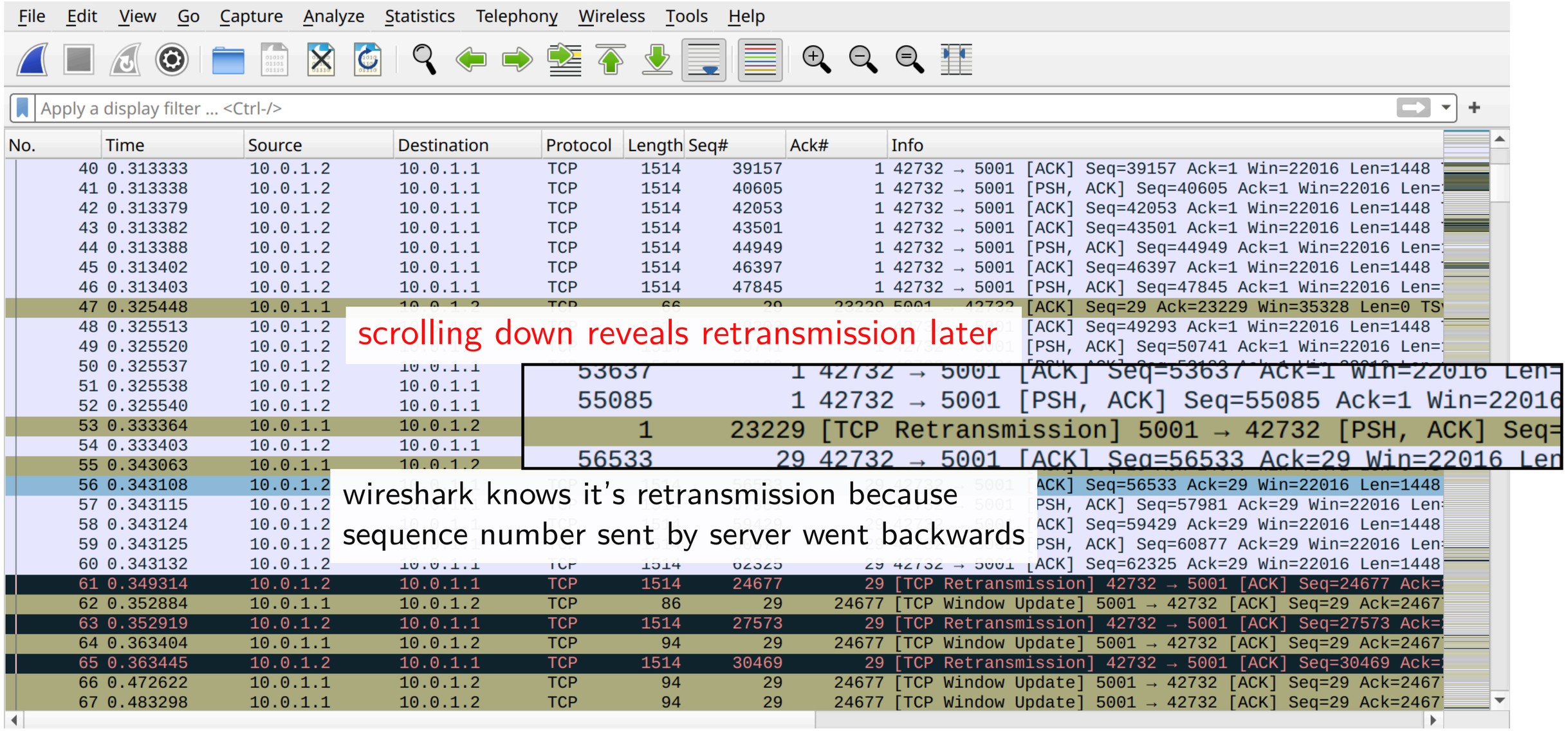
a TCP connection



a TCP connection

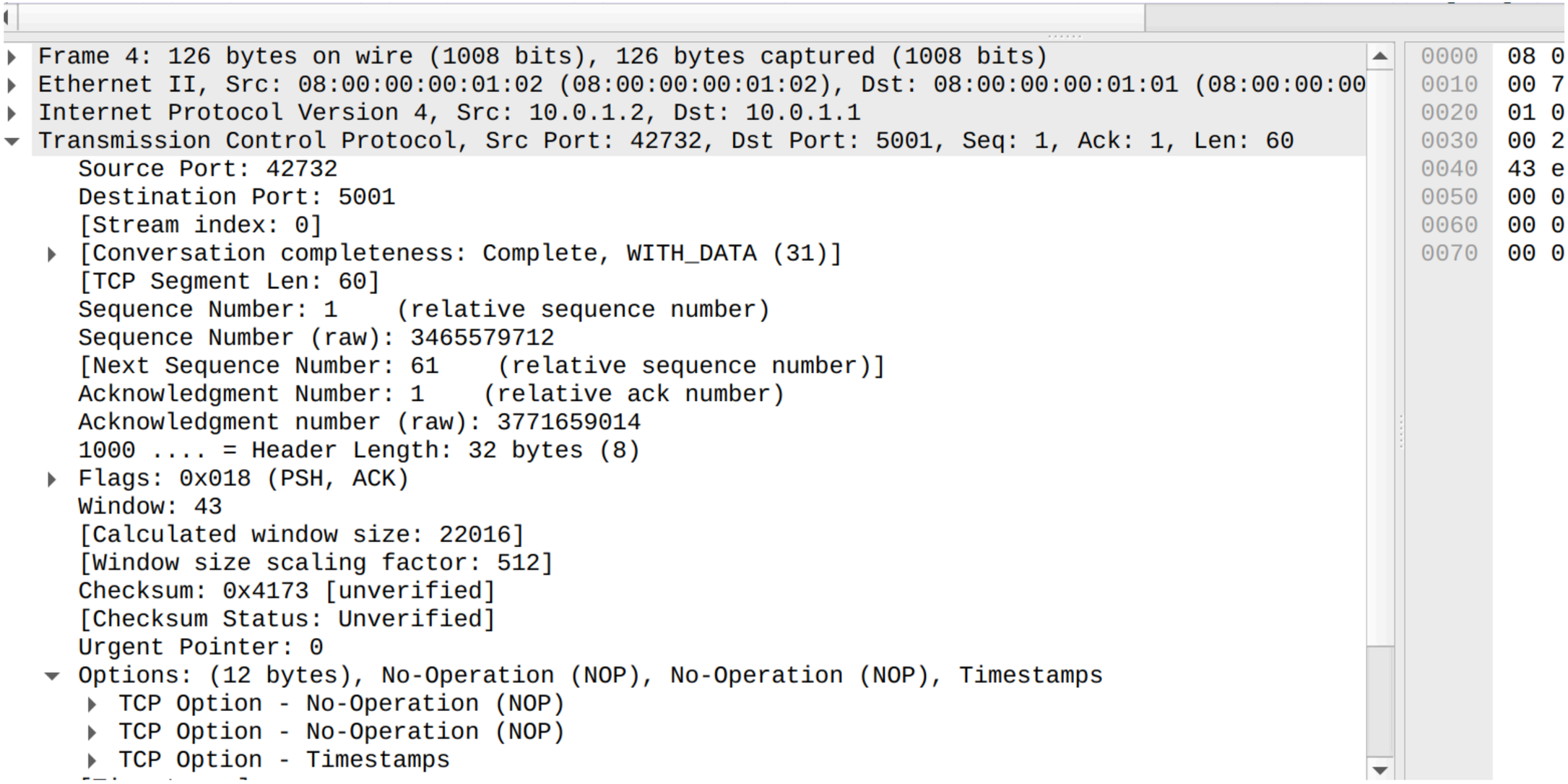


a TCP connection

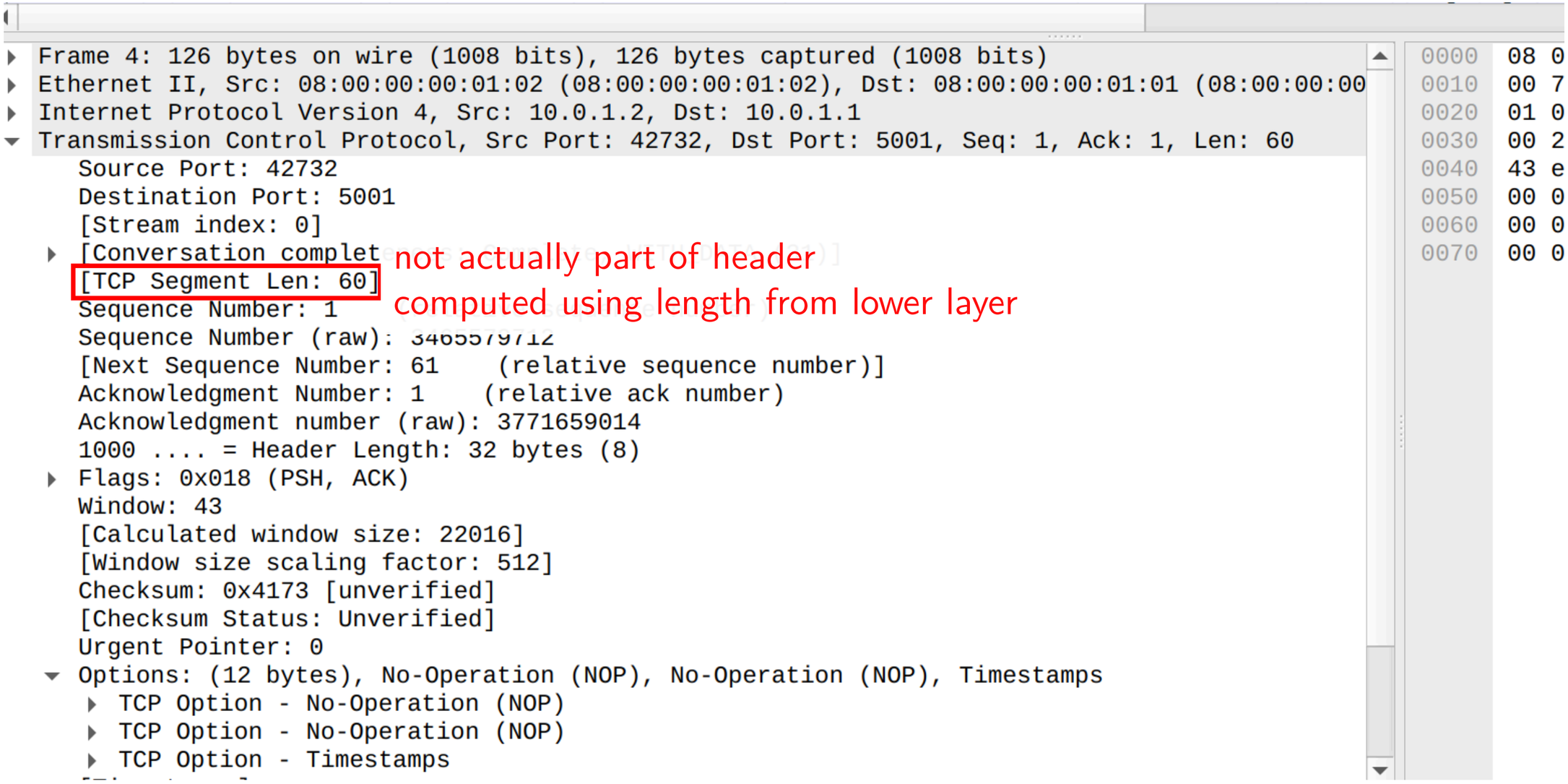


first data packet

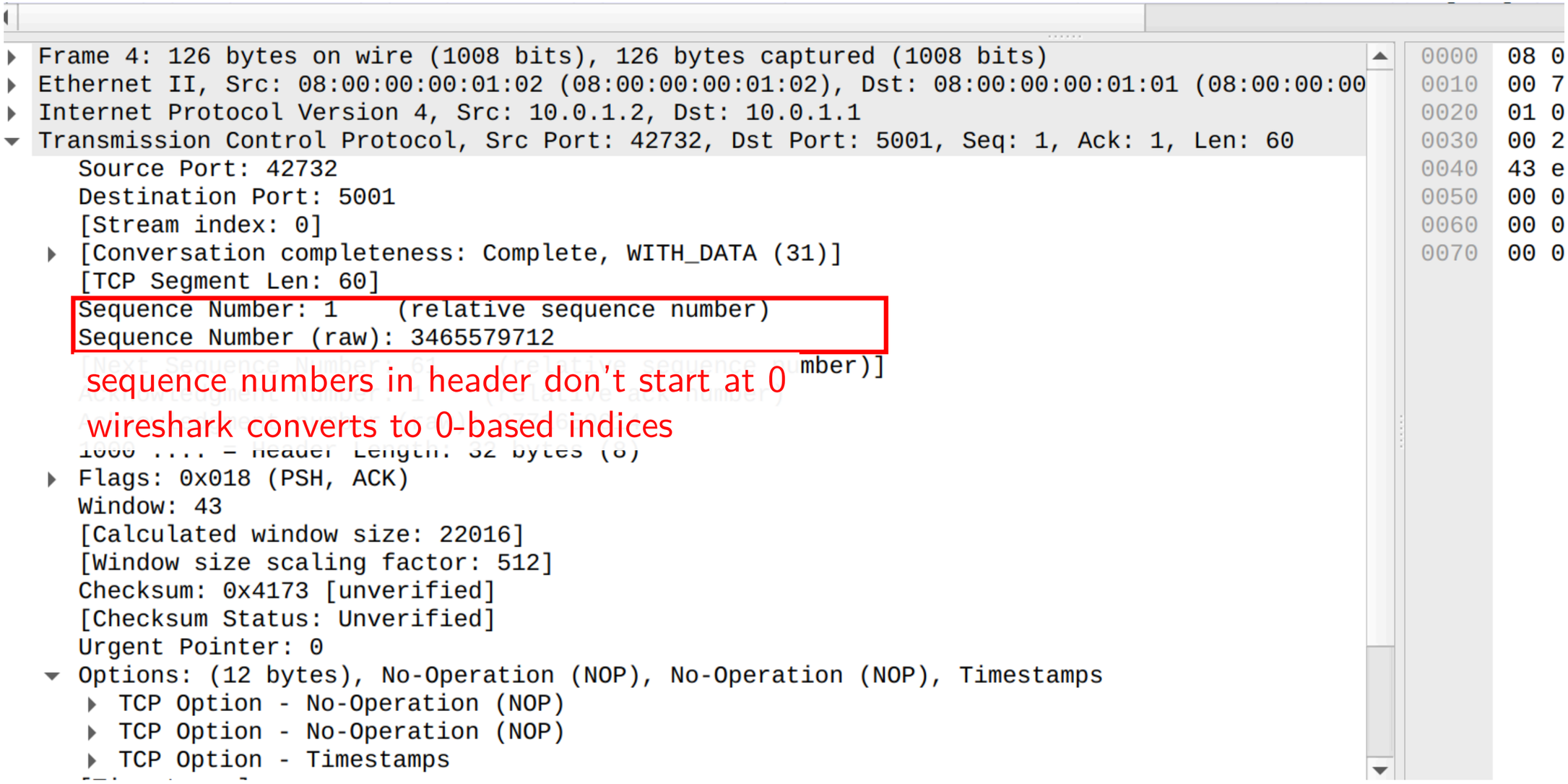
first data packet



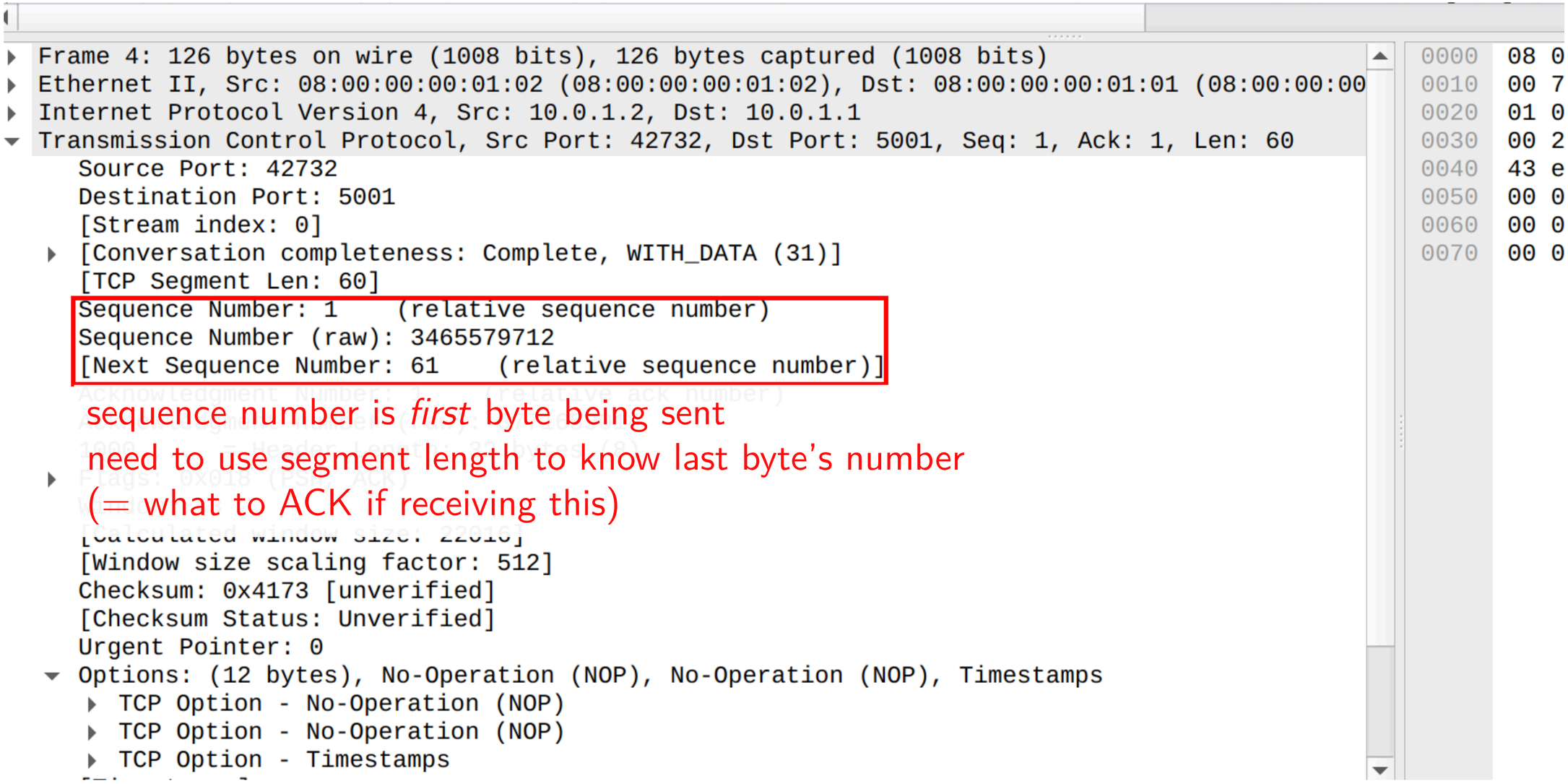
first data packet



first data packet



first data packet

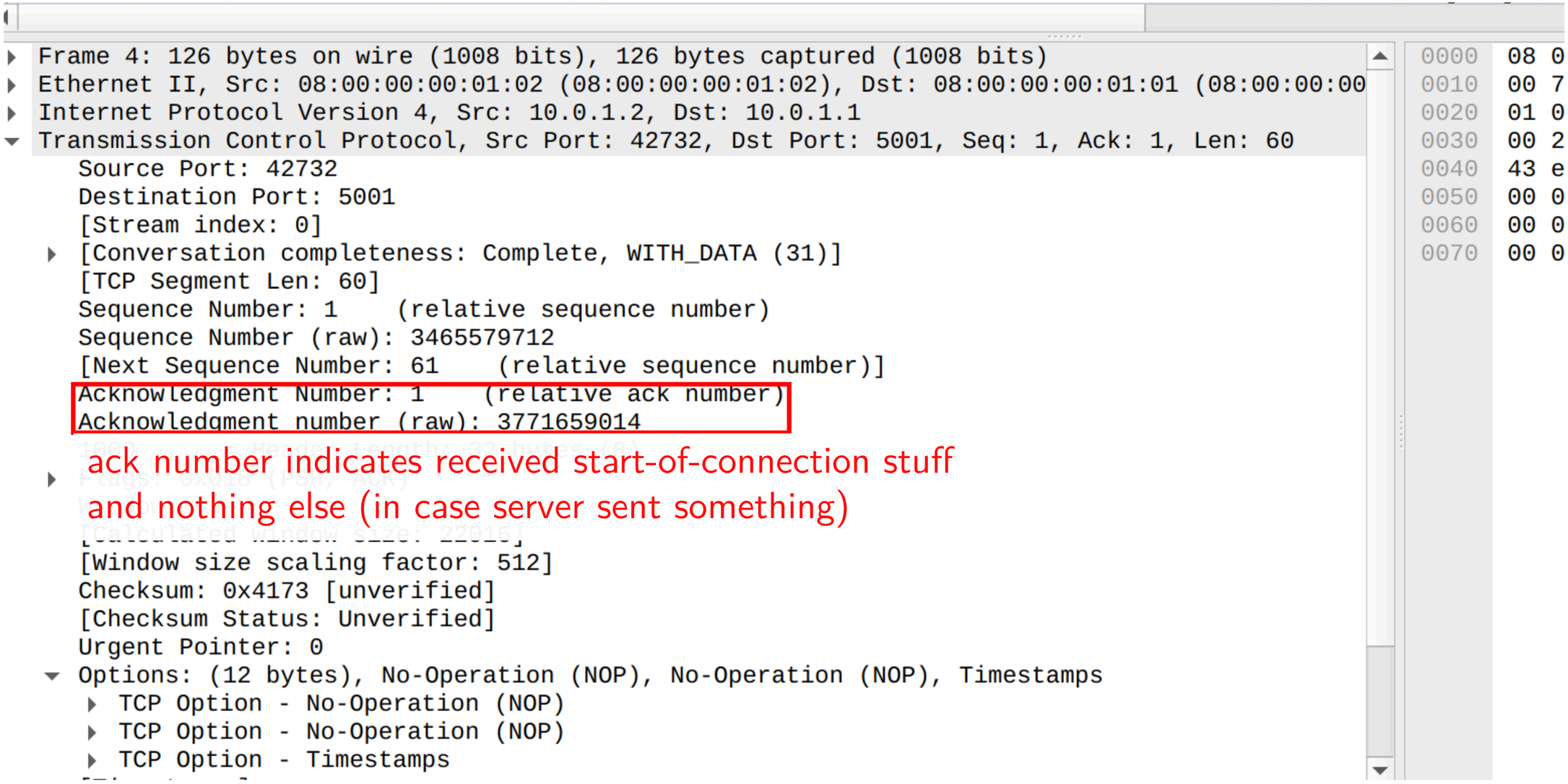


sequence number is *first* byte being sent

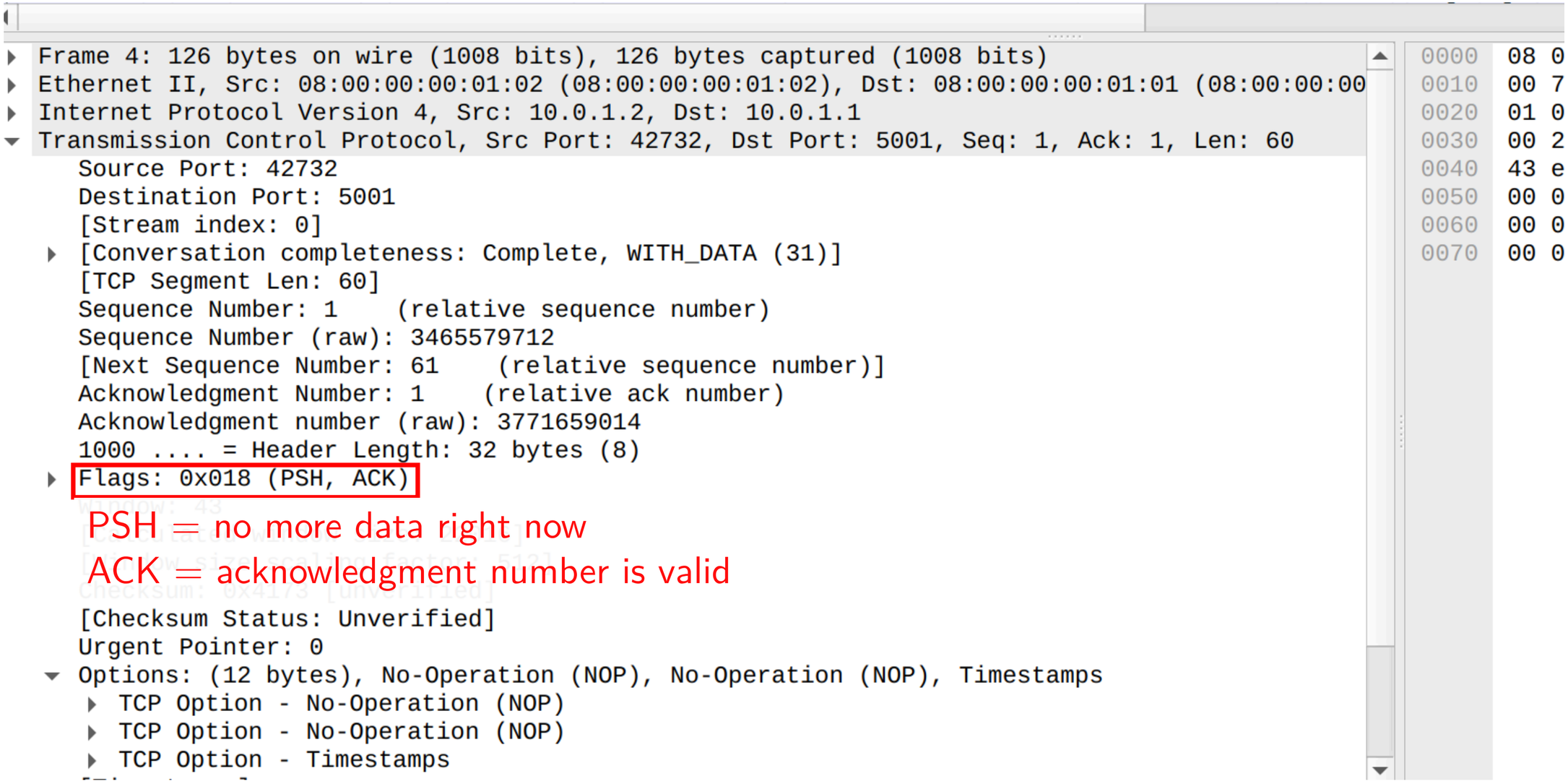
need to use segment length to know last byte's number

(= what to ACK if receiving this)

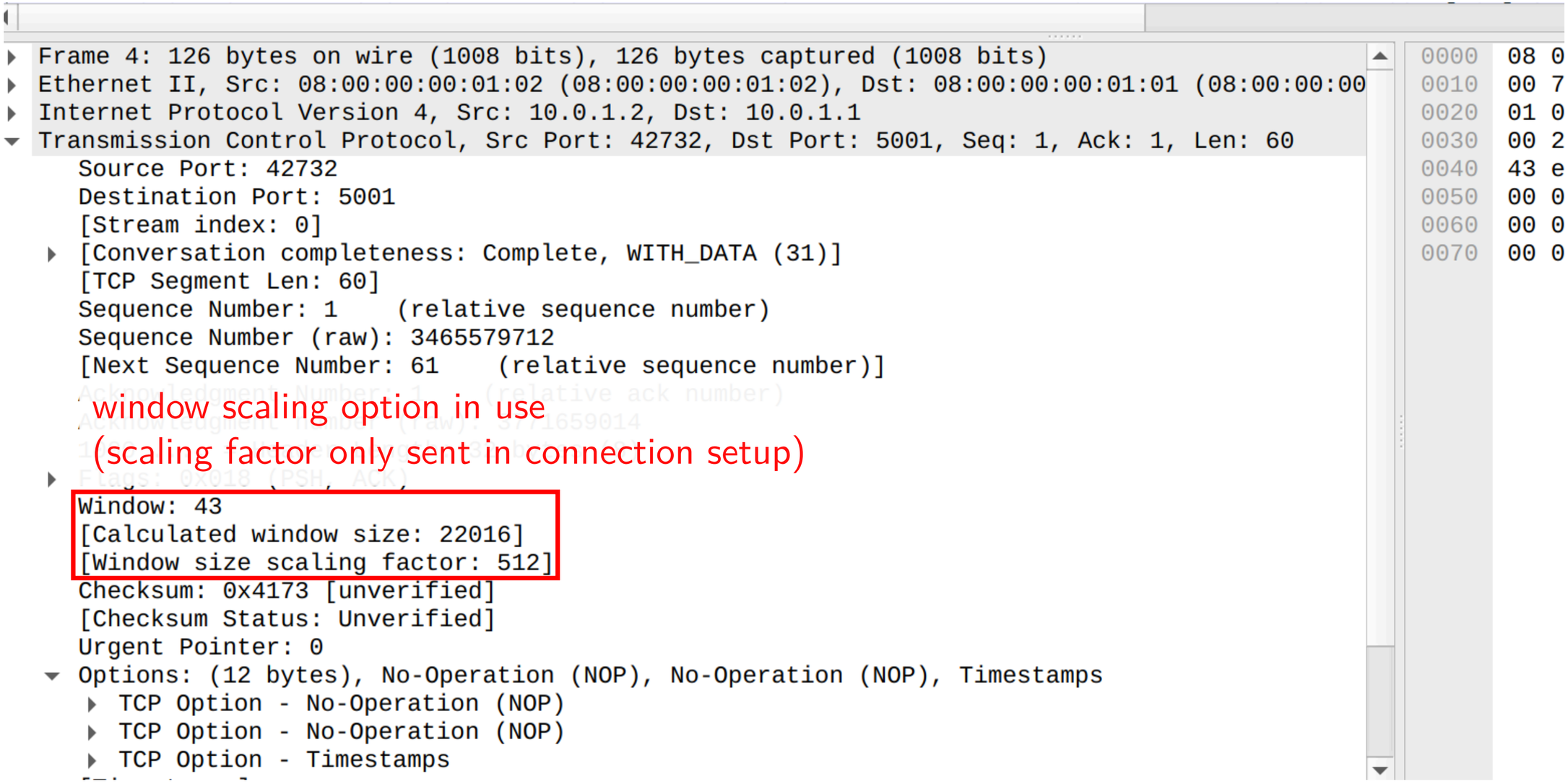
first data packet



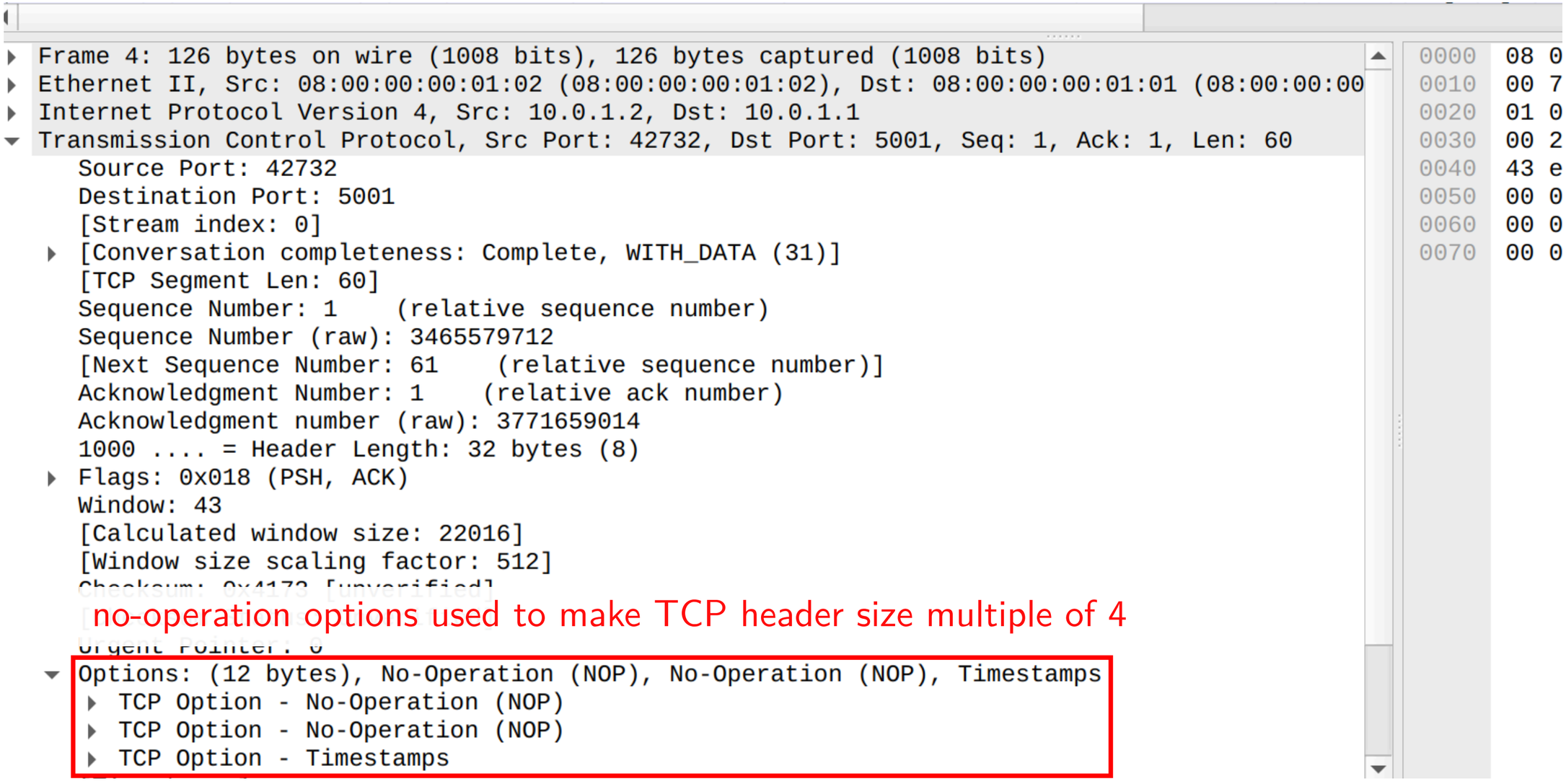
first data packet



first data packet

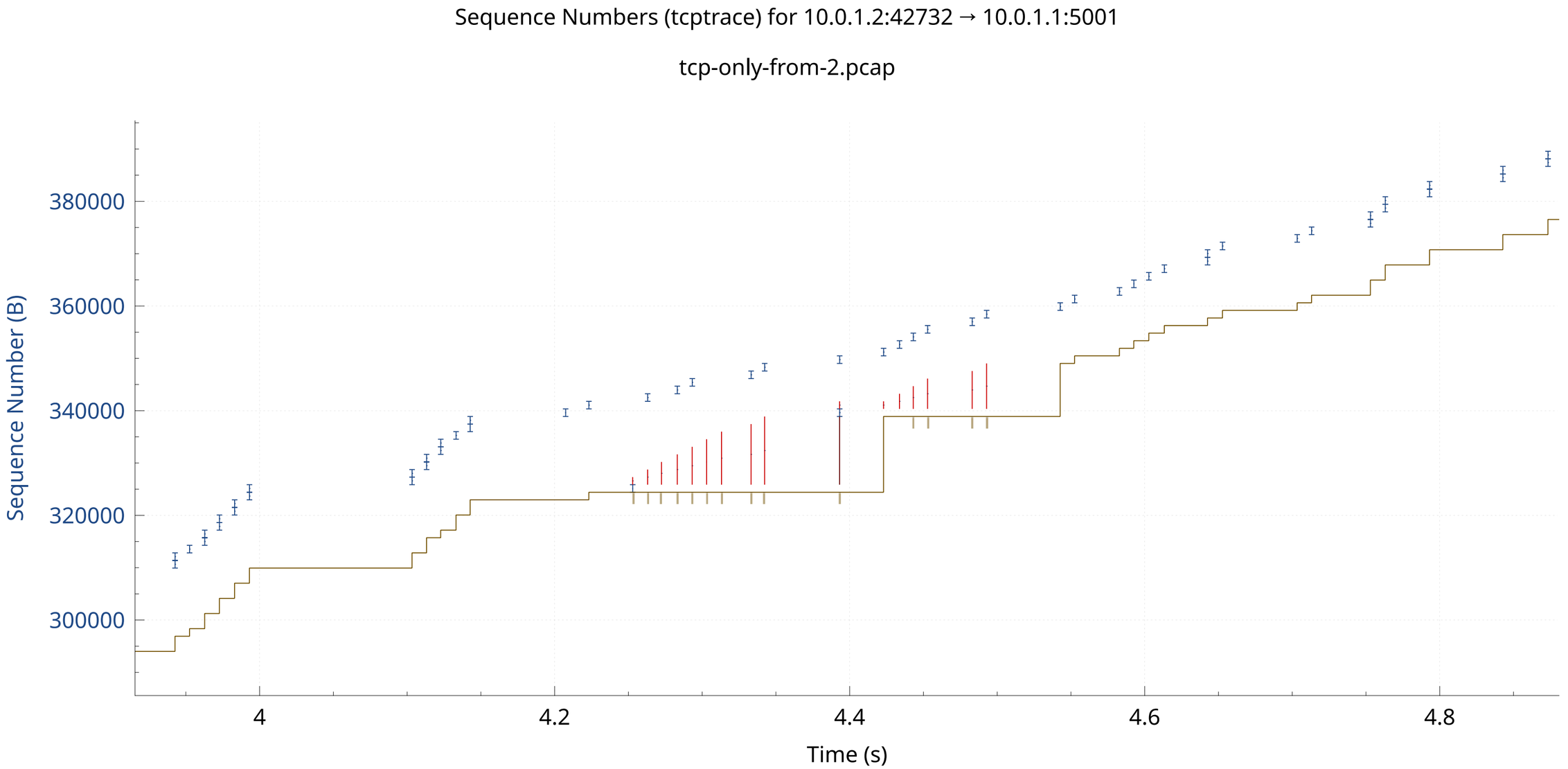


first data packet



no-operation options used to make TCP header size multiple of 4

sequence numbers graph



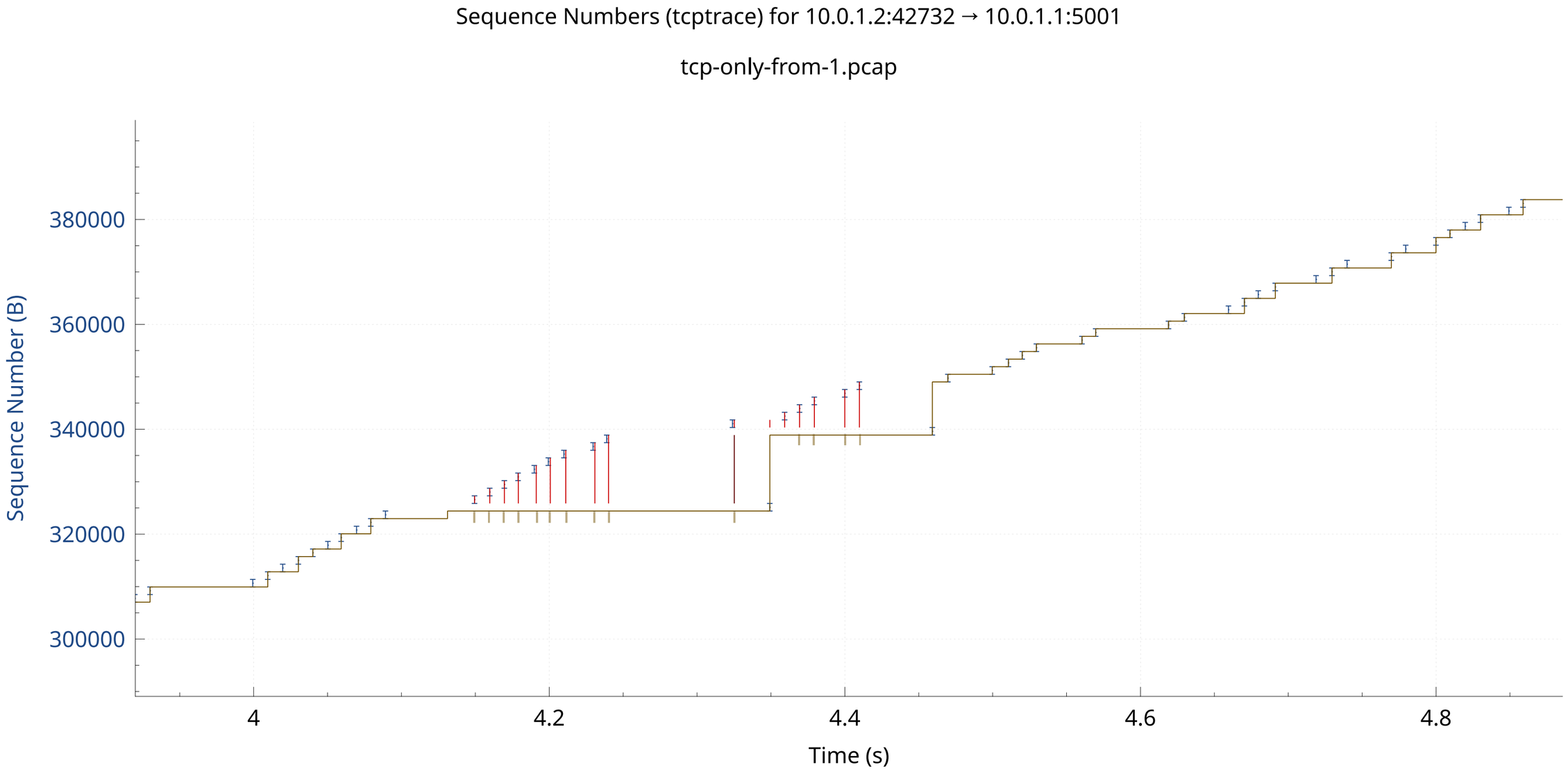
reading thigs graph

bottom line = last ack number

notches on bottom line = duplicate acks

red lines = selective ACK info

diff. timing in opposite direction



backup slides

TCP

transmission control protocol (TCP)

implements reliable streams of bytes

similar mechanism to what we've described

TCP extras/differences

bidirectional —

- separate sequence numbers in each direction
- can combine data (from A to B) with acknowledgment (from B to A)

sequence numbers are byte numbers —

- can retransmit data in different sized packets
- sequence numbers = index of *first byte* sent
- acknowledgment numbers = 1 + index of *last byte* acknowledged

dynamic/variable window sizes

- we'll discuss strategies later

offical name for packets = *segments*

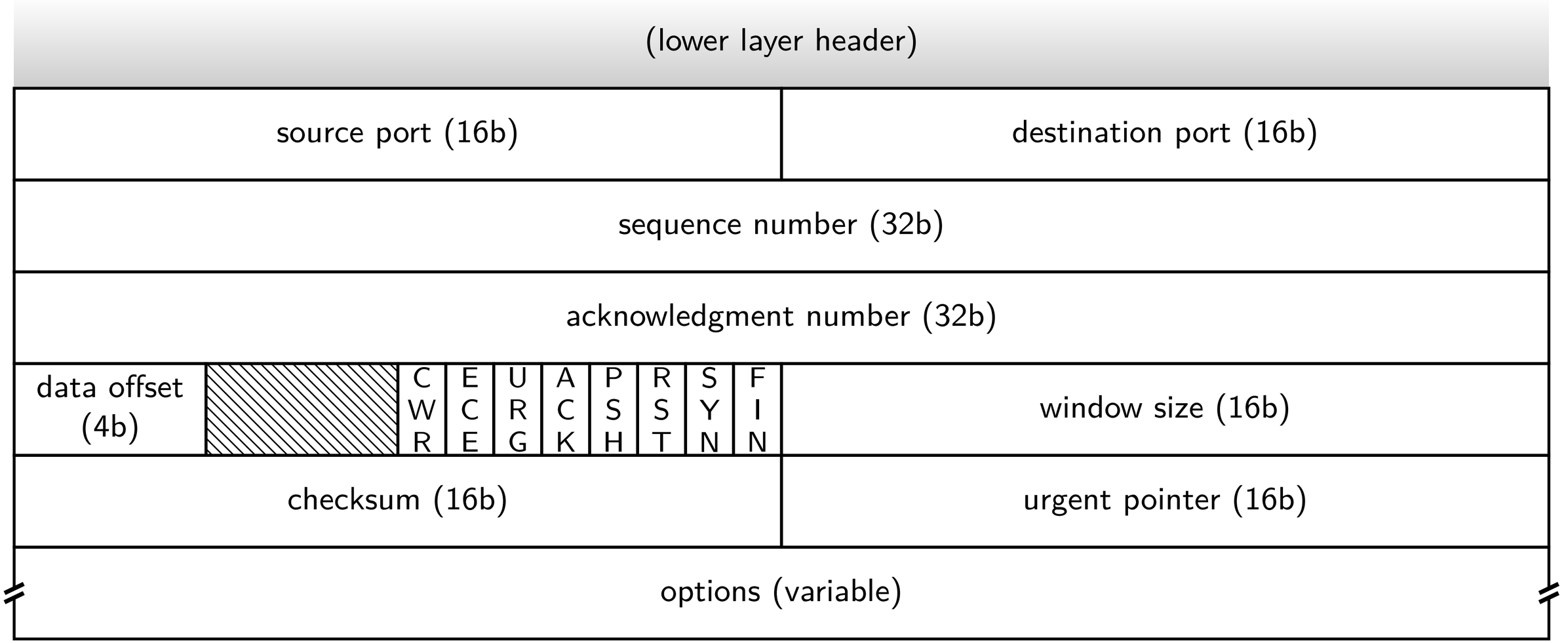
exercise (1)

direction	seq	ack	data len
→	100	200	200
→	300	200	300
←	300	600	0
←	300	600	100
→	?	?	0

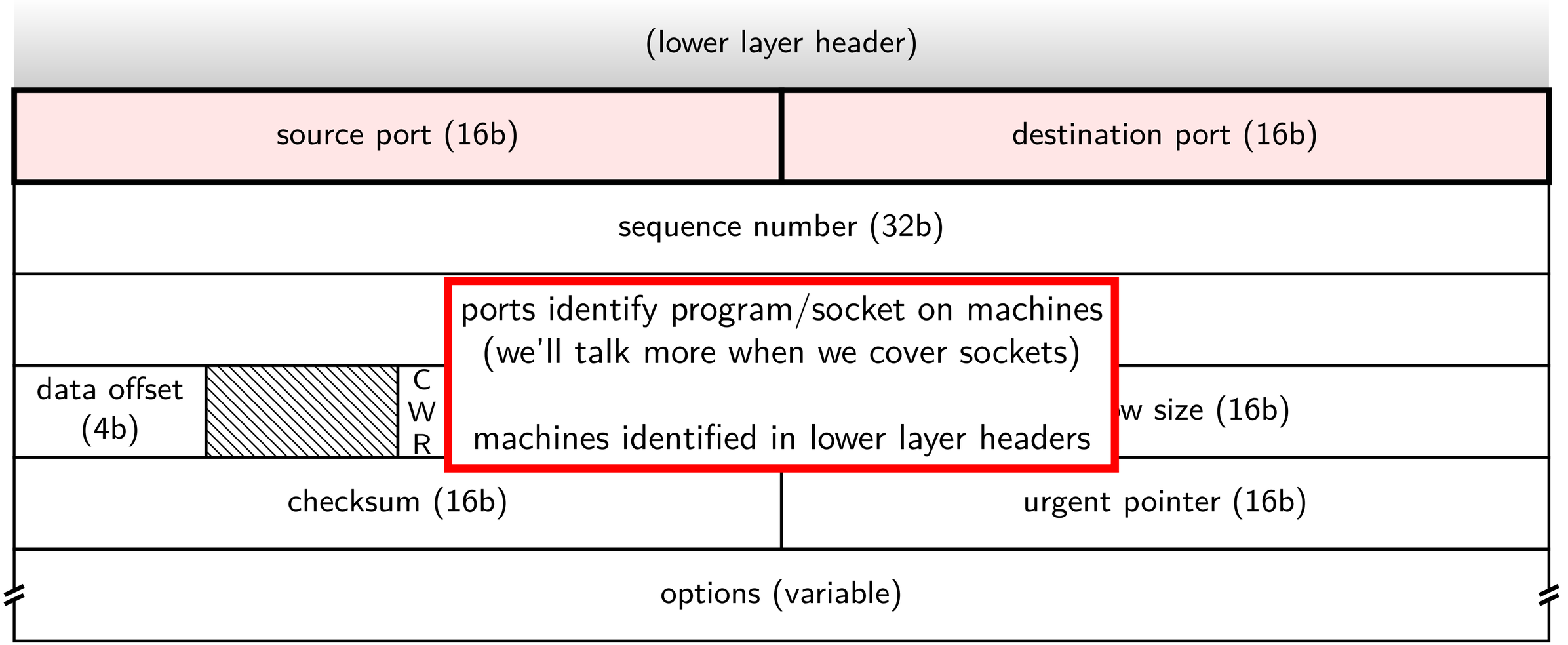
exercise (2)

direction	seq	ack	data len
→	100	200	200
→	300	200	300
←	?	?	100
→	600	300	0

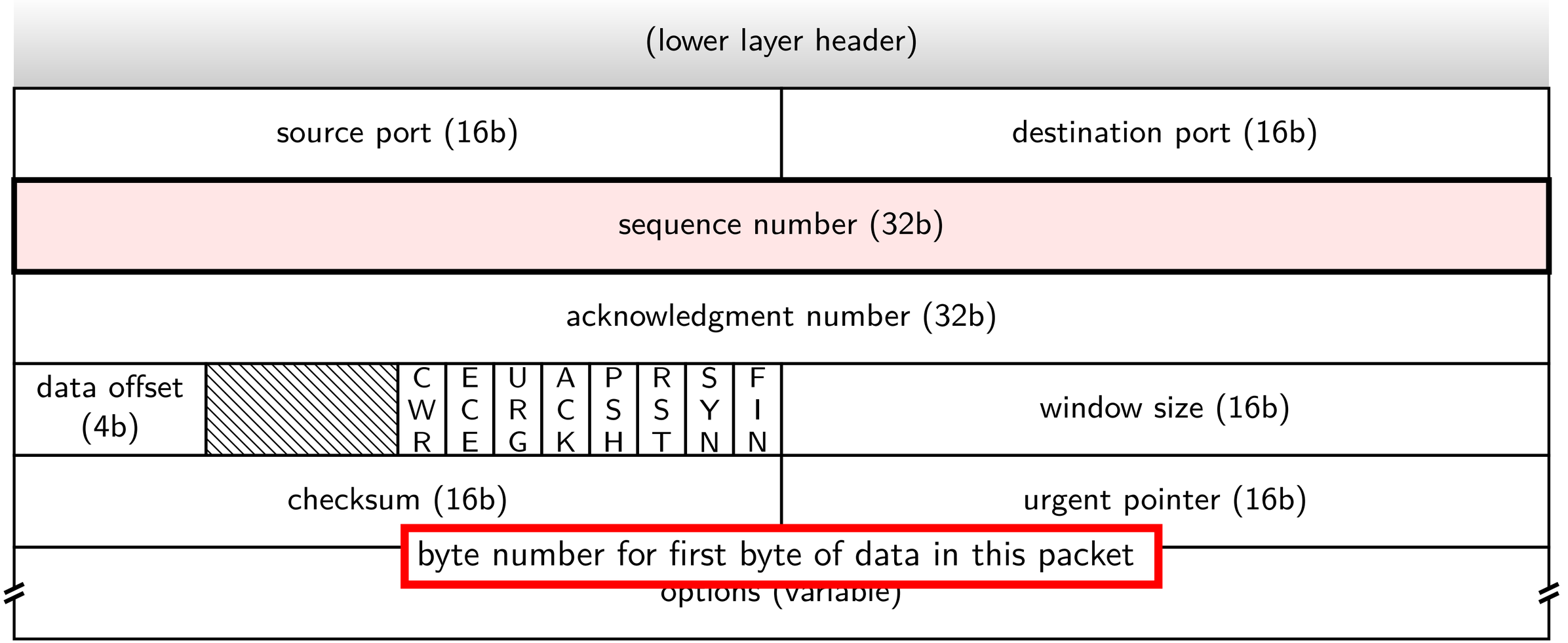
TCP segment format



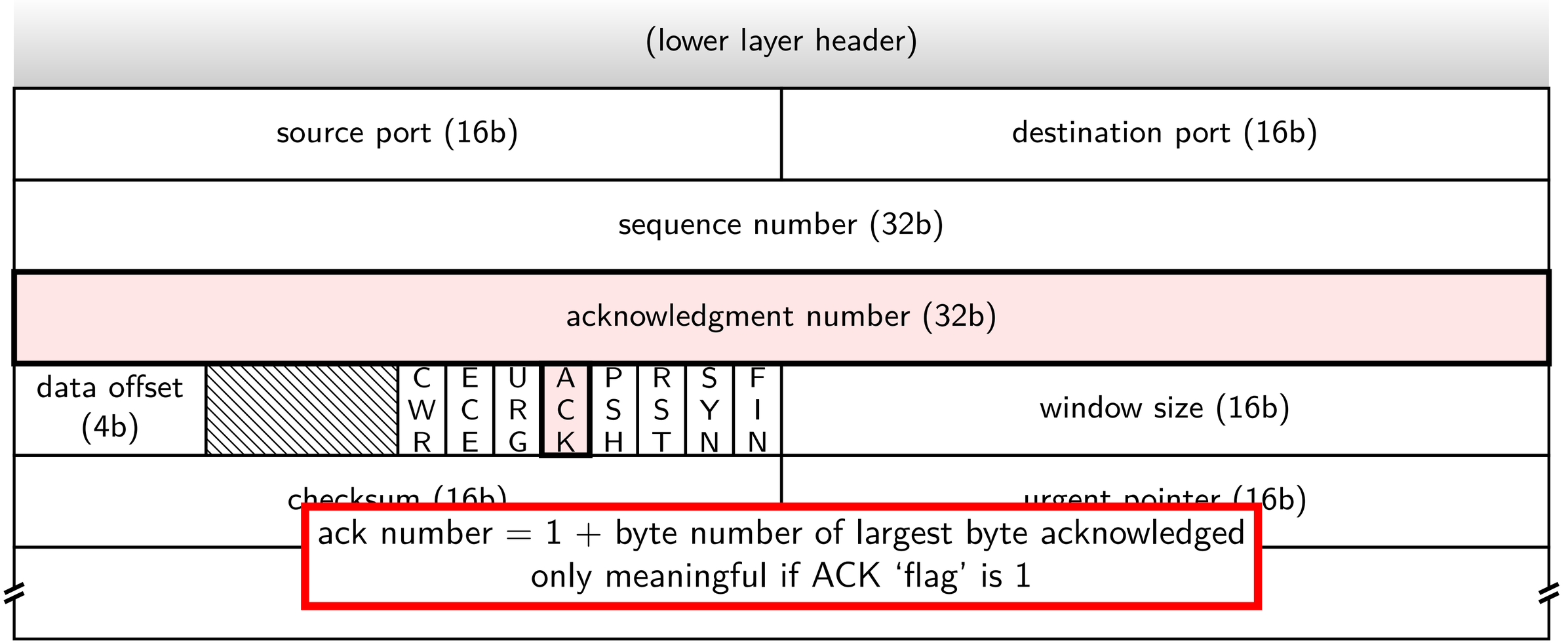
TCP segment format



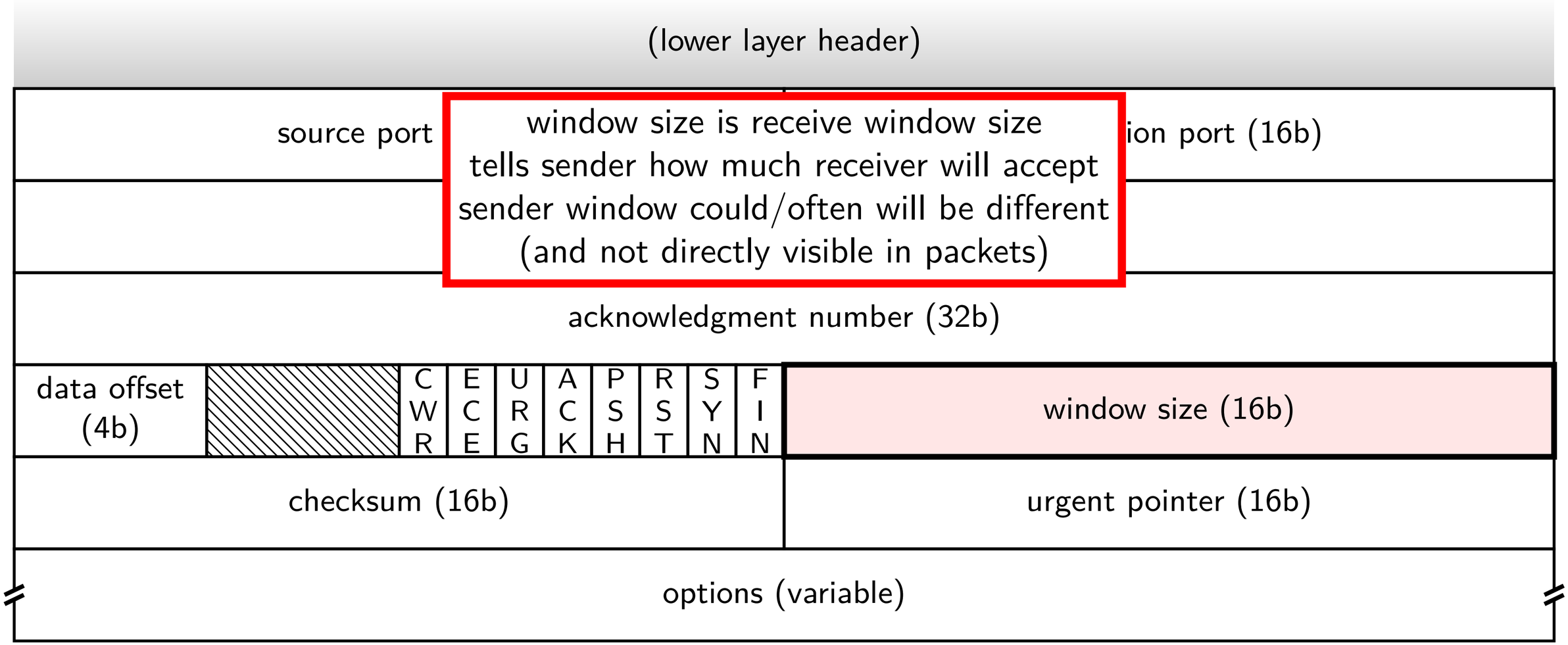
TCP segment format



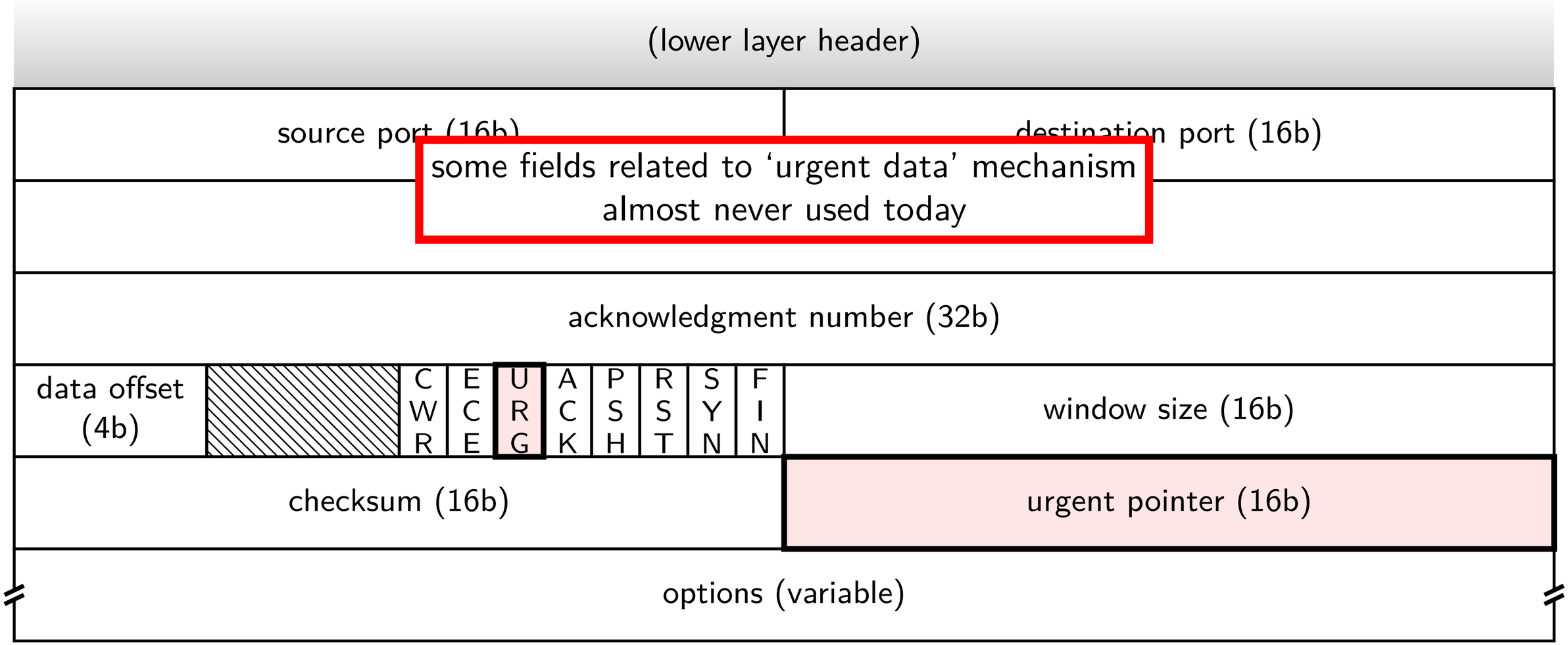
TCP segment format



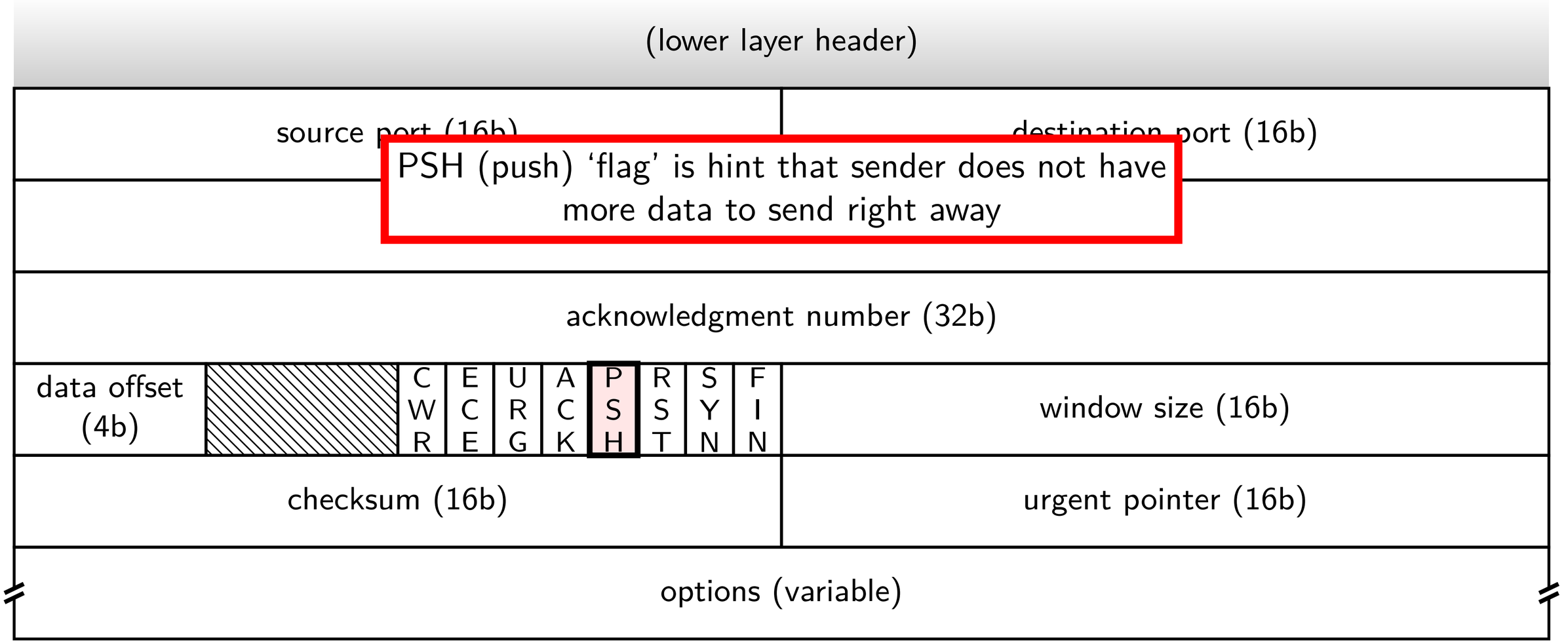
TCP segment format



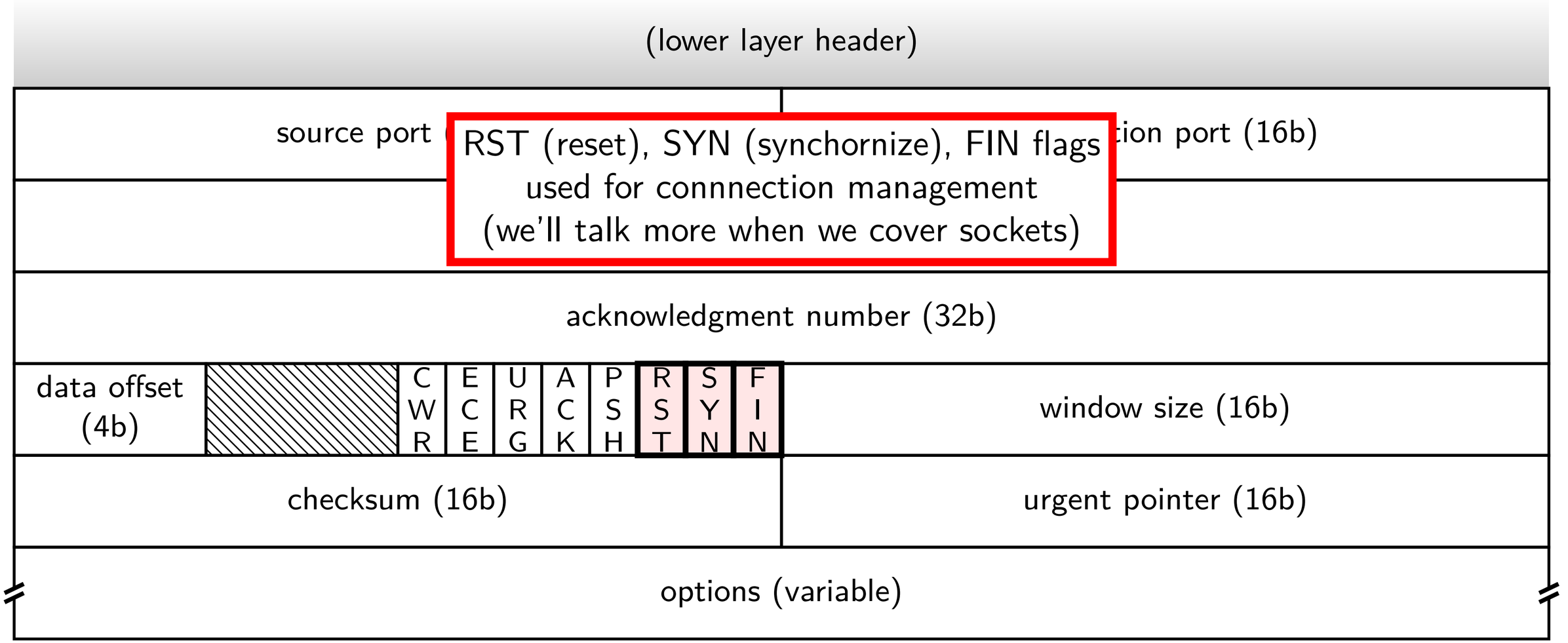
TCP segment format



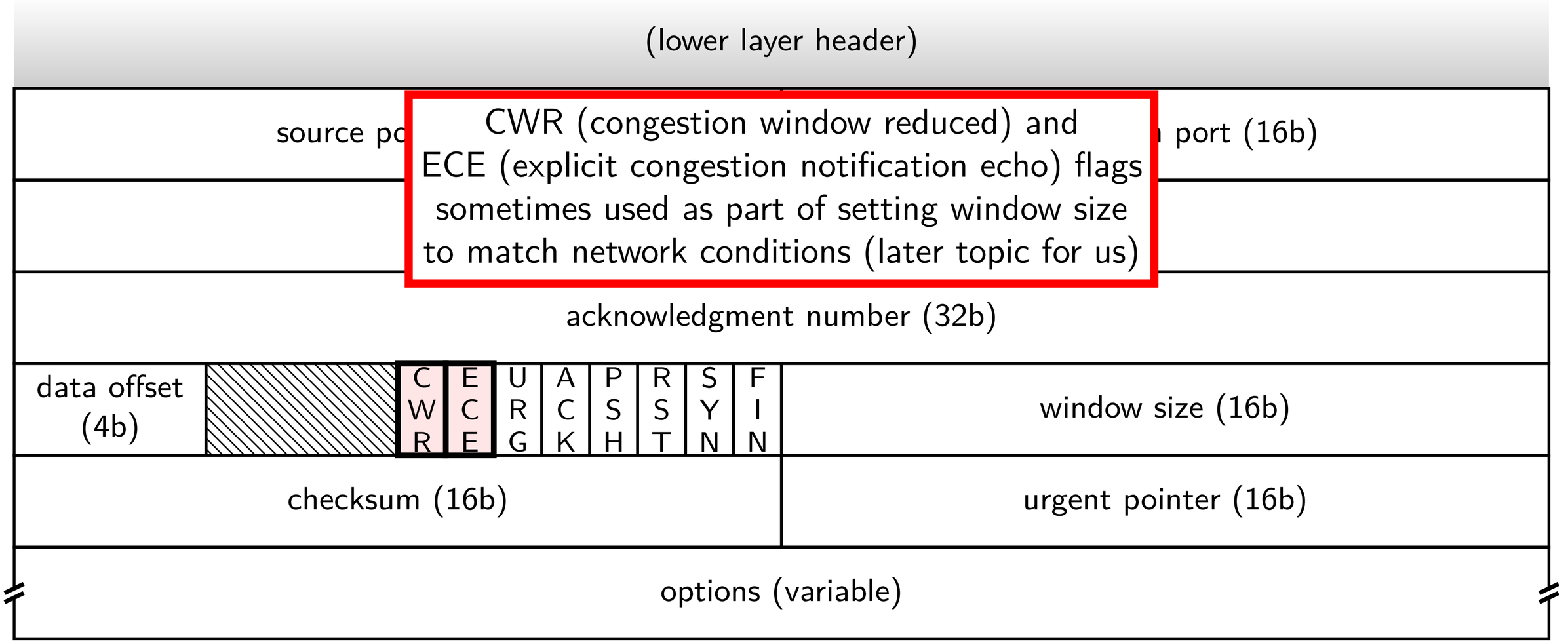
TCP segment format



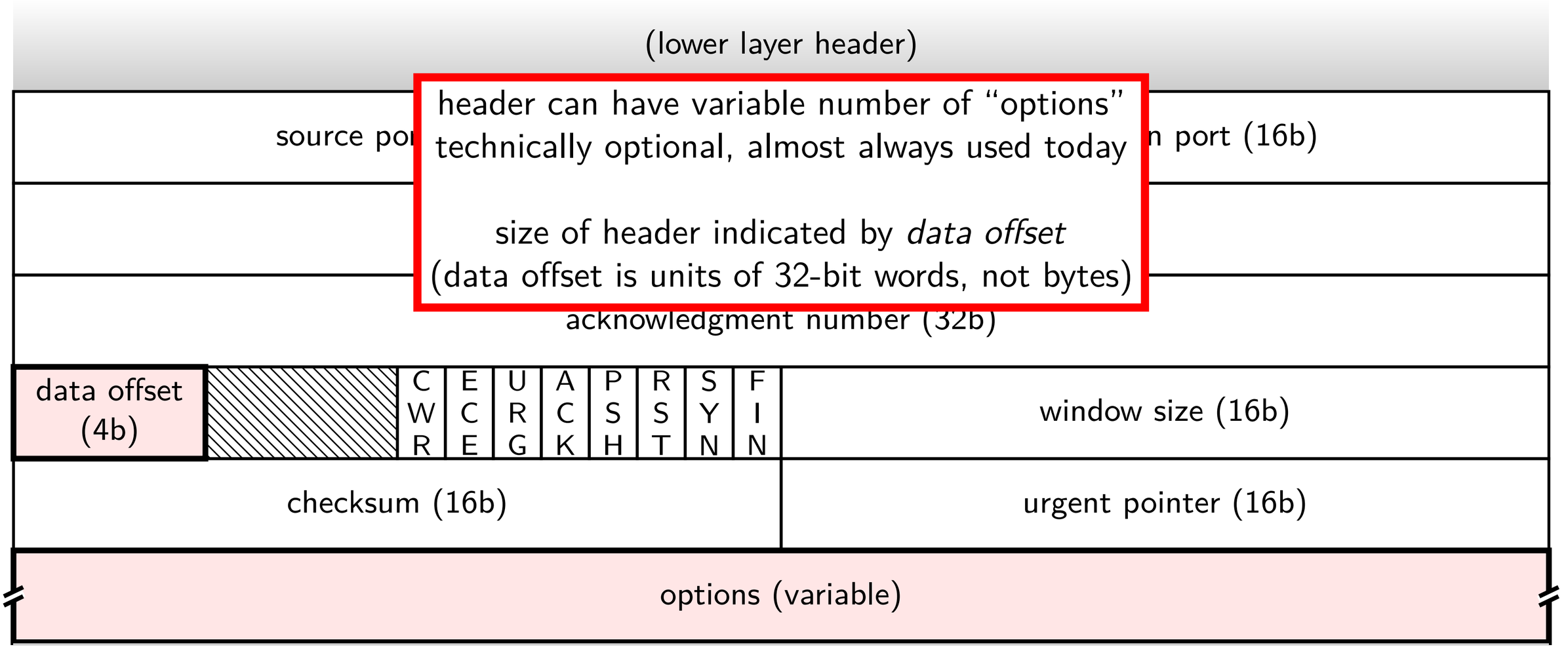
TCP segment format



TCP segment format



TCP segment format



exercise: maximum throughput

let's say we have a receiver window size of 65535 bytes

and a round-trip time of 100 ms

if we want to avoid sending data the receiver will reject as outside its window, maximum throughput?

A. around 32kbyte/sec
B. around 64kbyte/sec
C. around 128kbyte/sec
D. around 320kbyte/sec
E. around 640kbyte/sec
F. around 1280kbyte/sec
G. something else

selected TCP options

window size scale factor

- allow receiver window sizes greater than 64k
- needed to get reasonable bandwidth on modern networks

timestamps

- allow figuring out round trip time to estimate timeout
- extend 32-bit sequence number, which is too small for multi-gigabit networks

selective acknowledgements

- allow providing information about ‘holes’ in received data
- example: I got bytes 1–5000, 6000–7000, 8000–9000
- without it would only say 5000

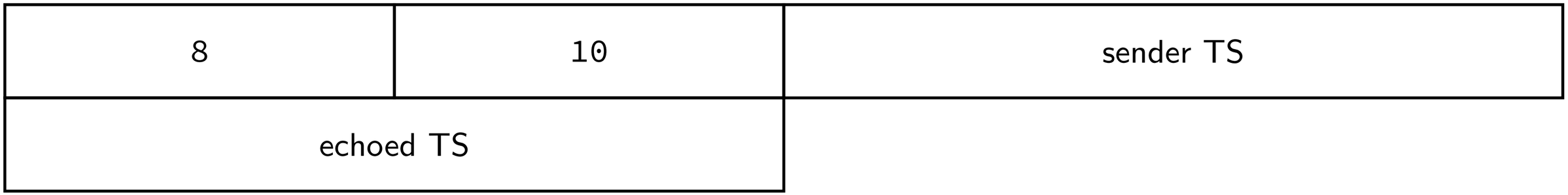
selected TCP option formats



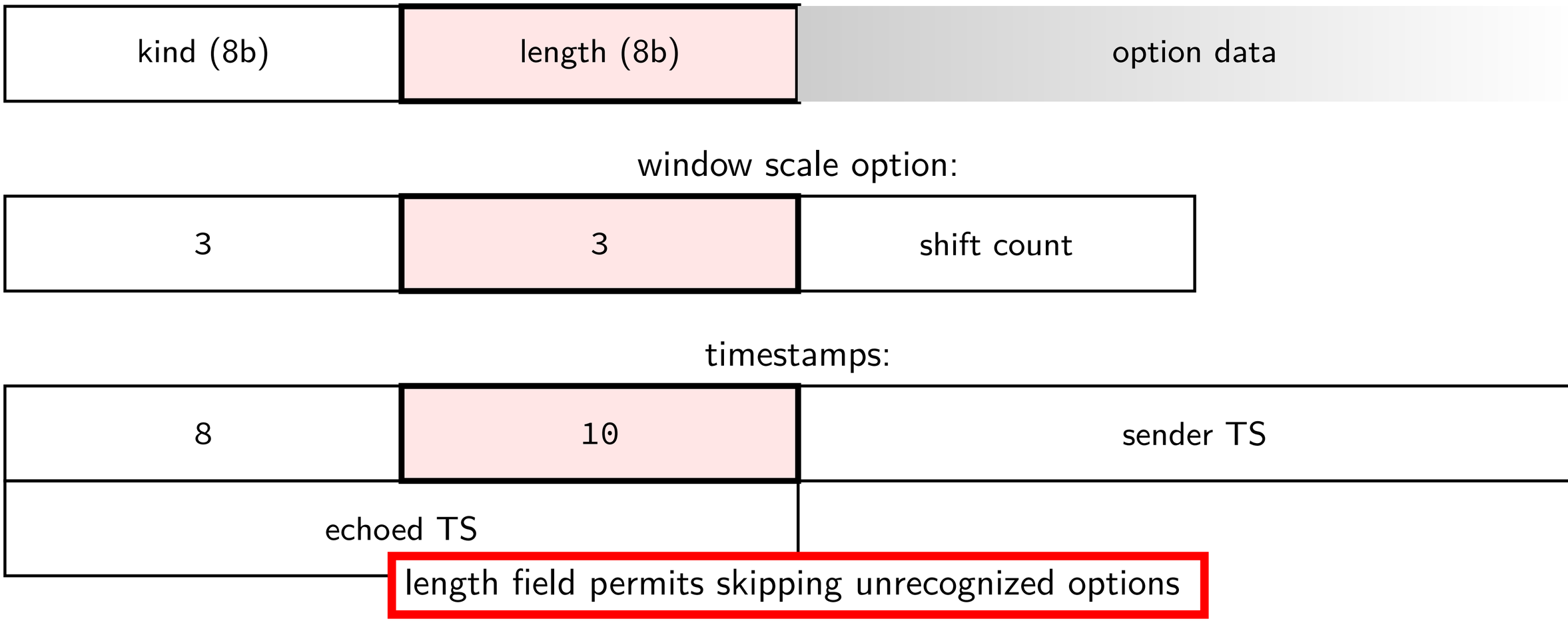
window scale option:



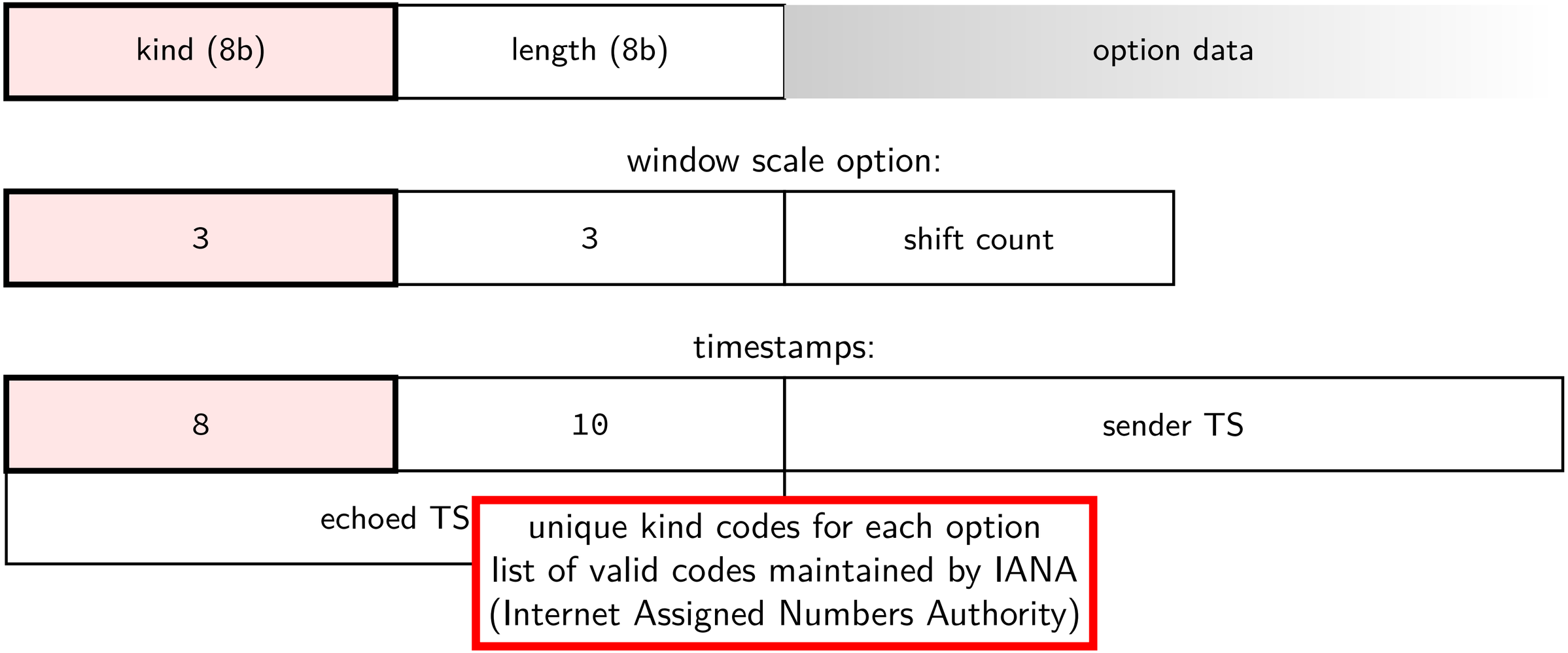
timestamps:



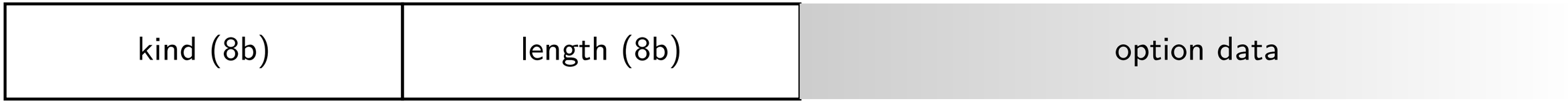
selected TCP option formats



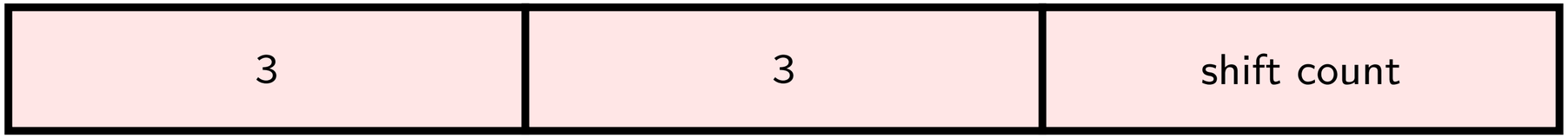
selected TCP option formats



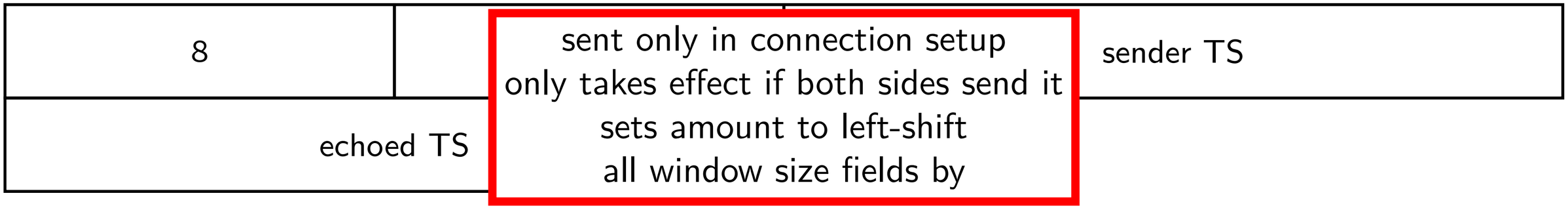
selected TCP option formats



window scale option:



timestamps:

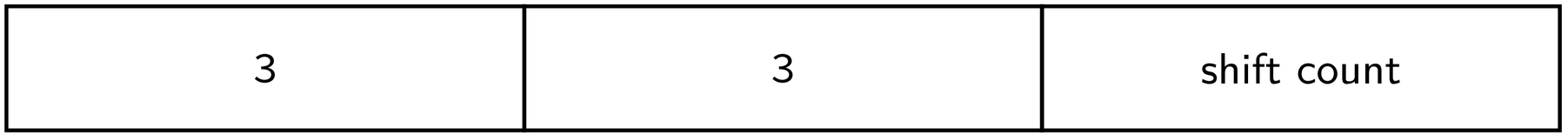


sent only in connection setup
only takes effect if both sides send it
sets amount to left-shift
all window size fields by

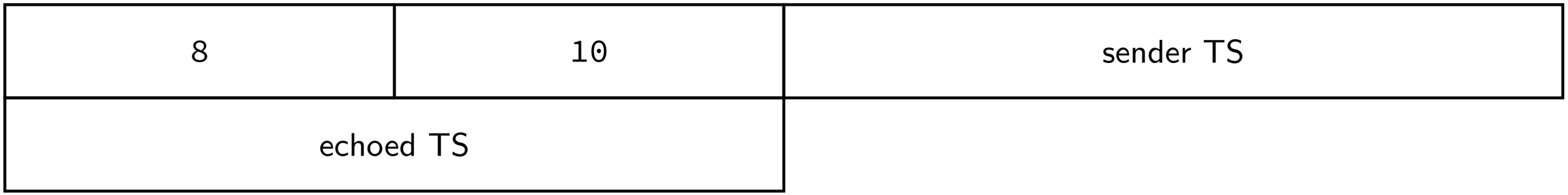
selected TCP option formats



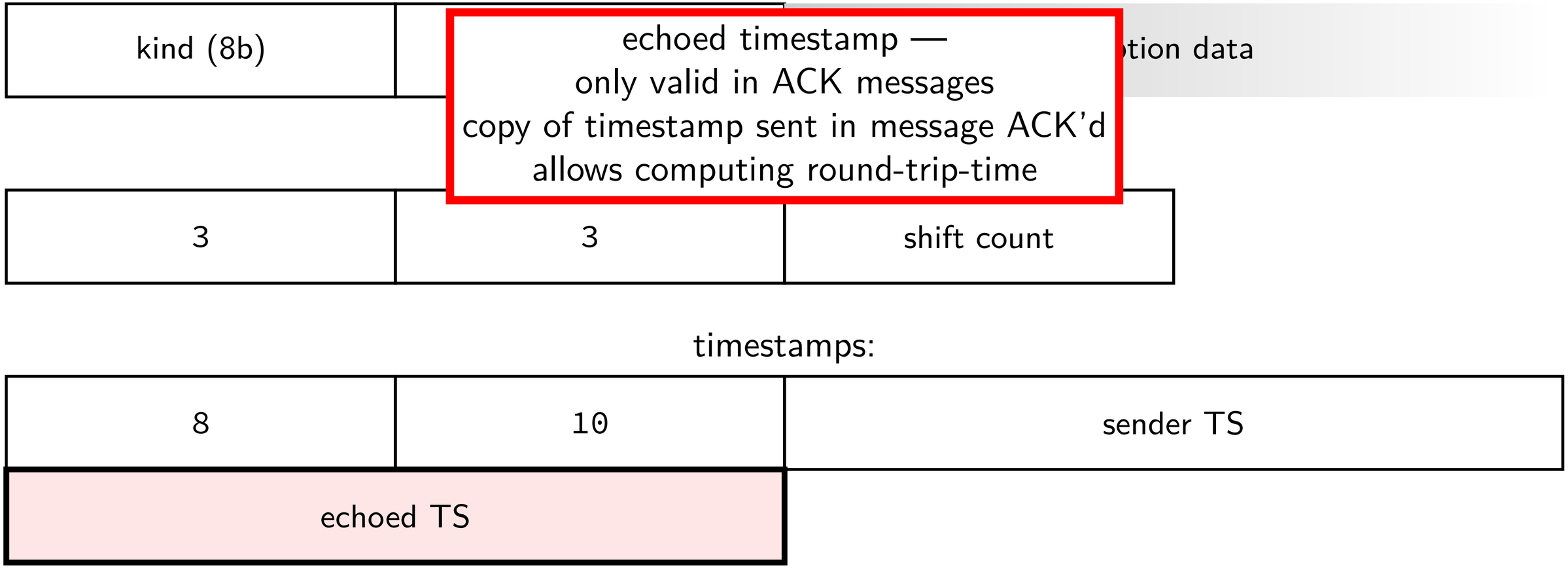
window scale option:



timestamps:



selected TCP option formats



TCP connection setup preview

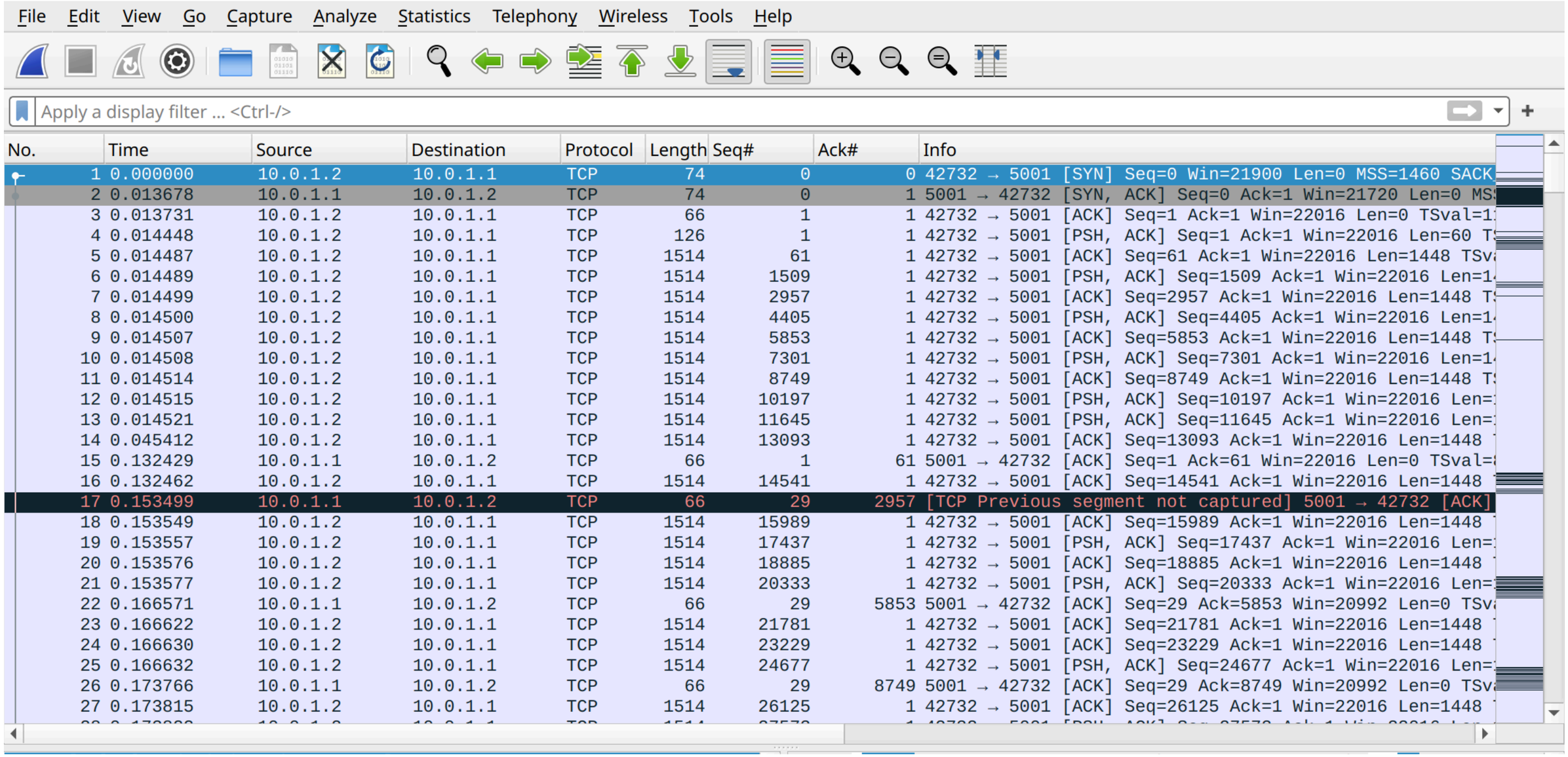
client→server: SYN flag + seq=randomA [no data]

server→client: SYN+ACK flags + seq=randomB ack=randomA+1 [no data]

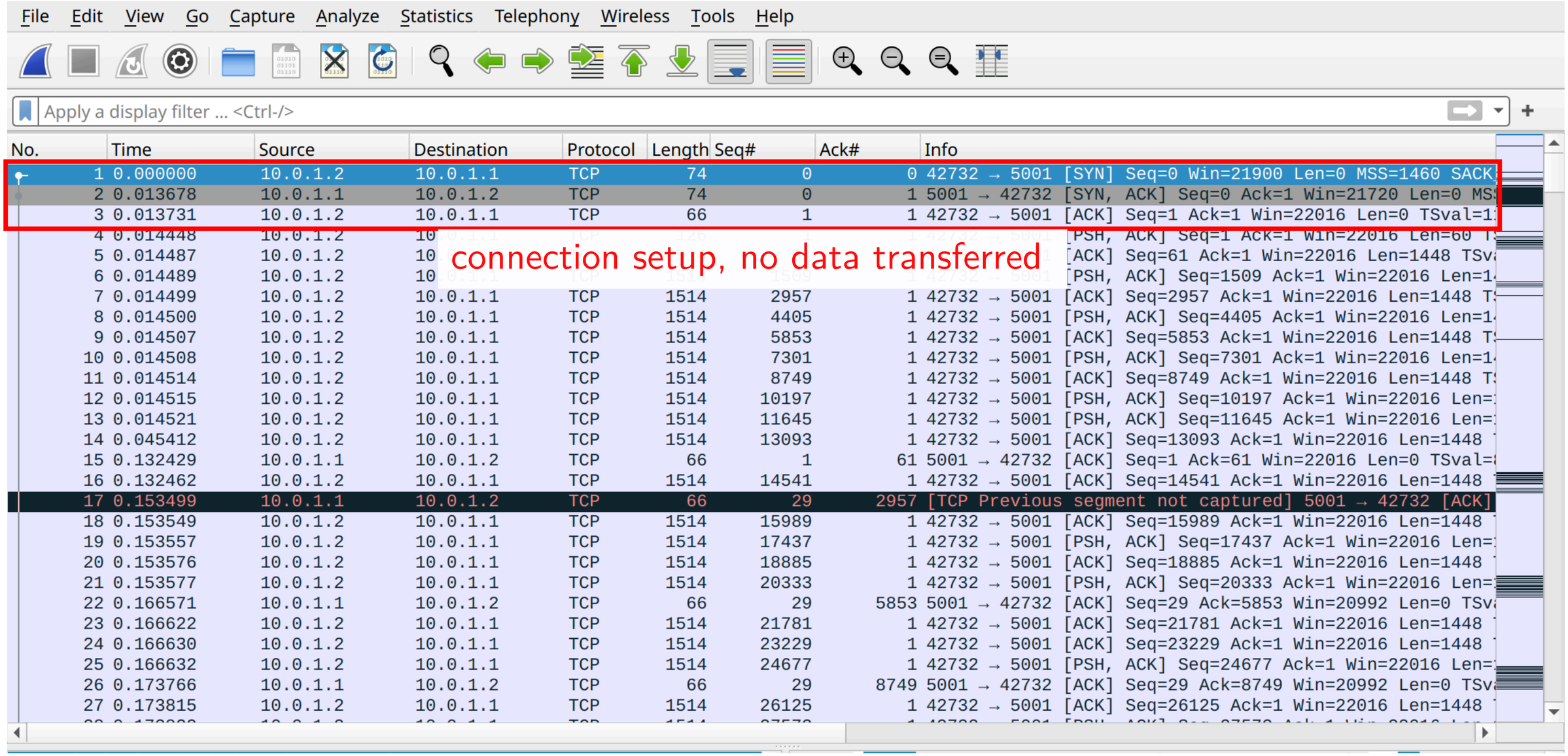
client→server: ACK flags + seq=randomA+1 ack=randomB+1 [no data]

a TCP connection

a TCP connection



a TCP connection



No.	Time	Source	Destination	Protocol	Length	Seq#	Ack#	Info
1	0.000000	10.0.1.2	10.0.1.1	TCP	74	0	0	42732 → 5001 [SYN] Seq=0 Win=21900 Len=0 MSS=1460 SACK
2	0.013678	10.0.1.1	10.0.1.2	TCP	74	0	1	5001 → 42732 [SYN, ACK] Seq=0 Ack=1 Win=21720 Len=0 MS
3	0.013731	10.0.1.2	10.0.1.1	TCP	66	1	1	42732 → 5001 [ACK] Seq=1 Ack=1 Win=22016 Len=0 TSval=1
4	0.014448	10.0.1.2	10					[PSH, ACK] Seq=1 Ack=1 Win=22016 Len=60 T
5	0.014487	10.0.1.2	10					[ACK] Seq=61 Ack=1 Win=22016 Len=1448 TSv
6	0.014489	10.0.1.2	10					[PSH, ACK] Seq=1509 Ack=1 Win=22016 Len=1
7	0.014499	10.0.1.2	10.0.1.1	TCP	1514	2957	1	42732 → 5001 [ACK] Seq=2957 Ack=1 Win=22016 Len=1448 T
8	0.014500	10.0.1.2	10.0.1.1	TCP	1514	4405	1	42732 → 5001 [PSH, ACK] Seq=4405 Ack=1 Win=22016 Len=1
9	0.014507	10.0.1.2	10.0.1.1	TCP	1514	5853	1	42732 → 5001 [ACK] Seq=5853 Ack=1 Win=22016 Len=1448 T
10	0.014508	10.0.1.2	10.0.1.1	TCP	1514	7301	1	42732 → 5001 [PSH, ACK] Seq=7301 Ack=1 Win=22016 Len=1
11	0.014514	10.0.1.2	10.0.1.1	TCP	1514	8749	1	42732 → 5001 [ACK] Seq=8749 Ack=1 Win=22016 Len=1448 T
12	0.014515	10.0.1.2	10.0.1.1	TCP	1514	10197	1	42732 → 5001 [PSH, ACK] Seq=10197 Ack=1 Win=22016 Len=
13	0.014521	10.0.1.2	10.0.1.1	TCP	1514	11645	1	42732 → 5001 [PSH, ACK] Seq=11645 Ack=1 Win=22016 Len=
14	0.045412	10.0.1.2	10.0.1.1	TCP	1514	13093	1	42732 → 5001 [ACK] Seq=13093 Ack=1 Win=22016 Len=1448
15	0.132429	10.0.1.1	10.0.1.2	TCP	66	1	61	5001 → 42732 [ACK] Seq=1 Ack=61 Win=22016 Len=0 TSval=
16	0.132462	10.0.1.2	10.0.1.1	TCP	1514	14541	1	42732 → 5001 [ACK] Seq=14541 Ack=1 Win=22016 Len=1448
17	0.153499	10.0.1.1	10.0.1.2	TCP	66	29	2957	[TCP Previous segment not captured] 5001 → 42732 [ACK]
18	0.153549	10.0.1.2	10.0.1.1	TCP	1514	15989	1	42732 → 5001 [ACK] Seq=15989 Ack=1 Win=22016 Len=1448
19	0.153557	10.0.1.2	10.0.1.1	TCP	1514	17437	1	42732 → 5001 [PSH, ACK] Seq=17437 Ack=1 Win=22016 Len=
20	0.153576	10.0.1.2	10.0.1.1	TCP	1514	18885	1	42732 → 5001 [ACK] Seq=18885 Ack=1 Win=22016 Len=1448
21	0.153577	10.0.1.2	10.0.1.1	TCP	1514	20333	1	42732 → 5001 [PSH, ACK] Seq=20333 Ack=1 Win=22016 Len=
22	0.166571	10.0.1.1	10.0.1.2	TCP	66	29	5853	5001 → 42732 [ACK] Seq=29 Ack=5853 Win=20992 Len=0 TSv
23	0.166622	10.0.1.2	10.0.1.1	TCP	1514	21781	1	42732 → 5001 [ACK] Seq=21781 Ack=1 Win=22016 Len=1448
24	0.166630	10.0.1.2	10.0.1.1	TCP	1514	23229	1	42732 → 5001 [ACK] Seq=23229 Ack=1 Win=22016 Len=1448
25	0.166632	10.0.1.2	10.0.1.1	TCP	1514	24677	1	42732 → 5001 [PSH, ACK] Seq=24677 Ack=1 Win=22016 Len=
26	0.173766	10.0.1.1	10.0.1.2	TCP	66	29	8749	5001 → 42732 [ACK] Seq=29 Ack=8749 Win=20992 Len=0 TSv
27	0.173815	10.0.1.2	10.0.1.1	TCP	1514	26125	1	42732 → 5001 [ACK] Seq=26125 Ack=1 Win=22016 Len=1448

a TCP connection

server+client sequence numbers advance by 1 to indicate where in setup

ngth	Seq#	Ack#	In
74	0	0	42
74	0	1	50
66	1	1	42

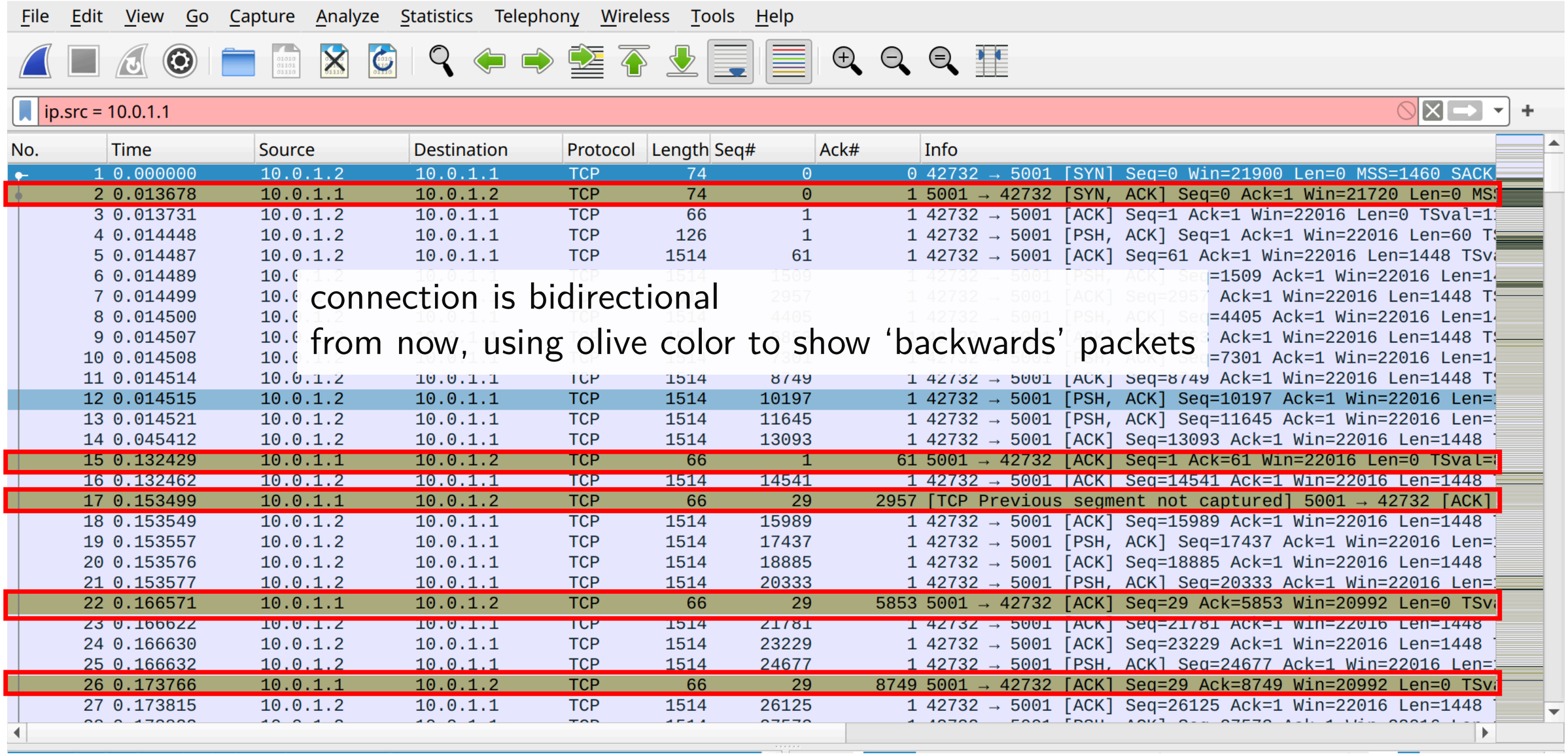
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

ip.src = 10.0.1.1

No.	Time	Source	Destination	Protocol	Length	Seq#	Ack#	Info
1	0.000000	10.0.1.2	10.0.1.1	TCP	74	0	0	42732 → 5001 [SYN] Seq=0 Win=21900 Len=0 MSS=1460 SACK
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4	0.014448	10.0.1.2	10					[PSH, ACK] Seq=1 Ack=1 Win=22016 Len=60 T
5	0.014487	10.0.1.2	10					[ACK] Seq=61 Ack=1 Win=22016 Len=1448 TSv
6	0.014489	10.0.1.2	10					[PSH, ACK] Seq=1509 Ack=1 Win=22016 Len=1
7	0.014499	10.0.1.2	10.0.1.1	TCP	1514	2957	1	42732 → 5001 [ACK] Seq=2957 Ack=1 Win=22016 Len=1448 T
8	0.014500	10.0.1.2	10.0.1.1	TCP	1514	4405	1	42732 → 5001 [PSH, ACK] Seq=4405 Ack=1 Win=22016 Len=1
9	0.014507	10.0.1.2	10.0.1.1	TCP	1514	5853	1	42732 → 5001 [ACK] Seq=5853 Ack=1 Win=22016 Len=1448 T
10	0.014508	10.0.1.2	10.0.1.1	TCP	1514	7301	1	42732 → 5001 [PSH, ACK] Seq=7301 Ack=1 Win=22016 Len=1
11	0.014514	10.0.1.2	10.0.1.1	TCP	1514	8749	1	42732 → 5001 [ACK] Seq=8749 Ack=1 Win=22016 Len=1448 T
12	0.014515	10.0.1.2	10.0.1.1	TCP	1514	10197	1	42732 → 5001 [PSH, ACK] Seq=10197 Ack=1 Win=22016 Len=
13	0.014521	10.0.1.2	10.0.1.1	TCP	1514	11645	1	42732 → 5001 [PSH, ACK] Seq=11645 Ack=1 Win=22016 Len=
14	0.045412	10.0.1.2	10.0.1.1	TCP	1514	13093	1	42732 → 5001 [ACK] Seq=13093 Ack=1 Win=22016 Len=1448
15	0.132429	10.0.1.1	10.0.1.2	TCP	66	1	61	5001 → 42732 [ACK] Seq=1 Ack=61 Win=22016 Len=0 TSval=
16	0.132462	10.0.1.2	10.0.1.1	TCP	1514	14541	1	42732 → 5001 [ACK] Seq=14541 Ack=1 Win=22016 Len=1448
17	0.153499	10.0.1.1	10.0.1.2	TCP	66	29	2957	[TCP Previous segment not captured] 5001 → 42732 [ACK]
18	0.153549	10.0.1.2	10.0.1.1	TCP	1514	15989	1	42732 → 5001 [ACK] Seq=15989 Ack=1 Win=22016 Len=1448
19	0.153557	10.0.1.2	10.0.1.1	TCP	1514	17437	1	42732 → 5001 [PSH, ACK] Seq=17437 Ack=1 Win=22016 Len=
20	0.153576	10.0.1.2	10.0.1.1	TCP	1514	18885	1	42732 → 5001 [ACK] Seq=18885 Ack=1 Win=22016 Len=1448
21	0.153577	10.0.1.2	10.0.1.1	TCP	1514	20333	1	42732 → 5001 [PSH, ACK] Seq=20333 Ack=1 Win=22016 Len=
22	0.166571	10.0.1.1	10.0.1.2	TCP	66	29	5853	5001 → 42732 [ACK] Seq=29 Ack=5853 Win=20992 Len=0 TSv
23	0.166622	10.0.1.2	10.0.1.1	TCP	1514	21781	1	42732 → 5001 [ACK] Seq=21781 Ack=1 Win=22016 Len=1448
24	0.166630	10.0.1.2	10.0.1.1	TCP	1514	23229	1	42732 → 5001 [ACK] Seq=23229 Ack=1 Win=22016 Len=1448
25	0.166632	10.0.1.2	10.0.1.1	TCP	1514	24677	1	42732 → 5001 [PSH, ACK] Seq=24677 Ack=1 Win=22016 Len=
26	0.173766	10.0.1.1	10.0.1.2	TCP	66	29	8749	5001 → 42732 [ACK] Seq=29 Ack=8749 Win=20992 Len=0 TSv
27	0.173815	10.0.1.2	10.0.1.1	TCP	1514	26125	1	42732 → 5001 [ACK] Seq=26125 Ack=1 Win=22016 Len=1448

connection setup, no data transferred

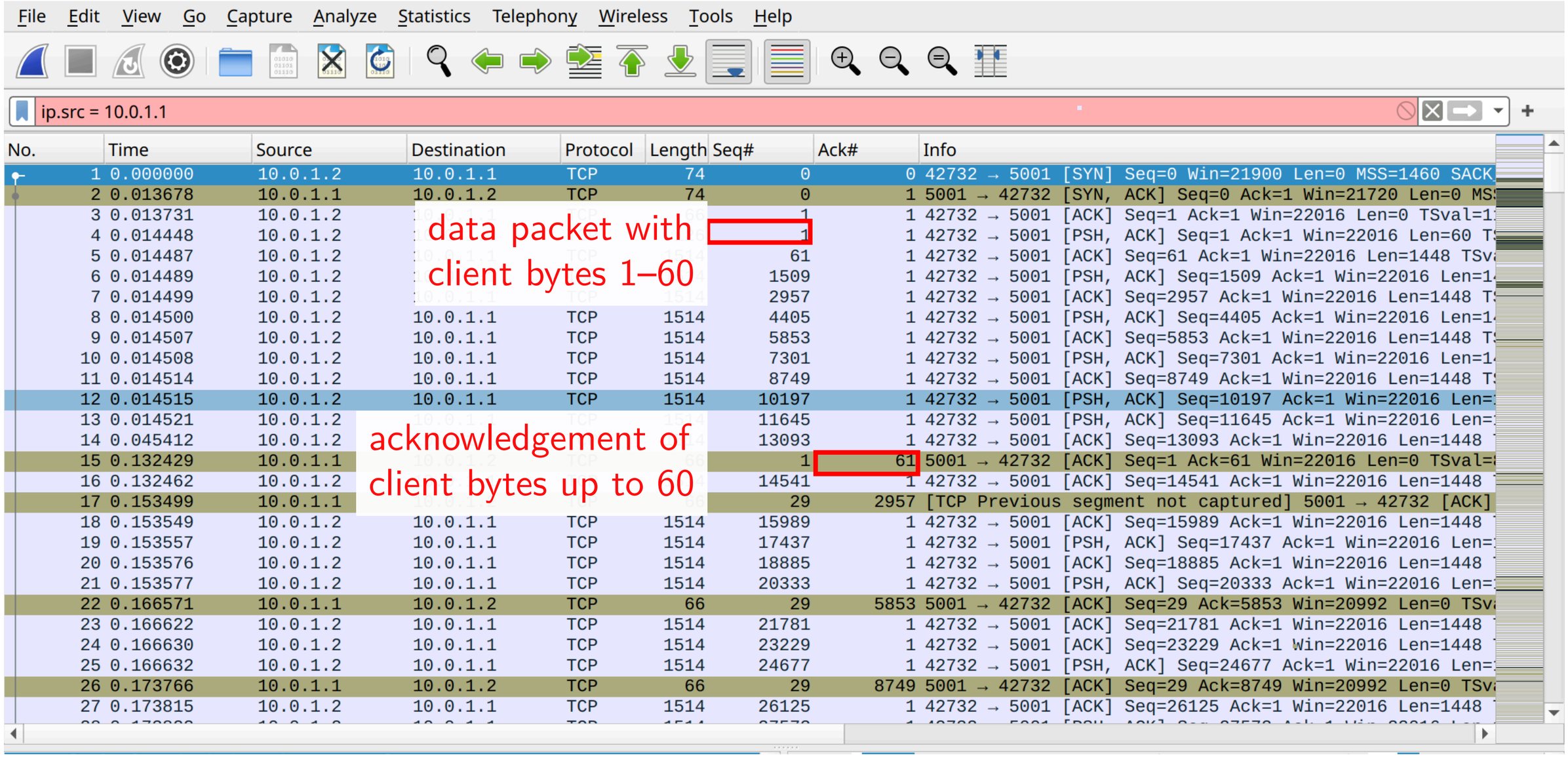
a TCP connection



connection is bidirectional

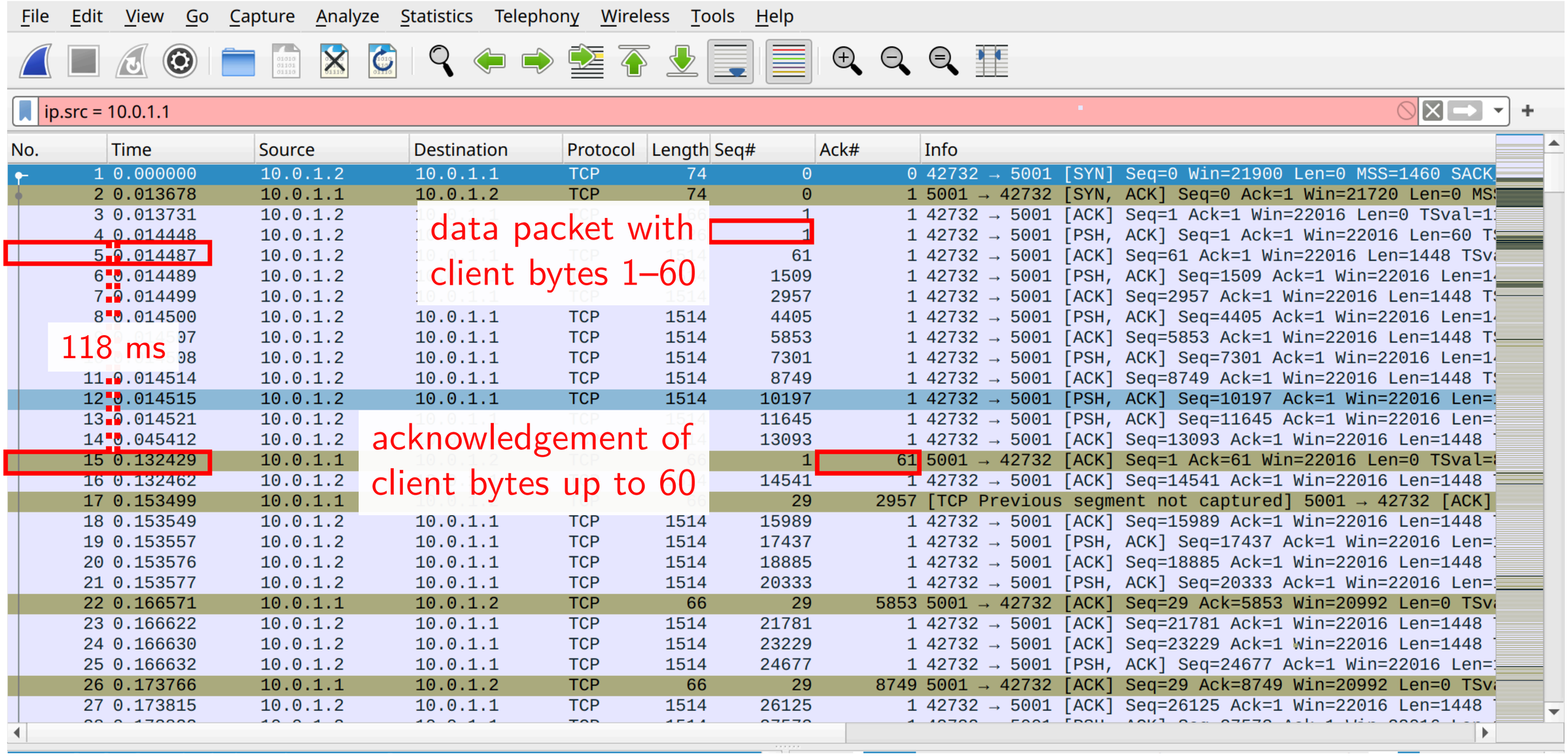
from now, using olive color to show ‘backwards’ packets

a TCP connection

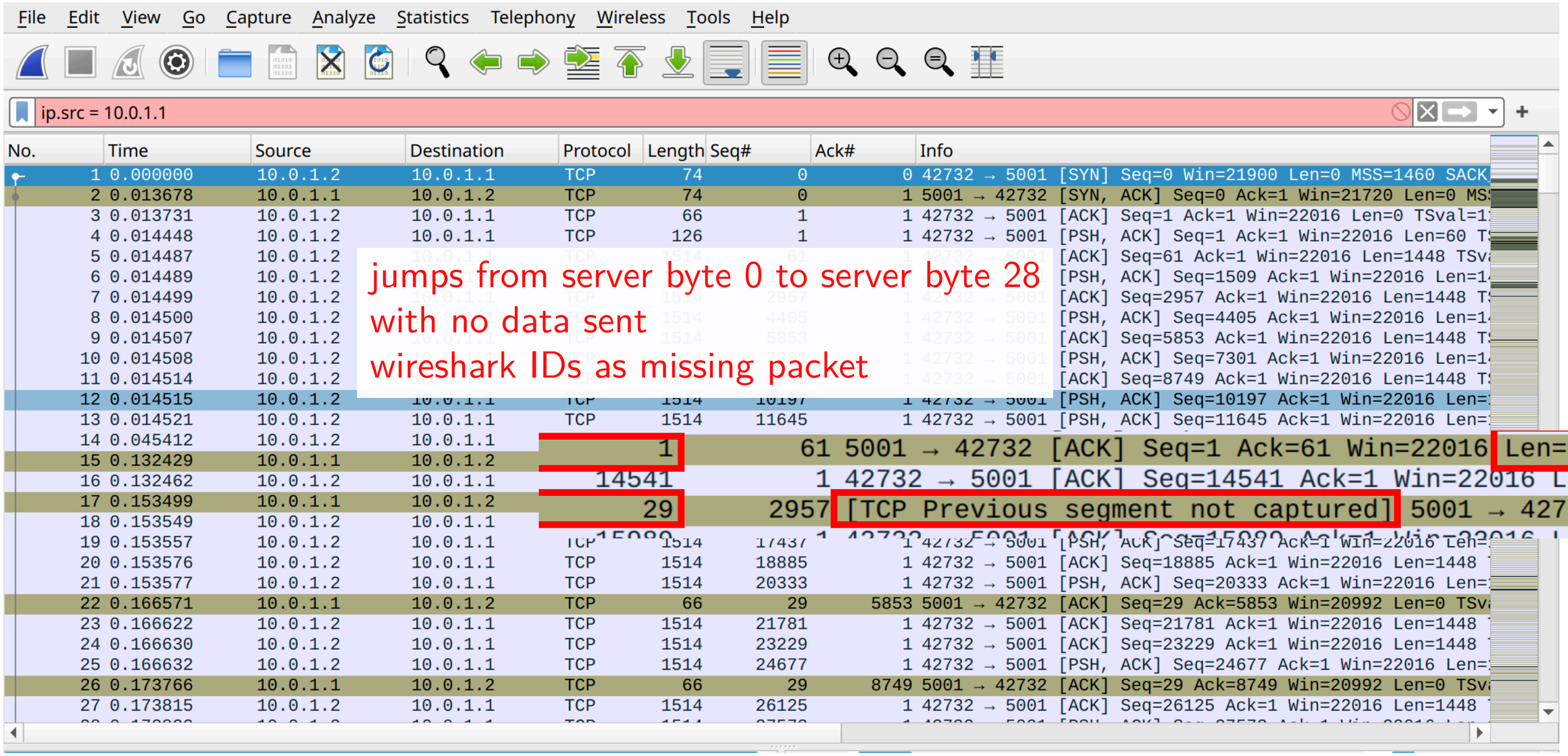


No.	Time	Source	Destination	Protocol	Length	Seq#	Ack#	Info
1	0.000000	10.0.1.2	10.0.1.1	TCP	74	0	0	42732 → 5001 [SYN] Seq=0 Win=21900 Len=0 MSS=1460 SACK
2	0.013678	10.0.1.1	10.0.1.2	TCP	74	0	1	5001 → 42732 [SYN, ACK] Seq=0 Ack=1 Win=21720 Len=0 MS
3	0.013731	10.0.1.2				1	1	42732 → 5001 [ACK] Seq=1 Ack=1 Win=22016 Len=0 TSval=1
4	0.014448	10.0.1.2				1	1	42732 → 5001 [PSH, ACK] Seq=1 Ack=1 Win=22016 Len=60 T
5	0.014487	10.0.1.2				61	1	42732 → 5001 [ACK] Seq=61 Ack=1 Win=22016 Len=1448 TSv
6	0.014489	10.0.1.2				1509	1	42732 → 5001 [PSH, ACK] Seq=1509 Ack=1 Win=22016 Len=1
7	0.014499	10.0.1.2				2957	1	42732 → 5001 [ACK] Seq=2957 Ack=1 Win=22016 Len=1448 T
8	0.014500	10.0.1.2	10.0.1.1	TCP	1514	4405	1	42732 → 5001 [PSH, ACK] Seq=4405 Ack=1 Win=22016 Len=1
9	0.014507	10.0.1.2	10.0.1.1	TCP	1514	5853	1	42732 → 5001 [ACK] Seq=5853 Ack=1 Win=22016 Len=1448 T
10	0.014508	10.0.1.2	10.0.1.1	TCP	1514	7301	1	42732 → 5001 [PSH, ACK] Seq=7301 Ack=1 Win=22016 Len=1
11	0.014514	10.0.1.2	10.0.1.1	TCP	1514	8749	1	42732 → 5001 [ACK] Seq=8749 Ack=1 Win=22016 Len=1448 T
12	0.014515	10.0.1.2	10.0.1.1	TCP	1514	10197	1	42732 → 5001 [PSH, ACK] Seq=10197 Ack=1 Win=22016 Len=
13	0.014521	10.0.1.2				11645	1	42732 → 5001 [PSH, ACK] Seq=11645 Ack=1 Win=22016 Len=
14	0.045412	10.0.1.2				13093	1	42732 → 5001 [ACK] Seq=13093 Ack=1 Win=22016 Len=1448
15	0.132429	10.0.1.1				1	61	5001 → 42732 [ACK] Seq=1 Ack=61 Win=22016 Len=0 TSval=
16	0.132462	10.0.1.2				14541	1	42732 → 5001 [ACK] Seq=14541 Ack=1 Win=22016 Len=1448
17	0.153499	10.0.1.1				29	2957	[TCP Previous segment not captured] 5001 → 42732 [ACK]
18	0.153549	10.0.1.2	10.0.1.1	TCP	1514	15989	1	42732 → 5001 [ACK] Seq=15989 Ack=1 Win=22016 Len=1448
19	0.153557	10.0.1.2	10.0.1.1	TCP	1514	17437	1	42732 → 5001 [PSH, ACK] Seq=17437 Ack=1 Win=22016 Len=
20	0.153576	10.0.1.2	10.0.1.1	TCP	1514	18885	1	42732 → 5001 [ACK] Seq=18885 Ack=1 Win=22016 Len=1448
21	0.153577	10.0.1.2	10.0.1.1	TCP	1514	20333	1	42732 → 5001 [PSH, ACK] Seq=20333 Ack=1 Win=22016 Len=
22	0.166571	10.0.1.1	10.0.1.2	TCP	66	29	5853	5001 → 42732 [ACK] Seq=29 Ack=5853 Win=20992 Len=0 TSv
23	0.166622	10.0.1.2	10.0.1.1	TCP	1514	21781	1	42732 → 5001 [ACK] Seq=21781 Ack=1 Win=22016 Len=1448
24	0.166630	10.0.1.2	10.0.1.1	TCP	1514	23229	1	42732 → 5001 [ACK] Seq=23229 Ack=1 Win=22016 Len=1448
25	0.166632	10.0.1.2	10.0.1.1	TCP	1514	24677	1	42732 → 5001 [PSH, ACK] Seq=24677 Ack=1 Win=22016 Len=
26	0.173766	10.0.1.1	10.0.1.2	TCP	66	29	8749	5001 → 42732 [ACK] Seq=29 Ack=8749 Win=20992 Len=0 TSv
27	0.173815	10.0.1.2	10.0.1.1	TCP	1514	26125	1	42732 → 5001 [ACK] Seq=26125 Ack=1 Win=22016 Len=1448

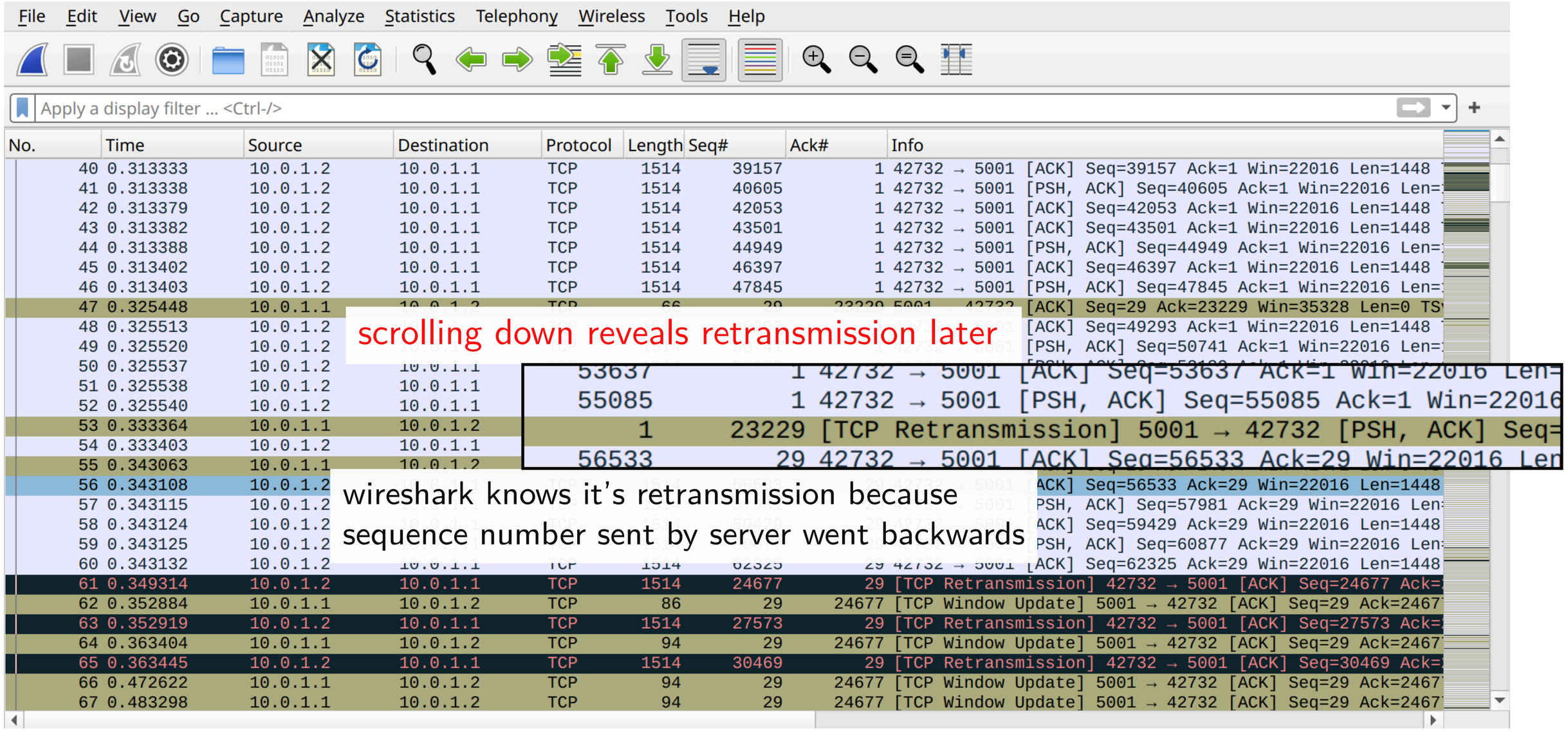
a TCP connection



a TCP connection

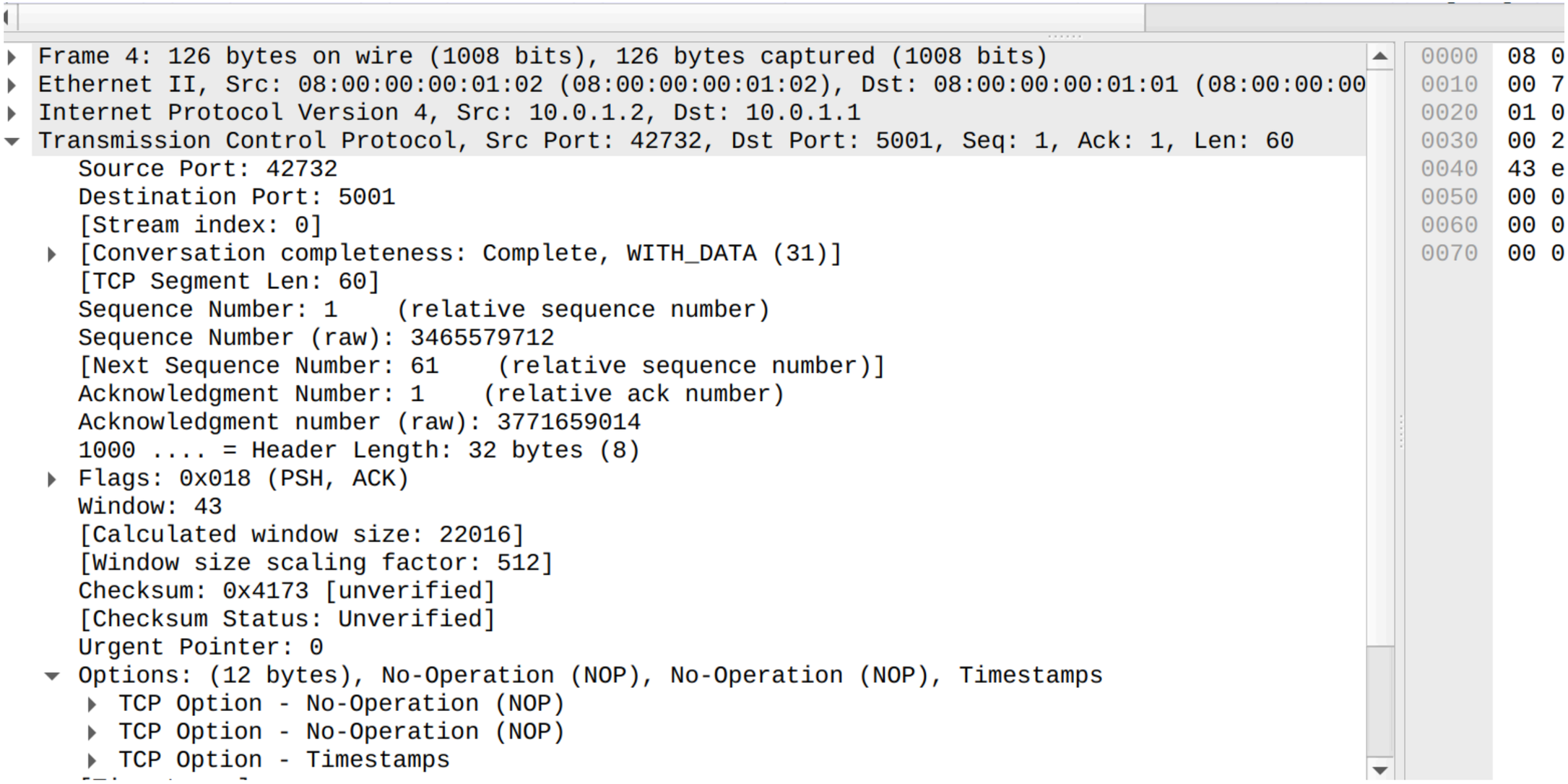


a TCP connection

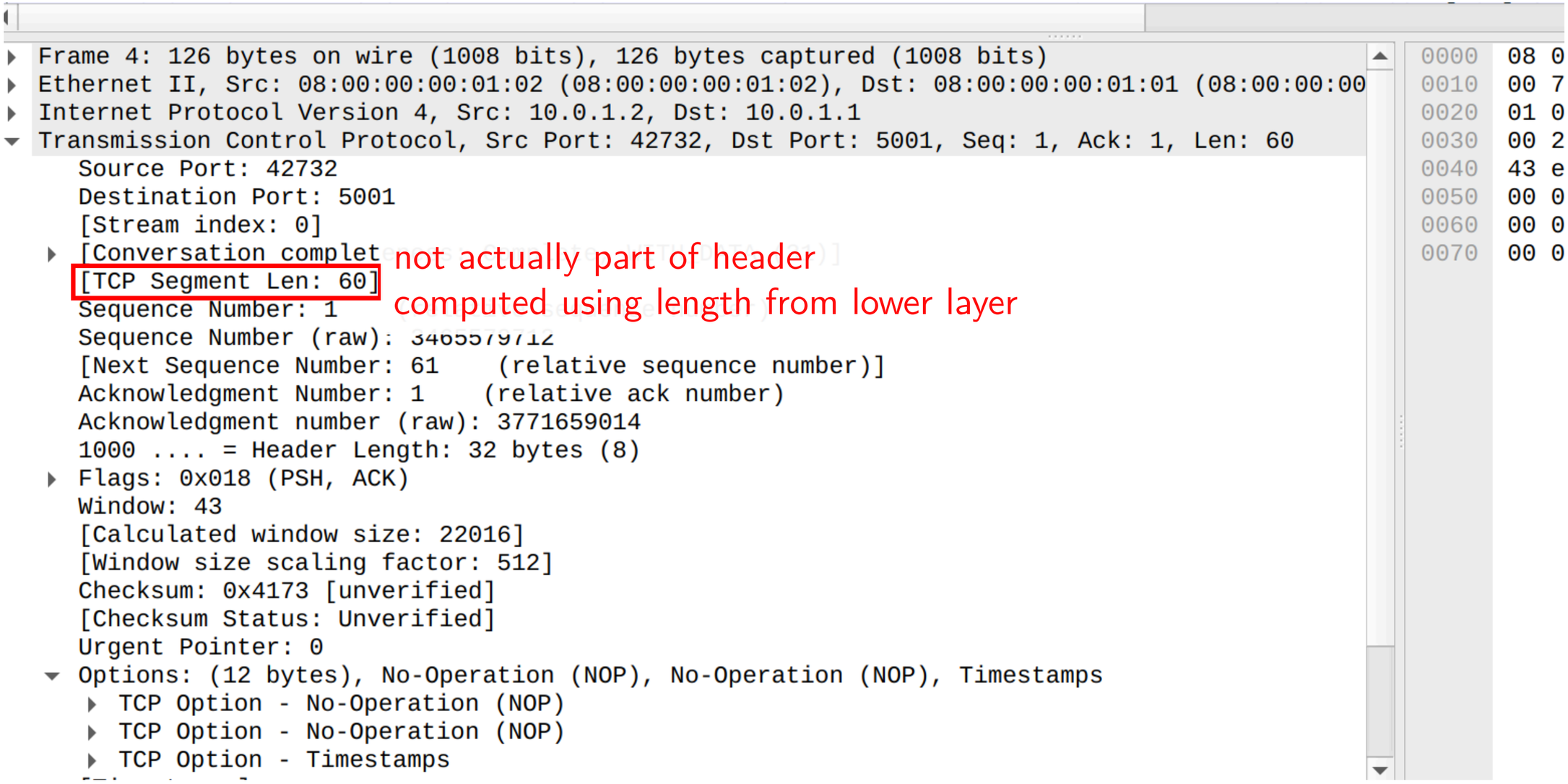


first data packet

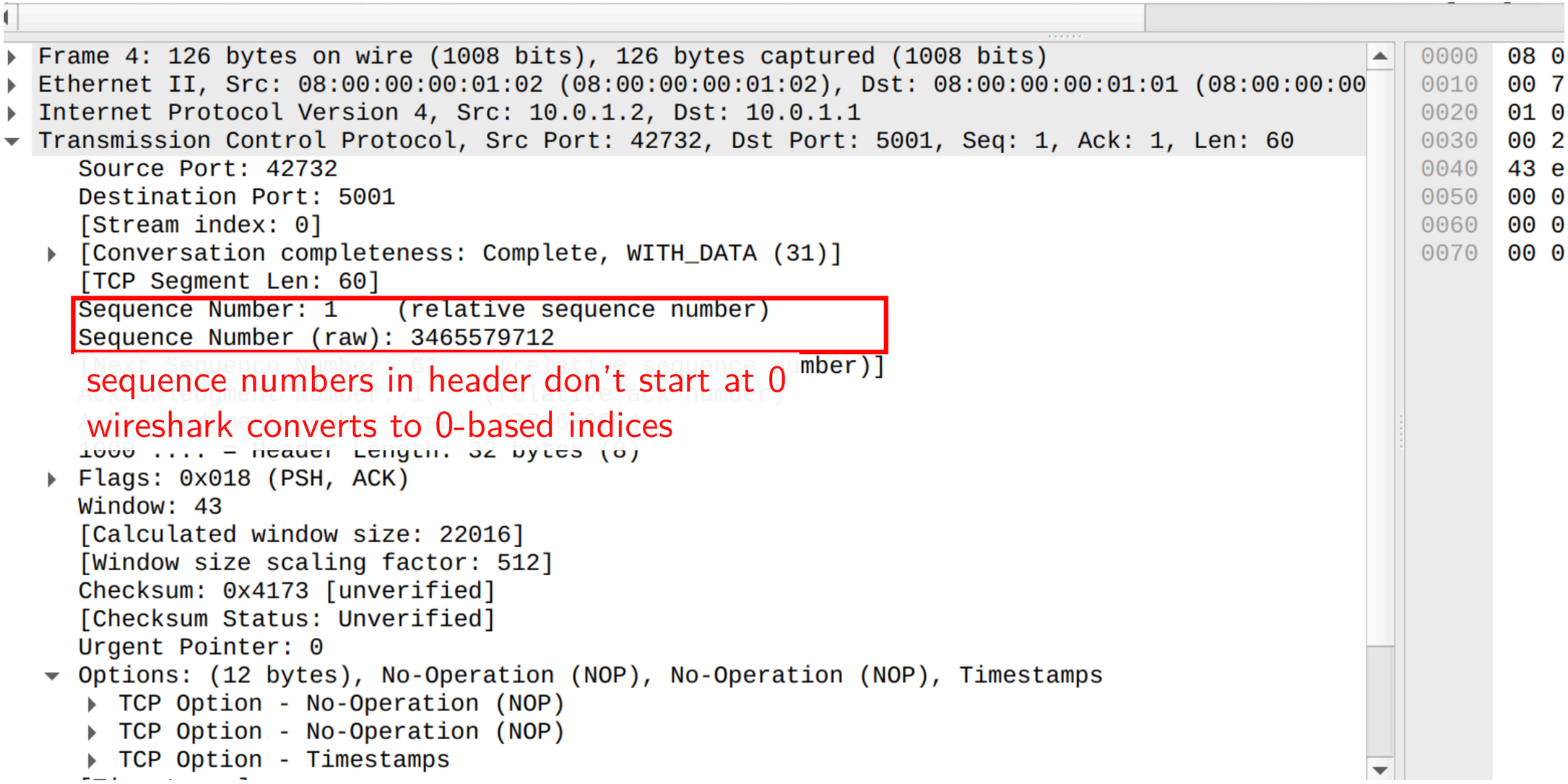
first data packet



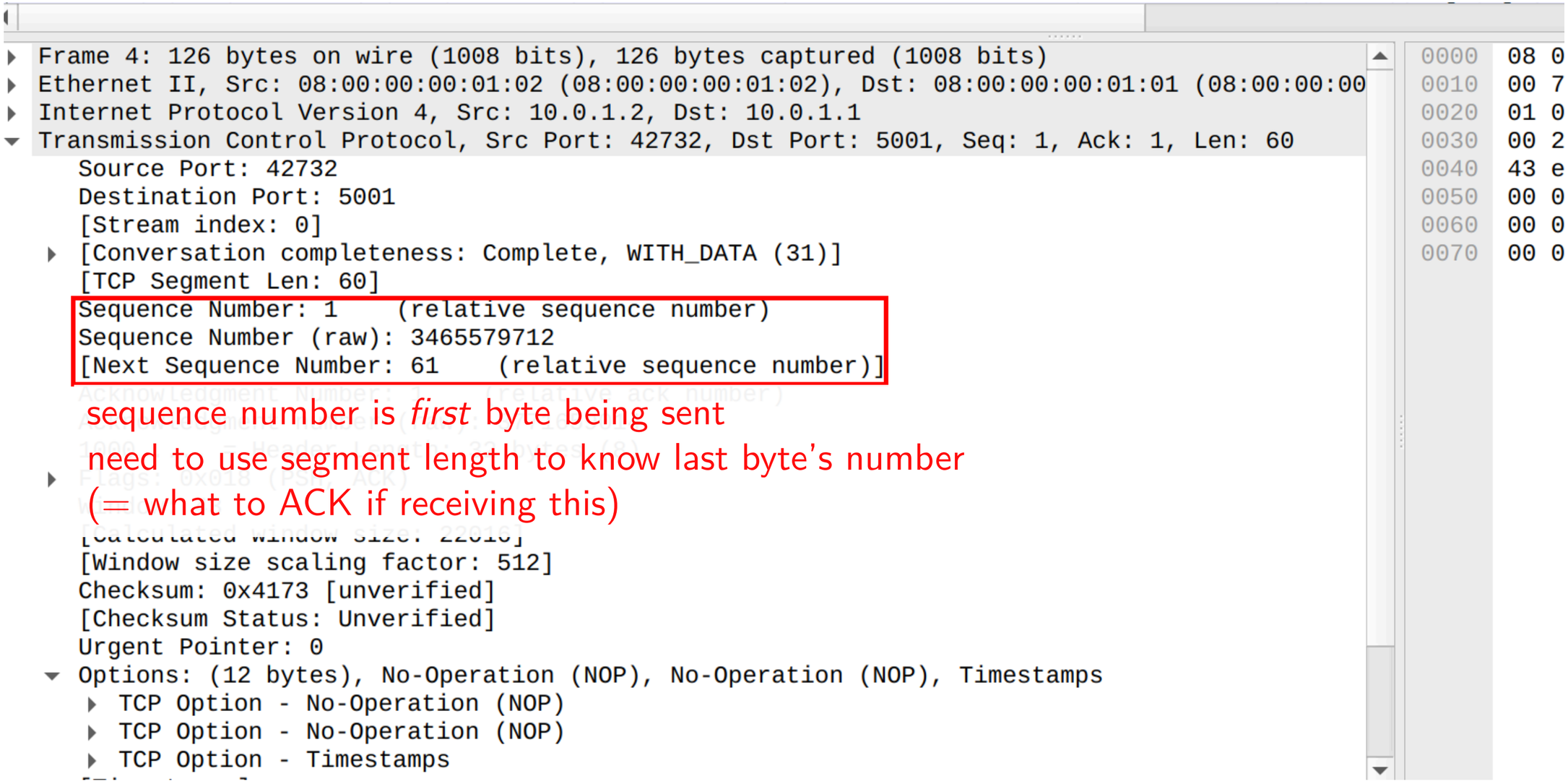
first data packet



first data packet



first data packet

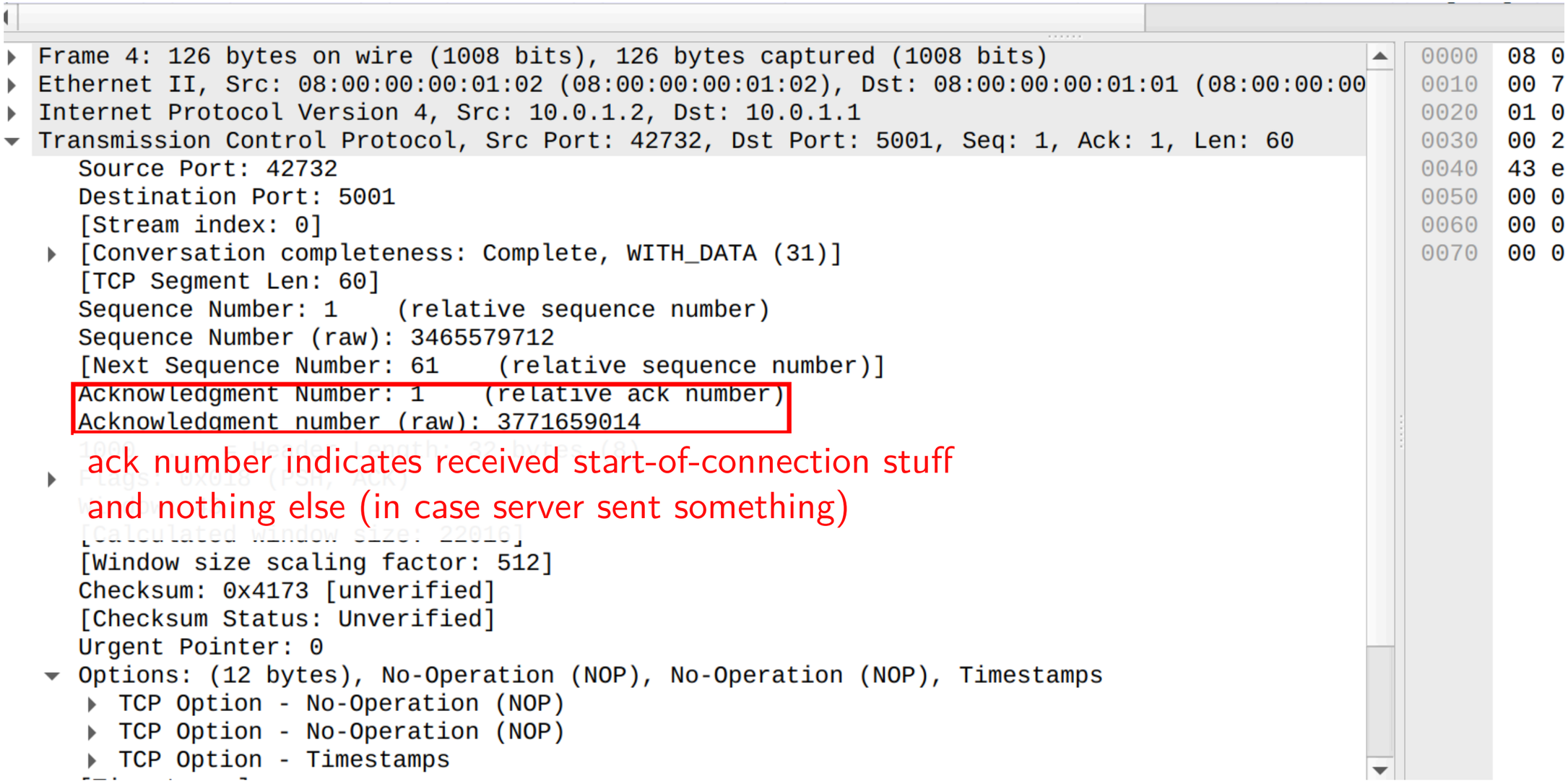


sequence number is *first* byte being sent

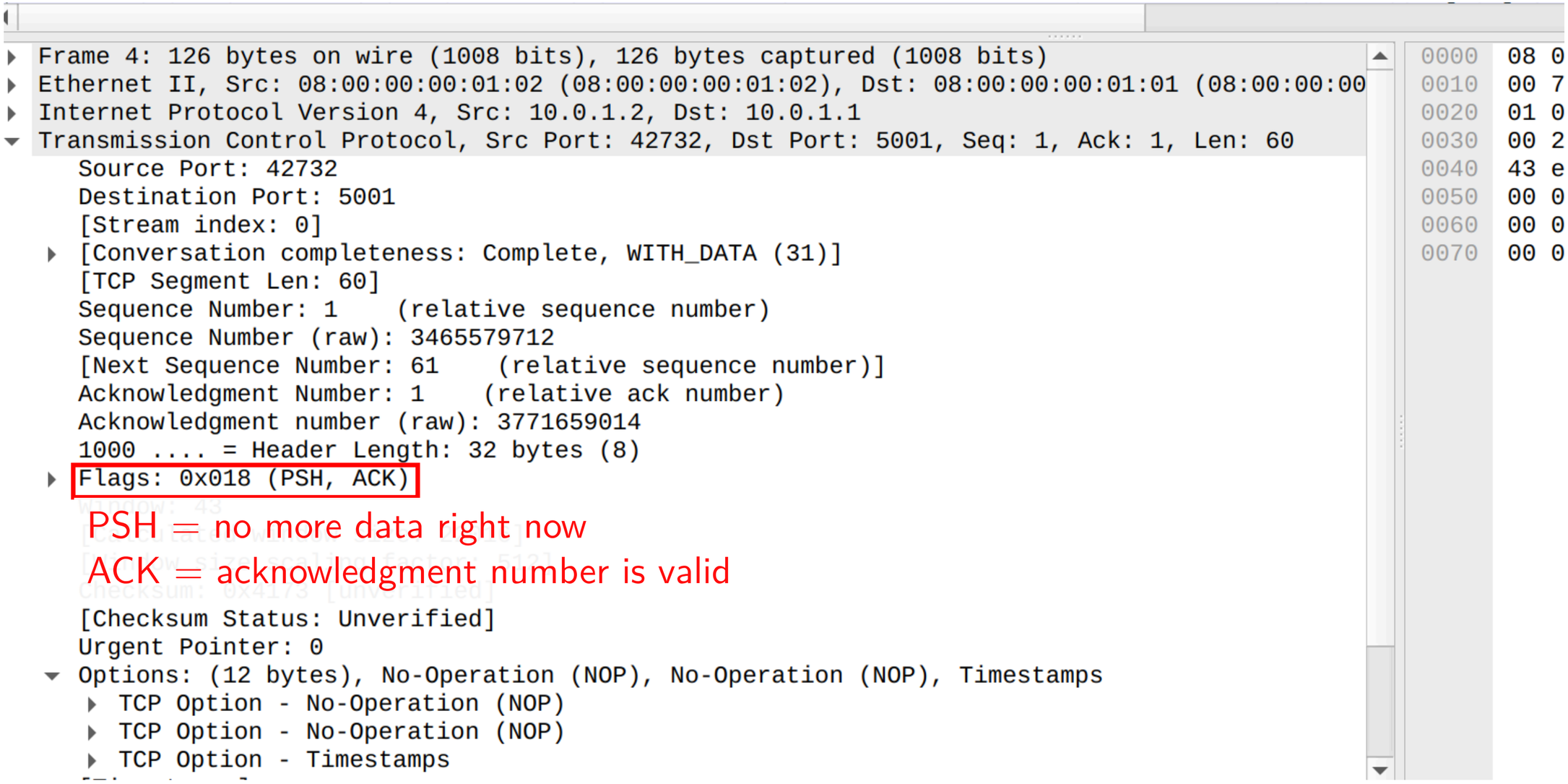
need to use segment length to know last byte's number

(= what to ACK if receiving this)

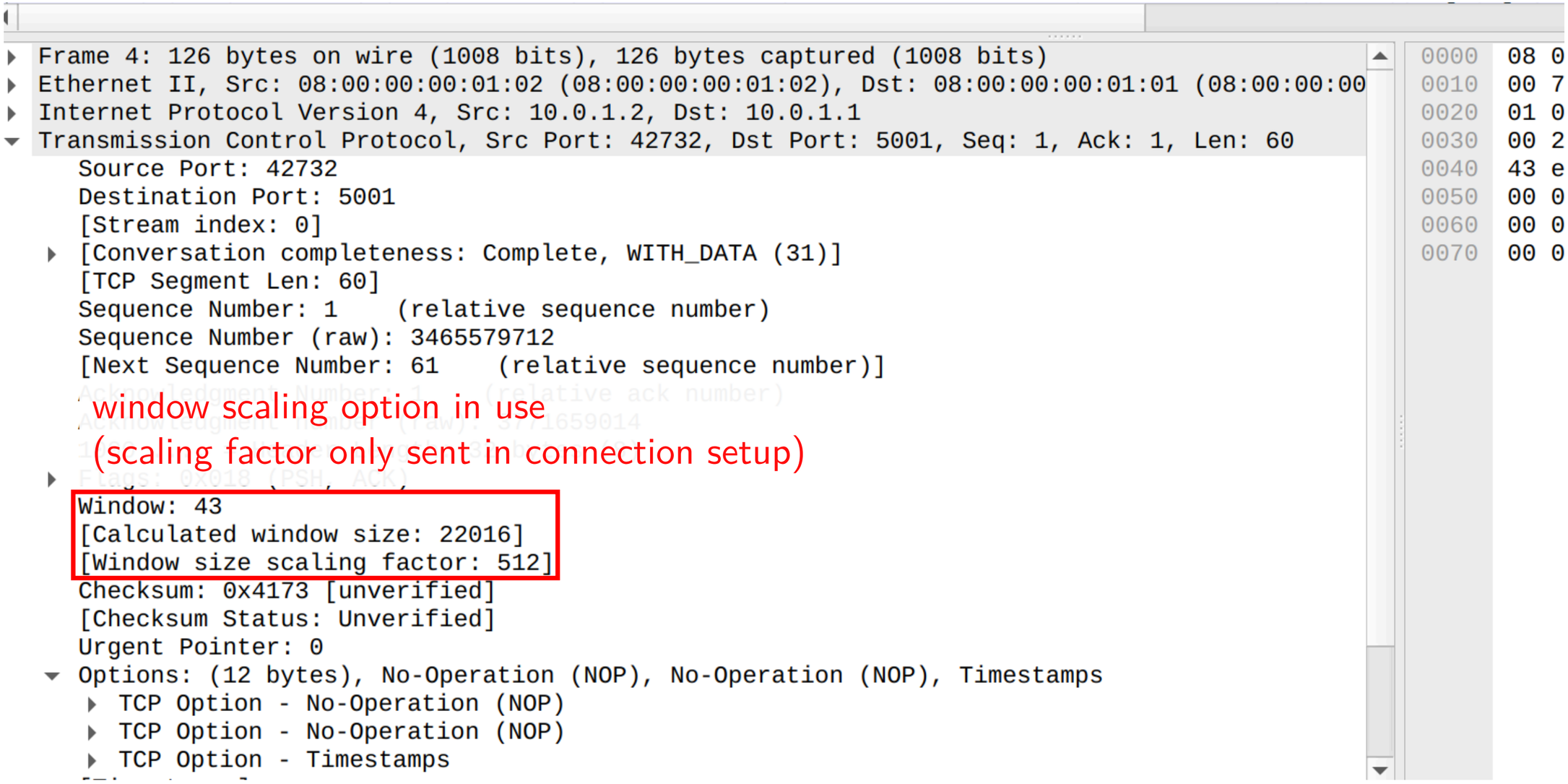
first data packet



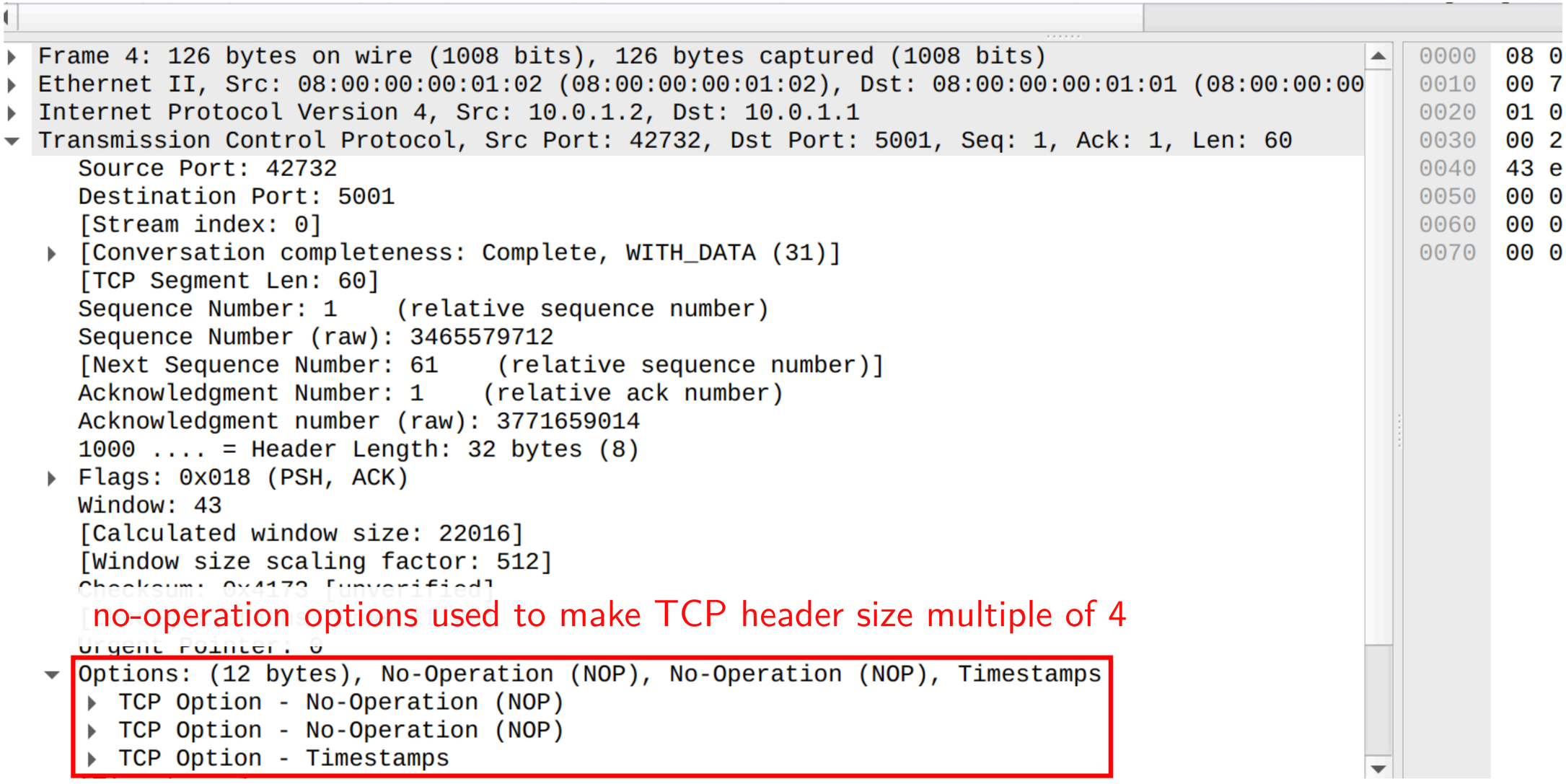
first data packet



first data packet

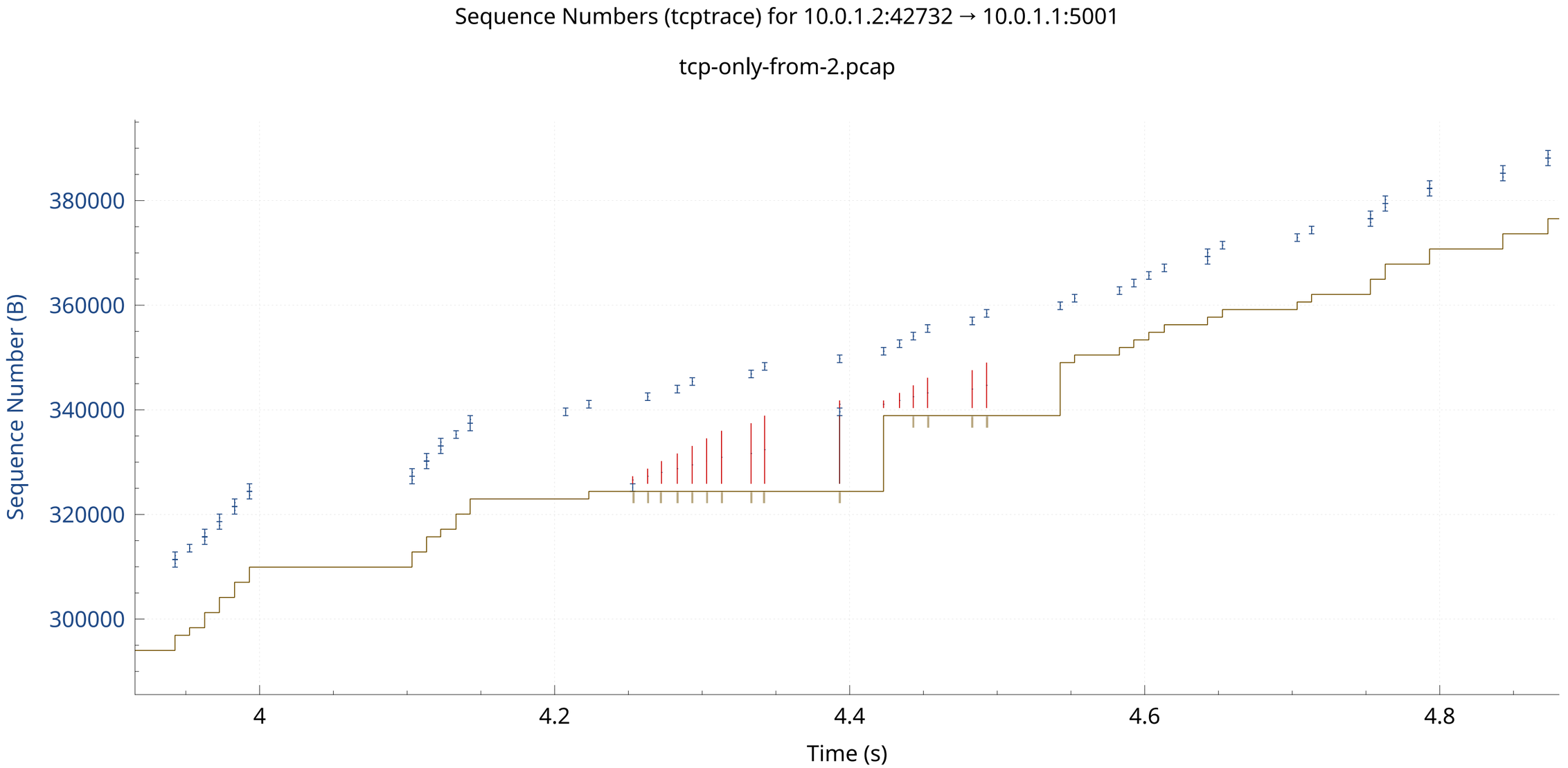


first data packet



no-operation options used to make TCP header size multiple of 4

sequence numbers graph



reading thigs graph

bottom line = last ack number

notches on bottom line = duplicate acks

red lines = selective ACK info

diff. timing in opposite direction

